Inapproximability of Treedepth, Treewidth, and Total Elimination Depth on Graphs of Bounded Degree

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Abstract

We prove approximation hardness for treedepth, treewidth, and total elimination depth. Unless P = NP, treedepth is hard to approximate within every fixed factor below 18/17, treewidth below 12/11, and total elimination depth below 63/62. Under the ordinary Unique Games Conjecture, the corresponding thresholds are 4/3, 4/3, and 8/7. Under the Small Set Expansion Hypothesis, every fixed constant factor is hard for all three parameters. These results hold on simple unweighted graphs of bounded maximum degree, with the degree bound depending only on the target factor. In particular, the treedepth result resolves a conjecture of Schirrmacher, Siebertz, and Vigny.

The three NP-hardness reductions reuse one bounded-occurrence parity constraint gap. The other reductions use independent-set and vertex-cover gaps and an SSE partition gap. To bound the degree, we replace clique and biclique factors by bounded-degree graphs built from expanders. A small exceptional set lets us extend sparse solutions to the original graph with additive cost below the hardness gap. Since these replacements delete edges, the completeness bounds remain valid. The general sparsification theorem does not require balanced biclique sides. Our results also imply hardness for trivially perfect completion, vertex integrity, fracture number, starwidth, average and maximum general customizable-label size, and average hierarchical customizable-label size. For hierarchical customizable labels and CCH upward vertex search spaces, the minimum total size equals total elimination depth, so their total and average sizes inherit all three hardness ranges as well. The treewidth gaps give exponential approximation lower bounds for the total table size of binary junction trees.

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1 Introduction

Treedepth is the minimum height of an *elimination forest*: a rooted forest on the graph's vertices in which the endpoints of every edge are comparable in the ancestor order. For connected graphs, every elimination forest is a tree. An optimal elimination tree can be chosen to encode a recursive vertex-query strategy: querying a vertex either finds the target or identifies the component containing it after that vertex is removed. Here the target must be queried even when it is the only remaining candidate; we call this the confirming convention. Treedepth is also characterized by vertex ranking and symbolic elimination [BDJ$^+$94], and it is studied directly in parameterized algorithms [RRSVS14]. Low-treedepth decompositions are used in the structural theory of sparse graphs. Nešetřil and Ossona de Mendez survey these methods, and their earlier work connects them to subgraph coloring and homomorphism bounds [NOdM15, NOdM06].

A *tree decomposition* consists of a tree of vertex sets, called bags, such that every graph edge has both endpoints in some bag and the bags containing any fixed vertex form a nonempty connected subtree. Treewidth is the minimum possible largest bag size minus one. Decompositions of small width support dynamic programming for many graph problems and have uses in graph-minor theory. For surveys of these applications, see [Bod93, Bod06]. We write $\mathrm{td}(G)$ and $\mathrm{tw}(G)$ for treedepth and treewidth respectively.

Bonnet, Neuen, and Sokołowski proved that approximating treedepth within factor 1.0003 is NP-hard [BNS25]. Bonnet proved the corresponding 1.00005 hardness bound for treewidth [Bon25]. Under the Small Set Expansion Hypothesis (SSEH), treewidth is hard to approximate within every fixed constant factor [WAPL15, Theorem 1.1]. On the algorithmic side, Anand gives an $O(\log n)$-approximation for treedepth [Ana26], while Feige, Hajiaghayi, and Lee give a polynomial-time $O(\sqrt{\log(\mathrm{tw}(G))})$-approximation for treewidth [FHL05]. We strengthen the hardness bounds and show that they persist on graphs of bounded maximum degree, resolving the conjecture of Schirrmacher, Siebertz, and Vigny [SSV25, Conjecture 1].

We also study *total elimination depth*, $\mathrm{totd}(G)$, the minimum sum of vertex depths in an elimination forest (roots have depth one). This is the vertex-partition-tree sum of [HBB$^+$21]. On a connected graph this is the min-sum cost of vertex-query search for a target under the confirming convention [HBB$^+$21, Szy26a].

1.1 Our results and techniques

Our results hold on simple unweighted graphs of bounded maximum degree. The table summarizes Corollaries 6.22, 6.23, and 6.25. Here UGC denotes the ordinary Unique Games Conjecture and SSEH the Small Set Expansion Hypothesis. The NP-hardness bounds exclude polynomial-time approximation unless P = NP.

Hardness	Treedepth	Treewidth	Total elimination depth
NP-hard	$1 \le r < 18/17$	$1 \le r < 12/11$	$1 \le r < 63/62$
Unique Games-hard	$1 \le r < 4/3$	$1 \le r < 4/3$	$1 \le r < 8/7$
SSE-hard	every fixed $r \ge 1$	every fixed $r \ge 1$	every fixed $r \ge 1$

The degree bound depends only on the excluded approximation factor. All reductions are deterministic. Figure 1 summarizes the source, construction, degree-reduction, and application routes.

The UGC reductions for treedepth and total elimination depth start from a bounded-degree form of the Khot–Regev independent-set gap [KR03], which we derive from Austrin, Khot, and Safra [AKS09]. We place copies of the source graph in clique layers. A large independent set gives a

shallow elimination forest by putting the other vertices on a common prefix and branching below it. For the lower bound, vertices in different child subtrees are incomparable, so no source edge can join vertices represented in different children. This limits the height saved by branching. For total depth, we also count the depth shift that the prefix imposes on each child, then bound the combined savings across the children.

For treewidth, a multipartite product first gives a vertex-cover gap on tripartite graphs with equal-sized parts. We replicate these source graphs before completing each global part to a clique. Bonnet's central bag argument then forces a cover from every copy into one bag, while the replication keeps the additional YES cost to one copy's scale [Bon25, Lemma 6].

Under SSEH, we transfer the Raghavendra–Steurer–Tulsiani partition gap [RST12] to treedepth and total depth through a line graph whose vertices represent source edges. The YES partition gives a shallow forest. In the NO case, a small balanced separator in the line graph would divide the surviving source edges into two large groups with disjoint endpoint sets. Their endpoints would give a balanced source cut whose boundary lies among the host edges represented by the separator, contradicting the source expansion bound. The same separator bounds yield the total depth gap. Wu, Austrin, Pitassi, and Liu already prove the corresponding SSE treewidth hardness [WAPL15, Theorem 1.1].

For treedepth under P $\neq$ NP, we refine the truth block and anchor construction of Bonnet, Neuen, and Sokołowski [BNS25] using the parity gap of Hazan, Safra, and Schwartz [HSS03, Theorem 10]. The two truth blocks of each variable encode its values. Each equation has four witnesses for its satisfying assignments, and star edges ensure that at most three can be selected together. Counting witnesses consistent with a partial assignment gives the independent-set bound. Balanced parts and the anchor graph then transfer this gap to treedepth. The same source gap transfers to treewidth by the replicated clique completion above.

For total depth, we reuse the normalized bounded-occurrence parity source of Theorem 5.1; Appendix A.5 derives its balance and divisibility conditions. A different tripartite construction uses literal blocks and auxiliary vertex pairs. Completing the parts to cliques and attaching three equally sized independent terminal sets makes the leading total-depth cost depend on three path lengths, by Lemmas 5.6 and 5.7. The pair gadgets encode the number of satisfied equations exactly (Proposition 5.8). Fixed block and padding sizes give hardness below 63/62 (Theorem 5.9); Corollary 6.25 preserves every fixed factor below it on bounded-degree graphs.

The unrestricted reductions use cliques and bicliques to enforce comparability and bag constraints, so their output degrees can be large. The bounded degree section replaces these dense factors with bounded degree graphs built from expanders. Their no hole condition says that every two sufficiently large disjoint subsets of a clique factor have an edge between them, and that sufficiently large subsets on opposite sides of a biclique factor also have an edge between them. Completion lemmas show how to extend a sparse solution to the original graph with an additive increase controlled by a small exceptional set. We choose the thresholds so this increase stays within the hardness gap. Edge deletion preserves the YES bounds, so the same reductions prove bounded degree hardness for all three parameters. The general sparsification theorems require bounded incidence of factors at each vertex and bounded degree of the retained edges, but no balance assumption on biclique sides. If a side is smaller than the no-hole threshold, that condition is vacuous and the factor can be deleted; otherwise its side ratio is bounded automatically.

1.2 Applications

Our main application is to *Trivially Perfect Completion*, where edges are added until the graph is a forest closure. Two distinct vertices are adjacent if and only if one is an ancestor of the other in a

rooted forest. The objective is the number of added edges. For every fixed $C \geq 1$, approximating this objective within factor C is SSE-hard on graphs of maximum degree bounded by a constant depending only on C (Theorem 7.2). The SSE proof uses the same line graph as our SSE treedepth reduction, but needs a bound on the *sum* of elimination depths. A balanced separator, whose deletion leaves every component with at most half the vertices, supplies this estimate, and an adjusted-cost argument handles the offset between total depth and the number of added edges. The identity relating these objectives is established in [HBB$^+$21]. Wu, Austrin, Pitassi, and Liu prove SSE hardness for chordal Minimum Fill-In at every fixed factor [WAPL15], but that result does not cover trivially perfect completions, which form a more restricted target class. The adjusted-cost argument also transfers both constant total-depth gaps to this completion objective, even on graphs of bounded maximum degree: every fixed factor below 63/62 is NP-hard, and every fixed factor below 8/7 is Unique Games-hard.

Section 7 derives the following consequences from these constructions and from established reductions:

- *Deletion and labeling objectives.* Vertex integrity minimizes the number of deleted vertices plus the largest remaining component. It inherits UGC hardness below 4/3 and SSE hardness at every fixed factor. The SSE separator gap also gives this hardness for fracture number, starwidth, average and maximum general customizable-label size, and average hierarchical customizable-label size, using the comparisons in [GHK$^+$24] and the label lower bound of Blum and Storandt [BS22]. The SSE line-graph construction also gives edge-ranking hardness at every fixed factor on simple bipartite graphs, via $\mathrm{er}(B) = \mathrm{td}(\mathcal{L}(B))$ [BDJ$^+$94]. For hierarchical customizable labels and CCH upward vertex search spaces, the minimum total size equals $\mathrm{totd}(G)$. Their total and average sizes therefore also inherit NP hardness below 63/62 and UGC hardness below 8/7, including on bounded-degree graphs (Corollary 7.5).

- *Automata and hypergraph widths.* Gruber's established transformations [Gru08] carry the treedepth gaps to star height of regular languages represented by trim binary bideterministic partial automata. Replacing each vertex by a pair of adjacent copies carries the treewidth gaps to fractional, generalized, and ordinary hypertree width on rank-two hypergraphs, following the treewidth-hardness strategy of [Mar09].

- *Junction-tree table size.* For n binary variables, the total state space is hard to approximate within $2^{\chi n}$ for every fixed $0 < \chi < 1/18$ under P $\neq$ NP, and for every fixed $0 < \chi < 1/6$ under UGC. These exponents follow from the linear width gaps. Constant-factor hardness was proved by Bodlaender and Fomin [BF02].

Established equivalent formulations of treedepth, including vertex ranking and minimum symbolic elimination height, inherit the same parameter bounds.

1.3 Related work and attribution

Bonnet's treewidth reduction [Bon25] introduced the truth-block and assignment-witness framework with three clique parts used in Bonnet, Neuen, and Sokołowski's treedepth reduction [BNS25]. Our NP-hardness proofs for treedepth and treewidth inherit that encoding. For treedepth, we also use their anchor graph and apply their anchor theorem as a black box, extending the anchor cliques only in the pairwise-incomparable case and analyzing comparable anchors separately. Our changes are the parity-witness stars and the partial-assignment analysis determining the block sizes. Appendix A.4 gives the detailed comparison. The treewidth argument also uses Bonnet's central-bag lower bound. Our replication lemma controls the completeness overhead for the source gaps used here.

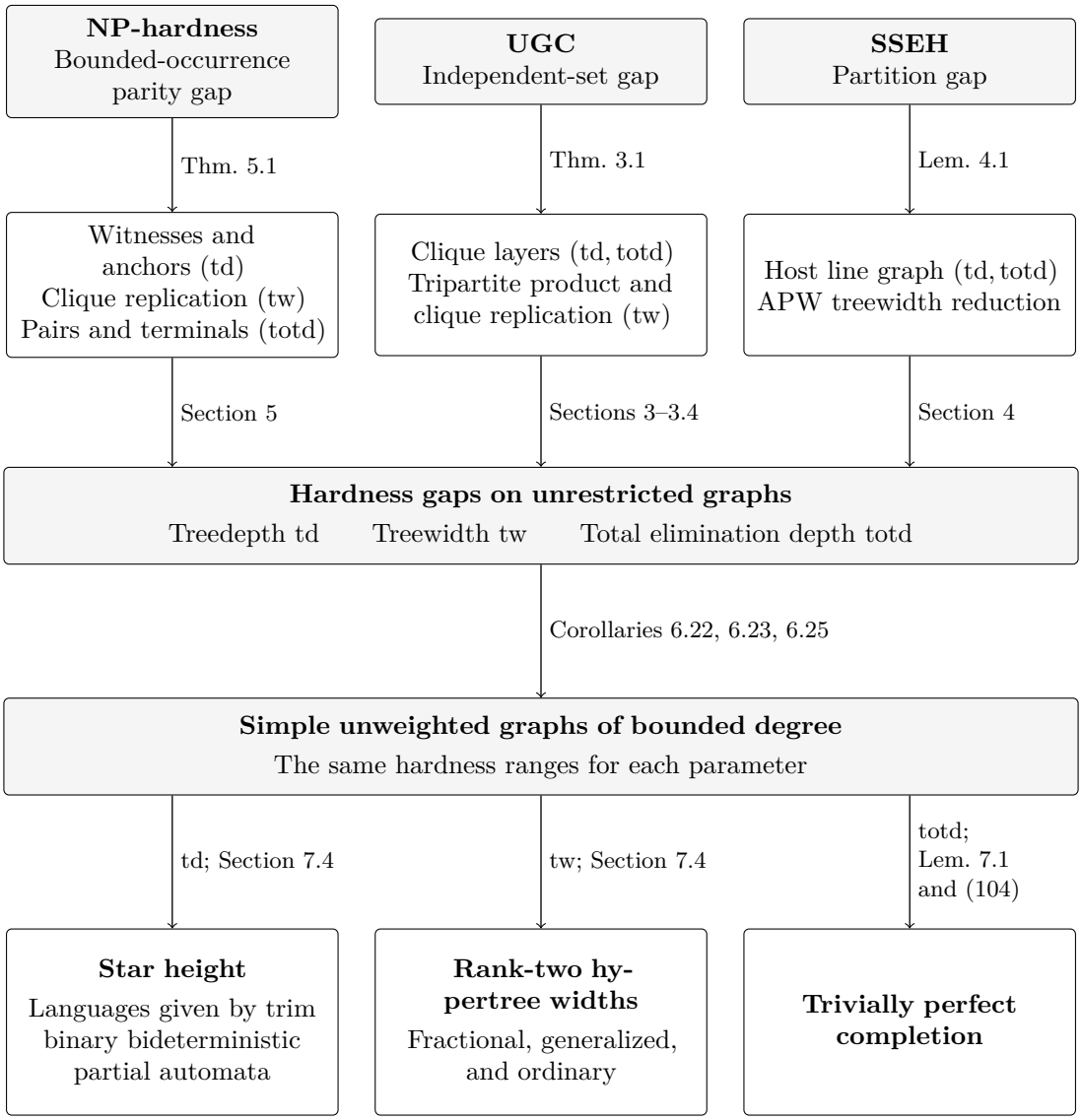
Khot and Regev [KR03] supply the independent-set gap underlying our ordinary-UGC reductions. We use its bounded-degree form throughout, derived from the Gaussian density source of Austrin, Khot, and Safra [AKS09]. Bérczi, Chandrasekaran, Király, Lee, and Xu [BCK+17, Sections 3.5–3.6] prove a weighted k-partite Vertex Cover gap with ratio approaching $2(k-1)/k$. Our treewidth transfer uses an equal-part unweighted gap, which we derive from the bounded-degree independent-set source by a multipartite product in Section 3.4. Our expander replacements use the standard Margulis–Gabber–Galil family, as presented in [HLW06, Construction 8.1 and Theorem 8.2].

Wu, Austrin, Pitassi, and Liu [WAPL15] use the expansion reductions of Raghavendra, Steurer, and Tulsiani [RST12] to prove SSE hardness for treewidth, pathwidth, and related objectives at every fixed factor.

For our bounded-degree result, Section 6 derives the degree bound from the weighted source by the same factor-completion tools used for NP and UGC hardness.

Høgemo, Bergougnoux, Brandes, Paul, and Telle [HBB+21] study VPT-sum, relate it to trivially perfect completion, and prove NP-hardness. The vertex-query interpretation described above gives the same objective as expected confirming search cost up to the factor $|V(G)|$. Under the alternative convention that stops once the target is uniquely identified, the summed optimum differs by at most $|V(G)|$. Section 7.2 proves this comparison and transfers our NP and UGC hardness ranges to average-case search under either convention, also on connected bounded-degree graphs. Szyfelbein [Szy26a] gives an $O(\sqrt{\log n})$-approximation for the more general search problem with nonuniform target weights and query costs, using the identification convention. The same asymptotic bound applies to our uniform confirming objective, as explained in Section 7.2. A subsequent preprint also studies target-dependent costs [Szy26b].

The edge-query analogue is Dasgupta's similarity-based hierarchical clustering objective [Das16]. For a hierarchy T with leaves $V(G)$, its cost is $\sum_{uv \in E(G)} w_{uv} \, |C_T(u,v)|$, where $C_T(u,v)$ is the leaf set below the least common ancestor of u and v. For connected graphs, Høgemo, Bergougnoux, Brandes, Paul, and Telle [HBB+21, Theorem 6] identify its unweighted optimum with EPT-sum, the minimum sum of leaf depths in an edge-partition tree. The unweighted problem is NP-hard by Høgemo, Paul, and Telle [HPT20]. Charikar and Chatziafratis [CC17] give an $O(\sqrt{\log n})$-approximation and prove hardness within every fixed constant factor under SSEH, including for unweighted graphs. Szyfelbein and Dereniowski [SD26] extend this SSE hardness to variants that stop at clusters inducing trees or graphs of bounded diameter. The total-depth bounds here concern vertex queries, whose feasible decompositions differ from those of edge queries.



The hardness ranges are listed in the introduction's table. The degree bound depends only on the fixed excluded approximation factor. Each application branch uses its indicated parameter under all three hardness assumptions. The SSE treewidth route uses [WAPL15] and its bounded-degree transfer in Corollary 6.23.

Figure 1: The main reduction routes. Each source column contains the constructions for the three parameters; the lower branches show three applications. Further applications appear in Section 7.

2 Preliminaries

All target graphs produced by our reductions are finite, simple, and unweighted. Source graphs may have weights, parallel edges, or loops when explicitly stated. An *independent set* contains no edge, including a loop. A *clique* has every pair of distinct vertices adjacent. A *vertex cover* meets every edge. For an unweighted graph G, let $\alpha(G)$ denote the maximum independent-set size and $\mathrm{vc}(G)$ the minimum vertex-cover size. Taking complements gives

$$\mathrm{vc}(G) = |V(G)| - \alpha(G). \tag{1}$$

An approximation algorithm for a minimization problem returns a feasible solution whose cost is at most the stated factor times the optimum. Our hardness ranges have open upper endpoints. It therefore suffices to prove them for rational factors: for a real factor below an endpoint, choose a larger rational factor still below that endpoint. An algorithm with the smaller guarantee also satisfies the larger one. In each reduction the factor and all auxiliary constants are fixed before the input instance, and the distinguishing cutoffs are computable from that instance.

We write $G[U]$ for the subgraph induced by U, $G - U$ for $G[V(G) \setminus U]$, $\Delta(G)$ for maximum degree, and $[q] = \{1, \dots, q\}$. A spanning subgraph retains all vertices and some of the edges. A *matching* is a set of edges with pairwise disjoint endpoints. A matching is *perfect* if it meets every vertex.

For $S \subseteq V(G)$, the *edge boundary* is

$$\delta_G(S) = \{uv \in E(G) : u \in S,\ v \in V(G) \setminus S\}.$$

In a multigraph, this is a multiset, counting parallel edge occurrences separately.

In a rooted forest, two vertices are *comparable* if one is an ancestor of the other. We count a vertex as its own ancestor and descendant. A *proper* ancestor is a different vertex. The subtree rooted at a vertex consists of that vertex and all its descendants. The path from the root of a component to one of its vertices is an *ancestor path*. We also allow the empty path. The *restriction* of a forest to a vertex subset deletes all other vertices and attaches each survivor to its nearest surviving proper ancestor, or makes it a root if none remains. It preserves the ancestor order between survivors and never increases their depths.

An *elimination forest* F of a graph H is a rooted forest on $V(H)$ such that the endpoints of every edge of H are comparable in the ancestor order. Its height $h(F)$ is the maximum number of vertices on a root-to-leaf path, with $h(\varnothing) = 0$. The *treedepth* $\mathrm{td}(H)$ is the minimum height of an elimination forest of H. We set $\mathrm{td}(\varnothing) = 0$. For disconnected graphs, treedepth is the maximum treedepth of a component.

The *total depth* of an elimination forest F is $\mathrm{totd}(F) = \sum_{v \in V(F)} \mathrm{depth}_F(v)$, with roots at depth one. Let $\mathrm{totd}(G)$ be the minimum total depth of an elimination forest of G. This objective is also called *VPT-sum* [HBB+21]. For disconnected graphs, totd is the sum of its values on the components. Restricting a forest to each component proves one inequality, and their disjoint union proves the other.

A *tree decomposition* of H consists of a tree T and bags $B_t \subseteq V(H)$ for $t \in V(T)$. For each $v \in V(H)$, write $T_v = \{t \in V(T) : v \in B_t\}$ for the set of nodes whose bags contain v. We require each T_v to be nonempty and to induce a connected subtree of T, and $T_u \cap T_v \neq \varnothing$ for every edge $uv \in E(H)$. Its width is $\max_{t \in V(T)} |B_t| - 1$, and the *treewidth* $\mathrm{tw}(H)$ is the minimum such width.

Lemma 2.1 (Minor monotonicity of treedepth and treewidth)**.** *If H' is obtained from H by vertex deletions, edge deletions, and edge contractions, then* $\mathrm{td}(H') \leq \mathrm{td}(H)$ *and* $\mathrm{tw}(H') \leq \mathrm{tw}(H)$.

Proof. For treedepth, vertex deletion is handled by restricting an elimination forest, and edge deletion leaves it valid. For an edge contraction uv, its endpoints are comparable in any elimination forest. Suppose u is above v. Restrict the forest to $V(H) \setminus \{v\}$, using u for the contracted vertex. All edges not incident with v retain comparable endpoints. Every ancestor of v is comparable with u, and every descendant of v is a descendant of u, so the edges transferred from v to u also have comparable endpoints. Restriction does not increase height.

For treewidth, delete vertices from every bag or leave the bags unchanged when deleting edges. To contract uv, replace u and v by the contracted vertex in every bag. The node sets T_u, T_v intersect because uv is an edge, so their union is connected. All edges remain covered and no bag grows. □

Lemma 2.2 (An ancestor in a connected set)**.** *Let F be an elimination forest of H, and let $\varnothing \neq U \subseteq V(H)$ induce a connected subgraph. The vertex of U with minimum depth in F is unique and is an ancestor of every vertex of U.*

Proof. Choose a minimum-depth vertex $r \in U$. If a path in $H[U]$ starting at r left its forest subtree, the first edge leaving it would join a descendant of r to a proper ancestor of r, since its endpoints are comparable. This contradicts the choice of r. Thus all of U lies below r, which also proves uniqueness. □

3 Hardness under the Unique Games Conjecture

We prove ordinary-UGC hardness below 4/3 for treedepth and treewidth, and below 8/7 for total elimination depth. The reductions start from the same bounded-degree independent-set promise.

3.1 Independent-set source

We use the following bounded-degree form of the Khot–Regev independent-set gap [KR03].

Theorem 3.1 (Bounded-degree independent-set source)**.** *For every fixed rational $0 < \eta < 1/10$, there is a constant d_η such that the following promise problem is Unique Games-hard under deterministic polynomial-time reductions. Given a simple unweighted graph G with $|V(G)| = n \geq 1$ and $\Delta(G) \leq d_\eta$, distinguish:*

YES: $\alpha(G) \geq (1/2 - \eta)n$.

NO: $\alpha(G) < \eta n$.

We derive this bounded-degree form in Appendix A.1, using the AKS density source in Theorem A.2 and BSS spectral sparsification in Theorem A.6.

3.2 Treedepth

For a fixed integer q, the construction replaces the source graph by q clique layers. A large independent set gives a shallow elimination forest: its copies can be placed in separate branches after the remaining vertices have been removed. For soundness, we prove that branching can save few levels when the source independence number is small. The first-branch lemmas will also be used for total depth.

3.2.1 The layered construction and first-branch notation

Let G be a finite simple graph on n vertices and let $q \geq 1$ be an integer. Define $S_q(G)$ on $V(G) \times [q]$, writing $v_i = (v, i)$. For distinct output vertices u_i, v_j, set

$$u_i v_j \in E(S_q(G)) \quad \iff \quad \big(i = j \text{ and } u \neq v\big) \text{ or } \big(i \neq j \text{ and } uv \in E(G)\big).$$

Thus each layer $V(G) \times \{i\}$ is a clique, and the displayed rule specifies all edges between layers. Its size is

$$|V(S_q(G))| = qn, \qquad |E(S_q(G))| = q\binom{n}{2} + q(q-1)|E(G)|,$$

so the construction has polynomial size for fixed q. For $n \geq 1$, its maximum degree is

$$\Delta(S_q(G)) = n - 1 + (q-1)\Delta(G).$$

The clique layers therefore give a dense output even for a bounded-degree source. Corollaries 6.22 and 6.25 replace these layers by bounded-degree expanders while retaining all cross-layer source edges.

Writing Λ for the Unique Games input, the reduction has the following form for the fixed integer q chosen in the gap proof:

$$\Lambda \xrightarrow[\text{independent-set source}]{\text{Theorem 3.1}} G \xrightarrow[\text{clique layers}]{\text{Lemma 3.2}} S_q(G).$$

Lemma 3.2 and Theorem 3.9 analyze its treedepth gap; Lemma 3.11 and Theorem 3.13 analyze its total-depth gap.

Bonnet's construction [Bon25, Section 3.1] completes three parts to cliques. Here the q parts are copies of one source graph. The first-branch analysis below gives a treedepth lower bound for arbitrary q.

Notation at the first branch. Fix a graph G on $n \geq 1$ vertices and let $q \geq 2$. Let F be an elimination forest of $S_q(G)$ that is not a single chain. Vertices in different rooted trees, or in different child subtrees of one node, are nonadjacent. We use this property in the first-branch analysis below.

We define its initial chain P and child trees as follows. If F has at least two roots, let P be empty and let $F_1, \ldots, F_\nu$ be its rooted trees. Otherwise, start at the unique root and follow the unique child until the first vertex with at least two children. Such a vertex exists because F is not a single chain. Let P include this vertex, and let $F_1, \ldots, F_\nu$ be the trees rooted at its children. In both cases $\nu \geq 2$. Measure the depths in each F_j from its own root at depth one.

For each $j \in [\nu]$, put $N_j = |V(F_j)|$ and define

$$I_j = \{i \in [q] : V(F_j) \cap (V(G) \times \{i\}) \neq \varnothing\}, \quad q_j = |I_j|,$$
$$U_j = \{v \in V(G) : \exists i \in [q],\ v_i \in V(F_j)\}, \qquad n_j = |U_j|.$$

Thus U_j is the set of source vertices represented in child F_j, without counting repeated copies. Of the $q_j n_j$ pairs in $U_j \times I_j$, the child contains N_j pairs. Write $M_j = q_j n_j - N_j$ for the number of missing pairs, and put

$$D = \{v \in V(G) : |\{j \in [\nu] : v \in U_j\}| \geq 2\}.$$

We use this notation in the treedepth and total-depth lower-bound arguments.

3.2.2 Treedepth bounds

Lemma 3.2 (YES upper bound)**.** *For every graph G on n vertices and integer $q \geq 1$,*

$$\mathrm{td}(S_q(G)) \leq q(n - \alpha(G)) + \alpha(G) = qn - (q-1)\alpha(G).$$

Proof. Fix a maximum independent set I of G. Put the q copies of every vertex of $V(G) \setminus I$ on one chain. Beneath this chain, put one chain containing the copies of I in each layer. Each lower chain contains one layer's copies of I. Vertices in different lower chains lie in different layers, and an edge between them would require two vertices of I to be adjacent in G. Every edge incident to a vertex on the top chain has comparable endpoints. This is an elimination forest of height $q(n-|I|)+|I|$. If $n = 0$, it is the empty forest. If the top chain is empty and $n > 0$, use the q lower chains as separate rooted trees. □

Write $A_{q,\eta} := (q+1)/2 + (q-1)\eta$. In a YES instance of Theorem 3.1, Lemma 3.2 gives

$$\begin{aligned} \mathrm{td}(S_q(G)) &\overset{\text{Lem. 3.2}}{\leq} qn - (q-1)\alpha(G) \\ &\overset{\text{Thm. 3.1}}{\leq} A_{q,\eta} n. \end{aligned} \tag{2}$$

The structural lower bound. At the first branch, a clique layer cannot occur in two child subtrees, since vertices in different children are incomparable. Every child therefore uses fewer layers, allowing induction on q. To combine the child bounds, we also need to control how much of the source graph they share. Source vertices repeated across children will be isolated in the union of their supports. Consequently, the child vertex-cover numbers sum to at most $\mathrm{vc}(G)$. The following inequality will convert this bound into a bound on the height of the whole forest, whose longest path enters only one child.

Lemma 3.3 (Branching inequality)**.** *For a positive integer ν, positive reals $q_1, \ldots, q_\nu$, and nonnegative reals $n_1, \ldots, n_\nu$,*

$$3\sum_{j=1}^{\nu} q_j n_j - 2\max_{j\in[\nu]}(q_j n_j) \leq \left(\sum_{j=1}^{\nu} q_j\right)\left(\sum_{j=1}^{\nu} n_j\right).$$

The proof is given in Appendix B.1.

Lemma 3.4 (Layers in different children)**.** *The sets I_j are pairwise disjoint. Moreover, $\nu \leq \sum_{j=1}^{\nu} q_j \leq q$ and $1 \leq q_j \leq q - \nu + 1 < q$ for every $j \in [\nu]$.*

Proof. Vertices in different child trees are incomparable, so a clique layer cannot occur in two children. Thus the I_j are pairwise disjoint. Each is nonempty, giving $\nu \leq \sum_{j=1}^{\nu} q_j \leq q$. The other $\nu - 1$ children use at least $\nu - 1$ layers, so $q_j \leq q - \nu + 1 < q$. □

Lemma 3.5 (Source vertices in different children)**.** *The set D is independent in G, and*

$$\sum_{j=1}^{\nu} n_j \leq n + (\nu-1)\alpha(G) \leq n + (q-1)\alpha(G) \leq qn. \tag{3}$$

Moreover, put $c = \mathrm{vc}(G)$ *and* $c_j = \mathrm{vc}(G[U_j])$ *for* $j \in [\nu]$*. The child resources satisfy*

$$\begin{aligned} &\sum_{j=1}^{\nu} q_j \le q, \\ &\sum_{j=1}^{\nu} c_j = \mathrm{vc}\Big(G[\bigcup_{j=1}^{\nu} U_j]\Big) \le c, \\ &\quad n_j \le c_j + \alpha(G) \qquad (j \in [\nu]). \end{aligned} \tag{4}$$

Proof. Let $i, j \in [\nu]$ be distinct, and let $u \in U_i$ and $v \in U_j$ be distinct source vertices. Choose copies $u_a \in F_i$ and $v_b \in F_j$. Since the layer sets I_i and I_j are disjoint, $a \neq b$. If uv were an edge of G, then $u_a v_b$ would be an edge of $S_q(G)$, contrary to the fact that vertices in different child trees are incomparable. Thus no source edge joins distinct vertices represented in different children.

Fix $u \in D$ and $v \in \bigcup_{j=1}^{\nu} U_j$ with $v \neq u$. Choose i with $v \in U_i$. Since u occurs in at least two supports, it belongs to some U_j with $j \neq i$. The preceding argument shows that uv is not an edge. Thus D is isolated in $G[\bigcup_{j=1}^{\nu} U_j]$, and in particular is independent, giving

$$|D| \le \alpha(G). \tag{5}$$

The sets $U_j \setminus D$ are pairwise disjoint and have no edges between them. Removing isolated vertices does not change a vertex-cover number, and vertex-cover numbers add over disjoint components. Therefore

$$\sum_{j=1}^{\nu} \mathrm{vc}(G[U_j]) = \mathrm{vc}\Big(G[\bigcup_{j=1}^{\nu} U_j]\Big).$$

A minimum vertex cover of G, restricted to $\bigcup_{j=1}^{\nu} U_j$, covers the induced graph on that union. This proves the remaining inequality $\mathrm{vc}(G[\bigcup_{j=1}^{\nu} U_j]) \le \mathrm{vc}(G)$. The layer sum is bounded by Lemma 3.4, and the last resource bound follows from

$$n_j \overset{(1)}{=} c_j + \alpha(G[U_j]) \le c_j + \alpha(G), \qquad j \in [\nu].$$

Each vertex in $\bigcup_{j=1}^{\nu} U_j$ is counted once in its first child support. A vertex in D may also be counted in other child supports, at most $\nu - 1$ additional times. Thus,

$$\begin{aligned} \sum_{j=1}^{\nu} n_j &= \Big|\bigcup_{j=1}^{\nu} U_j\Big| + \sum_{v \in D} \big(|\{j \in [\nu] : v \in U_j\}| - 1\big) \\ &\le n + (\nu - 1)|D| \\ &\overset{(5)}{\le} n + (\nu - 1)\alpha(G). \end{aligned}$$

The remaining inequalities in (3) follow from $\nu \overset{\text{Lem. 3.4}}{\le} q$ and $\alpha(G) \le n$. □

Lemma 3.6 (The full graph associated with a child)**.** *After relabeling the layers in* I_j *by* $[q_j]$*, the graph induced by* $V(F_j)$ *is obtained from* $S_{q_j}(G[U_j])$ *by deleting* M_j *vertices.*

Proof. All vertices of F_j lie in $U_j \times I_j$. On this set, the adjacency rule for $S_q(G)$ agrees with that of $S_{q_j}(G[U_j])$ after relabeling the layers. The child contains N_j of the $q_j n_j$ possible vertices, leaving $M_j = q_j n_j - N_j \ge 0$ missing. □

Lemma 3.7 (Prefix and child accounting)**.** *The prefix and the children satisfy*

$$|P| + \sum_{j=1}^{\nu} N_j = qn, \qquad h(F) = |P| + \max_{j\in[\nu]} h(F_j),$$

and

$$\operatorname{totd}(F) = \frac{|P|(|P|+1)}{2} + |P| \sum_{j=1}^{\nu} N_j + \sum_{j=1}^{\nu} \operatorname{totd}(F_j).$$

Proof. By the construction of P, its vertex set and the sets $V(F_j)$ partition $V(F)$, proving the first identity. A longest path in F consists of P followed by a longest path in one child, proving the height identity. The prefix vertices have depths $1, \dots, |P|$, and each child vertex has depth $|P|$ greater than its depth within that child. Summing these contributions proves the total-depth identity. □

Lemma 3.8 (Restoring deleted vertices)**.** *Let H be a graph on N vertices, let $X \subseteq V(H)$ have size d, and let F be an elimination forest of $H - X$. There is an elimination forest F' of H with*

$$h(F') = h(F) + d,$$
$$\operatorname{totd}(F') = \operatorname{totd}(F) + d(N-d) + \frac{d(d+1)}{2}.$$

In particular,

$$\operatorname{td}(H) \le \operatorname{td}(H - X) + d,$$
$$\operatorname{totd}(H) \le \operatorname{totd}(H - X) + d(N-d) + \frac{d(d+1)}{2}.$$

Proof. Place X on a chain above all roots of F. If F is empty, use the chain on X alone; if X is empty, keep F. Every new vertex is comparable with every vertex, so the restored forest is valid. Its height increases by d, as does the depth of each of the $N - d$ old vertices; the new vertices have depths $1, \dots, d$, with sum $d(d+1)/2$. These counts prove both identities. Minimizing separately over F gives the two inequalities. □

Theorem 3.9 (Structural treedepth lower bound)**.** *For every finite simple graph G on n vertices and every integer $q \ge 1$,*

$$\operatorname{td}(S_q(G)) \ge \frac{2q}{3} \operatorname{vc}(G). \tag{6}$$

Proof. We prove the statement by induction on q, with the induction hypothesis asserted for every finite simple source graph.

Base cases and the first branch. If $n = 0$, then $S_q(G)$ has no vertices and both sides of (6) are zero. For $q = 1$, $S_1(G)$ is a clique, so $\operatorname{td}(S_1(G)) = n \ge 2\operatorname{vc}(G)/3$.

Now fix $q \ge 2$, assume the statement for every smaller number of layers, and take an arbitrary elimination forest F of $S_q(G)$. We will lower-bound its height $h(F)$. A single chain has height $qn \ge 2q\operatorname{vc}(G)/3$, so assume that F is not a single chain.

Use the first-branch notation from Section 3.2.1. We have

$$|P| \overset{\text{Lem. 3.7}}{=} qn - \sum_{j=1}^{\nu} N_j, \qquad 1 \le q_j \overset{\text{Lem. 3.4}}{<} q.$$

Put $c = \operatorname{vc}(G)$ and $c_j = \operatorname{vc}(G[U_j])$ for $j \in [\nu]$. Their resource bounds are (4).

Induction within each child. By Lemma 3.6, the graph induced by child j is $S_{q_j}(G[U_j])$ with M_j vertices deleted. Lemma 3.8 restores them to obtain an elimination forest of $S_{q_j}(G[U_j])$ with height $h(F_j) + M_j$. Since $q_j < q$, induction therefore gives

$$h(F_j) \overset{\text{Lem. 3.8, induction}}{\geq} \frac{2}{3} q_j c_j - M_j. \tag{7}$$

Combining the child bounds. Choose $k \in [\nu]$ maximizing $q_j c_j$. A path through this child contains the prefix as well as a longest path in F_k. Substituting $N_j = q_j n_j - M_j$ into the prefix accounting and the child bound gives the following cancellation:

$$\begin{aligned} h(F) &\overset{\text{Lem. 3.7}}{\geq} qn - \sum_{j=1}^{\nu} N_j + h(F_k) \\ &\overset{(7)}{\geq} qn - \sum_{j=1}^{\nu} (q_j n_j - M_j) + \frac{2}{3} q_k c_k - M_k \\ &= qn - \sum_{j=1}^{\nu} q_j n_j + \frac{2}{3} q_k c_k + \sum_{j \in [\nu] \setminus \{k\}} M_j \\ &\geq qn - \sum_{j=1}^{\nu} q_j n_j + \frac{2}{3} q_k c_k. \end{aligned} \tag{8}$$

The prefix is long when few source vertices fit in the children. We obtain

$$\begin{aligned} qn - \sum_{j=1}^{\nu} q_j n_j &\overset{(1),(4)}{\geq} q(c + \alpha(G)) - \sum_{j=1}^{\nu} q_j (c_j + \alpha(G)) \\ &= qc - \sum_{j=1}^{\nu} q_j c_j + \alpha(G)\Big(q - \sum_{j=1}^{\nu} q_j\Big) \\ &\overset{\text{Lem. 3.4}}{\geq} qc - \sum_{j=1}^{\nu} q_j c_j. \end{aligned} \tag{9}$$

Combining these estimates and applying Lemma 3.3 to q_j, c_j yields

$$\begin{aligned} h(F) &\overset{(8),(9)}{\geq} qc - \frac{1}{3}\left(3 \sum_{j=1}^{\nu} q_j c_j - 2 \max_{j \in [\nu]} (q_j c_j)\right) \\ &\overset{\text{Lem. 3.3}}{\geq} qc - \frac{1}{3}\left(\sum_{j=1}^{\nu} q_j\right)\left(\sum_{j=1}^{\nu} c_j\right) \\ &\overset{\text{Lem. 3.4, (4)}}{\geq} \frac{2}{3} qc. \end{aligned}$$

This completes the induction. □

For $q \geq 2$, write $B_{q,\eta} := (2q/3)(1 - \eta)$. For a NO instance of Theorem 3.1, the structural bound gives

$$\begin{aligned} \mathrm{td}(S_q(G)) &\overset{\text{Thm. 3.9, (1)}}{\geq} \frac{2q}{3}(n - \alpha(G)) \\ &\overset{\text{Thm. 3.1}}{>} B_{q,\eta} n. \end{aligned} \tag{10}$$

3.2.3 The approximation gap

Dense treedepth gap. At zero source error, the ratio of the NO and YES coefficients is $B_{q,0}/A_{q,0} = 4q/(3(q+1))$, which approaches $4/3$ as q increases. Fix $1 \le r < 4/3$ and choose $q = \lfloor 3r/(4-3r) \rfloor + 1$. Put $\delta = [q(4-3r) - 3r]/6 > 0$ and $C = 2q/3 + r(q-1) > 0$. Choose a rational $0 < \eta < \min\{1/10, \delta/(2C)\}$. Then

$$\left(\frac{2q}{3} + r(q-1)\right)\eta < \frac{q(4-3r) - 3r}{6}. \tag{11}$$

Then

$$\begin{aligned} B_{q,\eta} - rA_{q,\eta} &= \frac{2q}{3}(1-\eta) - r\left(\frac{q+1}{2} + (q-1)\eta\right) \\ &= \frac{q(4-3r) - 3r}{6} - \left(\frac{2q}{3} + r(q-1)\right)\eta \overset{(11)}{>} 0. \end{aligned}$$

An r-approximation returning a feasible forest has height at most $rA_{q,\eta}n$ in a YES instance. In a NO instance every feasible forest has height greater than $B_{q,\eta}n > rA_{q,\eta}n$ by (10) and (11). Comparing the returned height with $rA_{q,\eta}n$ therefore distinguishes the cases. For an arbitrary real target factor, one may first increase it to a fixed rational factor below $4/3$, so the cutoff is rational. Since q and η depend only on r, this is a polynomial reduction. □

3.3 Total elimination depth

The same layered graph $S_q(G)$ gives a gap for total elimination depth, also called VPT-sum [HBB+21]. Here every child contributes to the objective, and each vertex of the common prefix increases the depth of every vertex below it. We reuse the first-branch structure and bound these contributions together.

For a rooted forest F, let $\iota(F)$ be the number of unordered pairs of distinct vertices that are incomparable in its ancestor order. This counts the savings in total depth relative to one chain.

Lemma 3.10 (Incomparable-pair accounting)**.** *For a rooted forest F on N vertices,*

$$\operatorname{totd}(F) = \frac{N(N+1)}{2} - \iota(F).$$

If $P, F_1, \ldots, F_\nu$ are its first-branch prefix and children, with $N_j = |V(F_j)|$, then

$$\iota(F) = \sum_{1 \le i < j \le \nu} N_i N_j + \sum_{j=1}^{\nu} \iota(F_j).$$

Adding a chain above all roots leaves $\iota(F)$ unchanged.

Proof. A vertex's depth counts itself and its proper ancestors. Thus total depth counts all N vertices and all comparable unordered pairs, proving the first identity, including $N = 0$. Every prefix vertex is comparable with every vertex, while vertices in different children are incomparable. This partitions the incomparable pairs as stated. New vertices on a chain above all roots are comparable with every vertex, so they create no incomparable pair. □

3.3.1 The YES upper bound

The elimination forest from Lemma 3.2 has a particularly simple depth sum. Its common prefix contains all copies of the vertices outside an independent set. The remaining copies form one chain per layer.

Lemma 3.11 (Total depth of the cover-first forest)**.** *For every graph G on n vertices and every integer $q \geq 1$,*

$$\operatorname{totd}(S_q(G)) \leq \frac{qn(qn+1) - q(q-1)\alpha(G)^2}{2}.$$

Proof. Choose a maximum independent set of G, and use the forest F of Lemma 3.2. Its only incomparable pairs have endpoints in different lower chains. There are q such chains, each with $\alpha(G)$ vertices, so there are $\binom{q}{2}\alpha(G)^2$ incomparable pairs. Therefore the forest has total depth

$$\operatorname{totd}(F) \overset{\text{Lem. 3.10}}{=} \frac{qn(qn+1)}{2} - \binom{q}{2}\alpha(G)^2 = \frac{qn(qn+1) - q(q-1)\alpha(G)^2}{2},$$

which gives the claimed upper bound, also when the common prefix is empty. □

3.3.2 A structural lower bound for total depth

Use the notation from Section 3.2.1. Lemma 3.10 accounts for the prefix through one pair-counting identity: only incomparable pairs save depth. Restoring a child's missing pairs in $U_j \times I_j$ above its roots preserves this saving and lets induction apply to a full layered graph.

The next inequality combines the squared child terms and their pairwise products. Cauchy–Schwarz will then bound its right-hand side using the total layer count and the sum of the child vertex-cover numbers.

Lemma 3.12 (Quadratic branching inequality)**.** *For a positive integer ν and nonnegative reals $W_1, \ldots, W_\nu$,*

$$\sum_{j=1}^{\nu} W_j^2 + 14 \sum_{1 \leq i < j \leq \nu} W_i W_j \leq \left(\sum_{j=1}^{\nu} \sqrt{W_j} \right)^4.$$

Its proof, by expansion into nonnegative terms, is given in Appendix B.2.

Theorem 3.13 (Structural total-depth lower bound)**.** *For every finite simple graph G on n vertices and every integer $q \geq 1$,*

$$\operatorname{totd}(S_q(G)) \geq \frac{3}{7} q^2 \operatorname{vc}(G)^2 + \frac{qn}{2}. \tag{12}$$

Proof. We induct on q, with the induction hypothesis asserted for every finite simple source graph. If $n = 0$, both sides are zero. Fix $n \geq 1$ and an arbitrary elimination forest F of $S_q(G)$. A single-chain forest has total depth $qn(qn+1)/2 \geq (3/7)q^2 \operatorname{vc}(G)^2 + qn/2$, since $\operatorname{vc}(G) \leq n$ and $1/2 \geq 3/7$. This includes $q = 1$, since $S_1(G)$ is a clique. Otherwise use the first-branch notation from Section 3.2.1, and put $c = \operatorname{vc}(G)$, $c_j = \operatorname{vc}(G[U_j])$, $W_j = q_j c_j$, $L_j = q_j n_j$, $S = \sum_{j=1}^{\nu} W_j$, $A = q\alpha(G)$, and $N = qn$. We have $n \overset{(1)}{=} c + \alpha(G)$ and therefore $N = qc + A$. The shared child resource bounds give

$$\left(\sum_{j=1}^{\nu} q_j \right) \left(\sum_{j=1}^{\nu} c_j \right) \overset{(4)}{\leq} qc. \tag{13}$$

Consequently,

$$\begin{aligned} S = \sum_{j=1}^{\nu} q_j c_j &\leq \Big(\sum_{j=1}^{\nu} q_j \Big) \Big(\sum_{j=1}^{\nu} c_j \Big) \overset{(13)}{\leq} qc, \\ \sum_{j=1}^{\nu} L_j = \sum_{j=1}^{\nu} q_j n_j &\overset{(4)}{\leq} \sum_{j=1}^{\nu} q_j c_j + \alpha(G) \sum_{j=1}^{\nu} q_j \\ &\overset{(4)}{\leq} S + A. \end{aligned} \tag{14}$$

Restore the $M_j = L_j - N_j$ missing vertices of child j above its roots, obtaining a forest $\widetilde{F}_j$ of $S_{q_j}(G[U_j])$ by Lemma 3.8. Since $q_j \overset{\text{Lem. 3.4}}{<} q$, induction bounds its total depth below by $3W_j^2/7 + L_j/2$. Restoration preserves incomparable pairs, so

$$\begin{aligned} \iota(F_j) &\overset{\text{Lem. 3.10}}{=} \frac{L_j(L_j+1)}{2} - \operatorname{totd}(\widetilde{F}_j) \\ &\overset{\text{induction}}{\leq} \frac{L_j^2}{2} - \frac{3}{7}W_j^2. \end{aligned} \tag{15}$$

The additional incomparable pairs lie in different children. Since $N_j \leq L_j$, their contribution and the squared terms in (15) satisfy

$$\begin{aligned} \sum_{1\leq i<j\leq \nu} N_i N_j + \frac{1}{2}\sum_{j=1}^{\nu} L_j^2 &\overset{\text{Lem. 3.6}}{\leq} \sum_{1\leq i<j\leq \nu} L_i L_j + \frac{1}{2}\sum_{j=1}^{\nu} L_j^2 \\ &= \frac{1}{2}\Big(\sum_{j=1}^{\nu} L_j\Big)^2 \overset{(14)}{\leq} \frac{(S+A)^2}{2}. \end{aligned} \tag{16}$$

Subtracting all incomparable pairs from the chain cost yields

$$\begin{aligned} \operatorname{totd}(F) &\overset{\text{Lem. 3.10; (15)}}{\geq} \frac{N^2+N}{2} - \sum_{1\leq i<j\leq \nu} N_i N_j - \frac{1}{2}\sum_{j=1}^{\nu} L_j^2 + \frac{3}{7}\sum_{j=1}^{\nu} W_j^2 \\ &\overset{(16)}{\geq} \frac{N^2+N}{2} - \frac{(S+A)^2}{2} + \frac{3}{7}\sum_{j=1}^{\nu} W_j^2. \end{aligned} \tag{17}$$

The terms involving only the W_j can be grouped into the left-hand side of Lemma 3.12. Applying it, then Cauchy–Schwarz to $\sqrt{q_j}, \sqrt{c_j}$, gives

$$\begin{aligned} \frac{S^2}{2} - \frac{3}{7}\sum_{j=1}^{\nu} W_j^2 &= \frac{1}{14}\left(\sum_{j=1}^{\nu} W_j^2 + 14 \sum_{1\leq i<j\leq \nu} W_i W_j\right) \\ &\overset{\text{Lem. 3.12}}{\leq} \frac{1}{14}\left(\sum_{j=1}^{\nu} \sqrt{W_j}\right)^4 \\ &= \frac{1}{14}\left(\sum_{j=1}^{\nu} \sqrt{q_j}\sqrt{c_j}\right)^4 \\ &\overset{\text{Cauchy–Schwarz}}{\leq} \frac{1}{14}\left(\sum_{j=1}^{\nu} q_j\right)^2 \left(\sum_{j=1}^{\nu} c_j\right)^2 \\ &\overset{(13)}{\leq} \frac{q^2c^2}{14}. \end{aligned} \tag{18}$$

Since $N = qc + A$ and $A \geq 0$, the bound $S \leq qc$ gives

$$N^2 - (S+A)^2 = q^2c^2 - S^2 + 2A(qc - S) \overset{(14)}{\geq} q^2c^2 - S^2.$$

Substituting this into (17) leaves only the quadratic estimate:

$$\begin{aligned}\operatorname{totd}(F) &\overset{(17),(14)}{\geq} \frac{N}{2} + \frac{q^2c^2}{2} - \left(\frac{S^2}{2} - \frac{3}{7}\sum_{j=1}^{\nu} W_j^2\right) \\ &\overset{(18)}{\geq} \frac{N}{2} + \frac{q^2c^2}{2} - \frac{q^2c^2}{14} = \frac{3}{7}q^2c^2 + \frac{N}{2}.\end{aligned}$$

□

3.3.3 Sharpness of the structural coefficient

The coefficient $3/7$ cannot be increased in a bound of the form (12) valid for every G and q. The following forests approach equality even when the source independence number is a vanishing fraction of its order.

Proposition 3.14. *Let $k \geq 0$ and $t \geq 2$ be integers, put $q = 2^k$, and let G be the disjoint union of q copies of K_t. Write $n = qt$ and $N = qn$. There is an elimination forest F of $S_q(G)$ with*

$$\operatorname{totd}(F) = \frac{N}{2} + \left(\frac{3}{7} + \frac{1}{14 \cdot 8^k}\right) N^2.$$

Consequently, if a constant β satisfies $\operatorname{totd}(S_q(G)) \geq \beta q^2 \operatorname{vc}(G)^2 + qn/2$ for every finite simple graph G and every integer $q \geq 1$, then $\beta \leq 3/7$.

Proof. We construct the forest recursively. For $k = 0$, the output is K_t, and a chain has total depth $t(t+1)/2$. For $k \geq 1$, partition the layer indices into equal sets I_1, I_2 and the source clique-components into equal groups, whose vertex sets are U_1, U_2. Put all vertices in $(U_1 \times I_2) \cup (U_2 \times I_1)$ on a common prefix. Its length is $N/2$. Below this prefix put separate forests on $U_1 \times I_1$ and $U_2 \times I_2$, each of order $N/4$. No output edge joins these two sets: their layers differ and no source edge joins U_1 to U_2. Each induced child graph has $q/2$ layers over a disjoint union of $q/2$ copies of K_t, so the recursive construction applies. Every edge incident with the prefix has comparable endpoints, proving that the forest is valid.

For fixed t, let T_k be its total depth and $N_k = 4^k t$ its order. The prefix contributes its own depths and $N_k/2$ to the depth of each of the $N_k/2$ child vertices. Thus

$$\begin{aligned}T_k &\overset{\text{Lem. 3.7}}{=} \frac{(N_k/2)(N_k/2+1)}{2} + \frac{N_k^2}{4} + 2T_{k-1} \\ &= \frac{3N_k^2}{8} + \frac{N_k}{4} + 2T_{k-1}.\end{aligned}$$

Writing $T_k = N_k/2 + a_k N_k^2$ and using $N_{k-1} = N_k/4$ gives

$$a_0 = \frac{1}{2}, \qquad a_k = \frac{3}{8} + \frac{a_{k-1}}{8}, \qquad a_k - \frac{3}{7} = \frac{1}{8}\left(a_{k-1} - \frac{3}{7}\right).$$

Iterating the last identity yields $a_k = 3/7 + 1/(14 \cdot 8^k)$, proving the formula.

Here $\alpha(G) = q$ and $\operatorname{vc}(G) = q(t-1)$. The proposed universal bound and the constructed forest would imply

$$\beta \overset{\text{assumed bound}}{\leq} \frac{\operatorname{totd}(S_q(G)) - N/2}{q^2 \operatorname{vc}(G)^2} \overset{\operatorname{totd}(S_q(G)) \leq \operatorname{totd}(F)}{\leq} \left(\frac{3}{7} + \frac{1}{14 \cdot 8^k}\right) \frac{t^2}{(t-1)^2}.$$

Letting t tend to infinity and then k tend to infinity gives $\beta \leq 3/7$. Also $\alpha(G)/n = 1/t$, as claimed. □

This sharpness statement concerns the universal structural bound; it does not rule out stronger hardness from additional source properties or a different construction.

3.3.4 The approximation gap

Dense total-depth gap. Fix $1 \le r < 8/7$. The source thresholds are

$$\begin{aligned} &\text{YES:} \quad \alpha(G) \overset{\text{Thm. 3.1}}{\ge} (1/2-\eta)n, \\ &\text{NO:} \quad \mathrm{vc}(G) \overset{(1)}{=} n - \alpha(G) \overset{\text{Thm. 3.1}}{>} (1-\eta)n. \end{aligned} \tag{19}$$

Define the quadratic coefficients $A^{\Sigma}_{q,\eta} = (q^2 - q(q-1)(1/2-\eta)^2)/2$ and $B^{\Sigma}_{q,\eta} = (3/7)q^2(1-\eta)^2$. The corresponding output satisfies

$$\begin{aligned} \text{YES:} \quad \mathrm{totd}(S_q(G)) &\overset{\text{Lem. 3.11}}{\le} \frac{qn(qn+1) - q(q-1)\alpha(G)^2}{2} \\ &\overset{(19)}{\le} A^{\Sigma}_{q,\eta} n^2 + qn/2, \\ \text{NO:} \quad \mathrm{totd}(S_q(G)) &\overset{(12)}{\ge} \frac{3}{7} q^2 \,\mathrm{vc}(G)^2 + \frac{qn}{2} \\ &\overset{(19)}{\ge} B^{\Sigma}_{q,\eta} n^2 + qn/2. \end{aligned}$$

At $\eta = 0$, the ratio of the quadratic NO and YES coefficients is $24q/(7(3q+1))$, which approaches $8/7$ as q increases. Choose a fixed integer q satisfying

$$q > \frac{7r}{24 - 21r}. \tag{20}$$

At $\eta = 0$,

$$\begin{aligned} A^{\Sigma}_{q,0} &= \frac{q^2 - q(q-1)/4}{2} = \frac{q(3q+1)}{8}, \\ B^{\Sigma}_{q,0} - r A^{\Sigma}_{q,0} &= \frac{3q^2}{7} - \frac{rq(3q+1)}{8} = \frac{q\big(24q - 7r(3q+1)\big)}{56} \overset{(20)}{>} 0. \end{aligned}$$

Indeed, $24q - 7r(3q+1) = (24-21r)q - 7r \overset{(20)}{>} 0$. To choose the source error explicitly, put

$$D_0 = \frac{q\big((24-21r)q - 7r\big)}{56}, \qquad K = \frac{6q^2}{7} + \frac{rq(q-1)}{2}, \qquad L = \frac{3q^2}{7} + \frac{rq(q-1)}{2}.$$

Here $D_0 > 0$ by (20), and $K, L > 0$. Choose a fixed rational $0 < \eta < \min\{1/10, D_0/(2K)\}$. Expanding the two coefficients gives

$$B^{\Sigma}_{q,\eta} - r A^{\Sigma}_{q,\eta} = D_0 - K\eta + L\eta^2 \overset{K\eta < D_0/2}{>} \frac{D_0}{2} > 0.$$

In particular,

$$B^{\Sigma}_{q,\eta} > r A^{\Sigma}_{q,\eta}. \tag{21}$$

For a source graph G_0 on $n_0 \ge 1$ vertices, put

$$m = 1 + \left\lfloor \frac{q(r-1)}{2(B^{\Sigma}_{q,\eta} - r A^{\Sigma}_{q,\eta})} \right\rfloor, \qquad G = \bigdot\bigcup_{j=1}^{m} G_0^{(j)},$$

where $G_0^{(1)}, \dots, G_0^{(m)}$ are disjoint copies of G_0. Then the order $n = mn_0 \geq m$ satisfies

$$n > \frac{q(r-1)}{2(B^{\Sigma}_{q,\eta} - rA^{\Sigma}_{q,\eta})}. \tag{22}$$

The denominator is positive by (21). Since there are no edges between the copies,

$$|V(G)| = m|V(G_0)|, \qquad \alpha(G) = m\alpha(G_0), \qquad \Delta(G) = \Delta(G_0).$$

Thus the source promise remains unchanged. The YES and NO bounds then have gap

$$\begin{aligned}\left(B^{\Sigma}_{q,\eta} n^2 + \frac{qn}{2}\right) - r\left(A^{\Sigma}_{q,\eta} n^2 + \frac{qn}{2}\right) \\ = \left(B^{\Sigma}_{q,\eta} - rA^{\Sigma}_{q,\eta}\right) n^2 - \frac{(r-1)qn}{2} \overset{(22)}{>} 0.\end{aligned}$$

The reduction has gap greater than r. All parameters, including m, depend only on r, so the reduction is polynomial. □

3.4 Treewidth

We prove UGC-hardness below 4/3 by transferring a balanced, unweighted tripartite Vertex Cover gap on graphs of bounded degree to treewidth. We derive this gap from Theorem 3.1 using a multipartite product. Bérczi, Chandrasekaran, Király, Lee, and Xu prove a related weighted k-partite Vertex Cover gap under UGC [BCK+17, Sections 3.5–3.6]. The same transfer gives the NP-hardness bound in Section 5.2.

The transfer uses Bonnet's completion of three parts to cliques and his central-bag lower bound [Bon25, Lemma 6]. We take disjoint source copies *before* completing the three global parts. This lets each of three paths attached to a central bag process the copies one at a time: the additional bag size is bounded by one part of one copy, while the cover contribution grows with the number of copies. Completing the global parts creates a dense intermediate graph. Corollary 6.23 obtains bounded-degree output by replacing these three cliques with bounded-degree expanders while retaining all source-copy edges.

3.4.1 A transfer from vertex cover to treewidth

Let G be a nonempty tripartite graph with specified independent parts $V(G) = A \dot\cup B \dot\cup C$, and put $n_{\max} = \max\{|A|, |B|, |C|\}$. Thus G has no edge within a part. For an integer $R \geq 1$, take copies $G_1, \dots, G_R$ on $V(G) \times \{1\}, \dots, V(G) \times \{R\}$, writing $v_j = (v, j)$. For $X \in \{A, B, C\}$, let $X_j = X \times \{j\}$ and $X^* = X \times [R]$. Define $K_R(G)$ on $V(G) \times [R]$ by writing $\binom{X^*}{2}$ for the set of unordered pairs of distinct vertices in X^*:

$$E(K_R(G)) = \bigcup_{X \in \{A,B,C\}} \binom{X^*}{2} \cup \bigcup_{j=1}^{R} \{\{u_j, v_j\} : uv \in E(G)\}.$$

These are all the edges: each global part is a clique, and vertices in different global parts are adjacent if and only if they belong to the same copy and their source vertices are adjacent in G.

Its vertex partition into three cliques makes $K_R(G)$ *co-tripartite*.

Lemma 3.15 (A bag covering the source edges)**.** *Every tree decomposition of $K_R(G)$ has a bag containing a vertex cover of $G_1 \dot\cup \cdots \dot\cup G_R$. In particular, that bag has size at least $R\,\mathrm{vc}(G)$.*

Proof. We use Bonnet's central-bag argument [Bon25, proof of Lemma 6]. Fix a tree decomposition $(T, (B_t)_{t\in V(T)})$. For each nonempty clique X^*, the subtrees T_x, $x \in X^*$, intersect pairwise. By the Helly property for subtrees of a tree, they have a common node u_X, so $X^* \subseteq B_{u_X}$ [Bon25, Lemma 4]. Choose u_X arbitrarily if $X^* = \varnothing$. Let u be the median of u_A, u_B, u_C, namely the unique common node of their three pairwise connecting paths. For every edge xy of a source copy G_j, with $x \in X^*$ and $y \in Y^*$, the subtrees T_x, T_y intersect. Their union is connected and contains u_X, u_Y, so it contains the u_X–u_Y path and hence u. Thus $u \in T_x \cup T_y$, which means that B_u contains x or y. Therefore B_u contains an endpoint of every source edge, so it contains a vertex cover of the disjoint source copies and has size at least $R\,\mathrm{vc}(G)$. □

Lemma 3.16 (A decomposition from a source cover)**.** *For every vertex cover S of G, the graph $K_R(G)$ has a tree decomposition whose bags have size at most $R\max\{|S|, n_{\max}\} + n_{\max}$.*

Proof. We adapt the three-path decomposition of [Bon25, Section 3.3] by processing copies separately, so the temporary bag increase is at most $n_{\max}$, independently of R. Put $I = V(G) \setminus S$. For each $j \in [R]$, put $S_j = S \times \{j\}$ and $I_j = I \times \{j\}$. Start with the central bag $S^* = \bigcup_{j=1}^{R} S_j$, where $|S^*| = R|S|$. Attach three paths to this node, one for each part X. The path to X ends with the bag X^*. On this path process the copies in order $j = 1, \dots, R$. At step j, first add all vertices of $I_j \cap X_j$, and then remove all vertices of $S_j \setminus X_j$.

After processing the first j copies, the bag is

$$B_j^X = \bigcup_{i=1}^{j} X_i \;\cup\; \bigcup_{i=j+1}^{R} S_i \qquad (0 \le j \le R).$$

Here $B_0^X = S^*$ and $B_R^X = X^*$. Between B_{j-1}^X and B_j^X insert the bag $B_{j-1}^X \cup (I_j \cap X_j)$. The bag sizes satisfy

$$|B_j^X| = j|X| + (R-j)|S| \le R\max\{|S|, |X|\},$$
$$|B_{j-1}^X \cup (I_j \cap X_j)| = |B_{j-1}^X| + |I \cap X| \le R\max\{|S|, n_{\max}\} + n_{\max}.$$

We verify the decomposition axioms. Every vertex is present in its part's final bag. A vertex in $S_j \cap X_j$ occurs at the center and throughout the X-path, and on each other path it is removed once. A vertex in $I_j \cap X_j$ occurs only on the X-path, from its addition to the end. Thus each T_v is nonempty and connected. Every edge within a global clique is covered at the corresponding final bag. Now consider an edge between two global parts. It comes from a single source copy, and its endpoints cannot both lie in that copy's independent set. If both endpoints lie in S^*, the center bag covers the edge. Otherwise, one endpoint lies in $I_j \cap X_j$ and the other in $S_j \setminus X_j$ for some j and part X. When the first endpoint is added on the X-path, the second is still in the bag, so the edge is covered there. All edges are covered, and the bag-size bound follows from the count above. □

Lemma 3.17 (Replicated co-tripartite completion)**.** *Let G be a nonempty tripartite graph with specified independent parts A, B, C, put $n_{\max} = \max\{|A|, |B|, |C|\}$, and let $R \ge 1$ be an integer. Then*

$$\mathrm{tw}(K_R(G)) \ge R\max\{\mathrm{vc}(G), n_{\max}\} - 1,$$
$$\mathrm{tw}(K_R(G)) \le R\max\{\mathrm{vc}(G), n_{\max}\} + n_{\max} - 1.$$

Proof. Lemma 3.15 gives a bag of size at least $R\,\mathrm{vc}(G)$, and its Helly argument gives one of size at least $Rn_{\max}$. Thus every decomposition has a bag of size at least $R\max\{\mathrm{vc}(G), n_{\max}\}$. For the upper bound, apply Lemma 3.16 to a minimum cover of G: its largest bag has size at most $R\max\{\mathrm{vc}(G), n_{\max}\} + n_{\max}$. Subtracting one from both bag-size bounds proves the claim. □

For fixed R, the construction has $R|V(G)|$ vertices and is polynomial in the source size.

3.4.2 The approximation gap

For the UGC reduction we need cover thresholds $(1/2+\zeta)N$ and $(2/3-\zeta)N$ on balanced tripartite graphs. We derive the balanced, unweighted form needed here from the bounded-degree independent-set promise using the product below. Its degree identity will also be used in Section 6.

Lemma 3.18 (Multipartite product)**.** *For a nonempty simple graph G on n vertices and a positive integer k, let $P_k(G)$ have vertex set $V(G)\times[k]$, and write $v_i=(v,i)$. Define its edges by*

$$u_iv_j \in E(P_k(G)) \quad\Longleftrightarrow\quad i\neq j \text{ and } uv\in E(G).$$

Its k layers are independent sets of size n. Moreover, $k\alpha(G)\le\alpha(P_k(G))\le n+(k-1)\alpha(G)$ and $\Delta(P_k(G))=(k-1)\Delta(G)$.

Proof. All k copies of an independent set of G form an independent set of $P_k(G)$, proving the first bound. For the upper bound, let I be any independent set in the product. Put $m_v=|\{i\in[k]: v_i\in I\}|$ for $v\in V(G)$ and $D=\{v\in V(G): m_v\ge 2\}$. If distinct $u,v\in D$ were adjacent in G, choose a copy $u_i\in I$. The set I contains at least two copies of v, so one has a layer index $j\neq i$. Then u_iv_j would be an edge inside I. Thus D is independent, so

$$|D|\le\alpha(G). \tag{23}$$

Since $m_v\le k$ for $v\in D$ and $m_v\le 1$ otherwise,

$$|I|=\sum_{v\in V(G)} m_v \le n+(k-1)|D| \overset{(23)}{\le} n+(k-1)\alpha(G).$$

For every $v\in V(G)$ and $i\in[k]$,

$$\deg_{P_k(G)}(v_i)=(k-1)\deg_G(v),$$

which proves the maximum-degree identity. □

Theorem 3.19 (Balanced bounded-degree multipartite cover gap)**.** *For every fixed integer $k\ge 3$ and fixed rational $0<\zeta<1/12$, there is a constant $D_{k,\zeta}$ such that the following promise problem is Unique Games-hard. The input is a simple unweighted k-partite graph G on $N\ge k$ vertices, of maximum degree at most $D_{k,\zeta}$, with a specified partition into k independent parts each of size N/k. Distinguish the following cases:*

YES: $\mathrm{vc}(G)\le(1/2+\zeta)N$.

NO: $\mathrm{vc}(G)\ge(1-1/k-\zeta)N$.

The reduction is deterministic and polynomial time.

The promise follows from Theorem 3.1 and Lemma 3.18, with $D_{k,\zeta}=(k-1)d_{\zeta/2}$. The independent-set source is already of bounded degree by the spectral sparsification in Appendix A.1. Appendix A.2 records the parameter substitution. Taking $k=3$ and applying the replication lemma proves the treewidth gap.

Dense treewidth gap. As the source error decreases and R increases, the leading NO and YES coefficients approach $2/3$ and $1/2$, giving ratio $4/3$. Fix $1 \le r < 4/3$ and put $\delta = 2/3 - r/2 > 0$. Choose a rational $0 < \zeta < \min\{1/12, \delta/[2(1+r)]\}$ and then an integer $R > 2r/(3\delta)$. These choices give $(1+r)\zeta + r/(3R) < \delta$, or equivalently

$$2/3 - \zeta > r(1/2 + \zeta + 1/(3R)). \tag{24}$$

Here ζ controls the source error and $1/(3R)$ controls the additional YES cost of the replication construction. Take the N-vertex source from Theorem 3.19 with $k = 3$. Its parts have size $N/3$. Since $\zeta < 1/12$, both cover thresholds, $(1/2+\zeta)N$ and $(2/3-\zeta)N$, exceed $N/3$. Thus the $n_{\max} = N/3$ term in the maximum of Lemma 3.17 does not weaken either case, and the lemma and source promise give

$$\begin{aligned}
\text{YES:}\quad \operatorname{tw}(K_R(G)) &\overset{\text{Lem. 3.17}}{\le} R\max\{\operatorname{vc}(G), N/3\} + N/3 - 1 \\
&\overset{\text{Thm. 3.19}}{\le} R(1/2+\zeta)N + N/3 - 1, \\
\text{NO:}\quad \operatorname{tw}(K_R(G)) &\overset{\text{Lem. 3.17}}{\ge} R\operatorname{vc}(G) - 1 \\
&\overset{\text{Thm. 3.19}}{\ge} R(2/3-\zeta)N - 1.
\end{aligned}$$

The choices of ζ, R make $\gamma := R(2/3-\zeta) - r\big(R(1/2+\zeta) + 1/3\big) \overset{(24)}{>} 0$. The NO lower bound minus r times the YES upper bound is $\gamma N + r - 1 \overset{(24)}{>} 0$. Adding one to both treewidth bounds gives the maximum-bag-size bounds, whose corresponding difference is $\gamma N \overset{(24)}{>} 0$. Thus both objectives have gap greater than r. More explicitly, let $U = R(1/2+\zeta)N + N/3 - 1$ be the displayed YES width bound. A factor-r approximation returning a tree decomposition has width at most rU in the YES case, whereas every decomposition has width greater than rU in the NO case. For maximum bag size, use the cutoff $r(U+1)$ instead. If the target factor is real, increase it first to a fixed rational factor below $4/3$, making these cutoffs rational. The construction is polynomial, and its three specified cliques each have size $RN/3$. Writing Λ for the Unique Games input, the complete flow is

$$\Lambda \xrightarrow[\text{tripartite source}]{\text{Theorem 3.19, } k=3} G \xrightarrow[\text{replicate and complete}]{\text{Lemma 3.17}} K_R(G).$$

□

4 Hardness under the Small Set Expansion Hypothesis

We prove that the Small Set Expansion Hypothesis rules out every fixed approximation factor for treedepth, using the global partition promise of Raghavendra, Steurer, and Tulsiani (RST) [RST12]. We call the source graph the *host*. Its line graph has one vertex per host edge, so both bounds are expressed in terms of the output order: the YES partition gives a shallow elimination forest, and the NO cut bound forces a large balanced separator. Appendix A.3 derives the unweighted host promise by rounding the weighted source.

4.1 The source partition promise

For a finite undirected weighted graph G, let $w_e \in \mathbb{Q}_{>0}$ be the weight of edge e. Parallel edges and loops are allowed. For a multiset F of edges, define

$$w(F) = \sum_{e \in F} w_e, \qquad W = w(E(G)) > 0, \qquad \deg_G(v) = \sum_{e \in E(G)} m_v(e) w_e,$$

where edge sums count occurrences, including parallel copies, and, for an edge e with endpoints x, y, $m_v(e) = \mathbf{1}_{\{x=v\}} + \mathbf{1}_{\{y=v\}}$; here $\mathbf{1}_{\{P\}}$ is one if P holds and zero otherwise. For $A \subseteq V(G)$, define $\mathrm{vol}_G(A) = \sum_{v\in A} \deg_G(v)$ and $\mu_G(A) = \mathrm{vol}_G(A)/(2W)$. When $\mathrm{vol}_G(A) > 0$, put $\Phi_G(A) = w(\delta_G(A))/\mathrm{vol}_G(A)$. A loop contributes twice its weight to the volume and nothing to a cut.

We use the following precise form of the hypothesis and of hardness [RST12, Hypothesis 5.4 and Definition 5.6]. For fixed constants $0 < \eta < 1/2$ and $0 < \delta < 1$, the promise problem $\mathrm{SSE}(\eta, \delta)$ has as input an unweighted regular graph of positive degree. Its YES case has a set A with $\mu_G(A) = \delta$ and $\Phi_G(A) \le \eta$; its NO case has $\Phi_G(A) > 1 - \eta$ for every set of that measure. The *Small Set Expansion Hypothesis* (SSEH) says that for every such η there is a constant $\delta \in (0, \eta)$ for which this promise problem is NP-hard. A promise problem is *SSE-hard* if, for some fixed $0 < \eta < 1/2$ and every fixed $\delta \in (0, \eta)$, there is a deterministic polynomial-time reduction from $\mathrm{SSE}(\eta, \delta)$ that maps its YES and NO cases to the respective cases of the target promise. The polynomial may depend on these fixed parameters. Consequently, an SSE-hard promise problem is NP-hard under SSEH.

Lemma 4.1 (Global SSE partition gap, normalized form)**.** *There are absolute constants $c > 0$ and $\varepsilon_0 > 0$ such that, for every fixed integer $q \ge 2$ and every fixed rational $0 < \varepsilon < \varepsilon_0$, the following promise on unweighted multigraphs G with $m > 0$ edges, allowing loops, is SSE-hard:*

YES: *There is a partition $V(G) = V_1 \dot\cup \cdots \dot\cup V_q$ such that $(1-\varepsilon)/q \le \mu_G(V_i) \le (1+\varepsilon)/q$ and $\Phi_G(V_i) \le 3\varepsilon$ for every i.*

NO: *Every $A \subseteq V(G)$ with $\mu_G(A) \in [1/4, 3/4]$ satisfies $|\delta_G(A)| \ge (c/2)\sqrt{\varepsilon}\, m$.*

The host degree is unrestricted. Section 6.5 later reduces the line graph's degree through incidence clique factors. The following lemma counts crossing and internal edge weights.

Lemma 4.2 (Crossing and internal edge weights)**.** *Let G be a finite undirected weighted multigraph with positive rational edge weights and total edge weight $W > 0$. Let $(V_1, \dots, V_q)$ be a partition of its vertices into parts of positive volume. Let D be the multiset of edges whose endpoints lie in different classes. For each i, let E_i be the multiset of edges with both endpoints in V_i, including loops at vertices of V_i. Each parallel edge is included in the corresponding multiset. Then $w(D) \le W \max_{i\in[q]} \Phi_G(V_i)$ and $w(E_i) \le \mathrm{vol}_G(V_i)/2$ for every i.*

Proof. Each edge in D belongs to the cuts of its two endpoint classes. Internal edges, including loops, belong to none. Hence

$$2w(D) = \sum_{i=1}^{q} w(\delta_G(V_i)) = \sum_{i=1}^{q} \Phi_G(V_i)\,\mathrm{vol}_G(V_i) \le \max_{i\in[q]} \Phi_G(V_i) \sum_{i=1}^{q} \mathrm{vol}_G(V_i) = 2W \max_{i\in[q]} \Phi_G(V_i).$$

Each internal edge, including a loop, contributes twice its weight to its class volume, whereas a crossing edge contributes once. Thus $2w(E_i) + w(\delta_G(V_i)) = \mathrm{vol}_G(V_i)$, which gives the second bound. □

4.2 The line graph and its separator lower bound

For an unweighted host multigraph G, define its *line graph* by

$$V(\mathcal{L}(G)) = E(G),$$
$$E(\mathcal{L}(G)) = \{\{e, f\} : e, f \in E(G),\ e \ne f,\ e \cap f \ne \varnothing\},$$

where $E(G)$ contains edge occurrences and $e \cap f \ne \varnothing$ means that the two occurrences share an endpoint. Parallel host edges therefore give distinct vertices. This is a simple unweighted graph,

even when the host has parallel edges or loops. In particular, $|V(\mathcal{L}(G))| = |E(G)| = m$. Testing each pair of distinct host-edge occurrences for a shared endpoint constructs this graph in $O(m^2)$ time and produces at most $\binom{m}{2}$ edges.

Lemma 4.3 (A treedepth upper bound from a partition)**.** *Let G be an unweighted host multigraph with $m > 0$ edges, counting parallel edges separately, and let $(V_1, \dots, V_q)$ be a partition of its vertices. If D is the multiset of edges joining different classes, then* $\mathrm{td}(\mathcal{L}(G)) \le |D| + \max_{i\in[q]} |E(G[V_i])|$.

Proof. Make one chain from the line-graph vertices corresponding to the edges in D. For each class V_i, make a second chain from the vertices corresponding to the internal edges of $G[V_i]$. If D is nonempty, attach the class chains below the last vertex of the D-chain. If D is empty, use the class chains as separate roots.

Vertices on the D-chain are comparable with all vertices of the forest. Within each class chain all vertices are comparable as well. The only remaining pairs lie on different class chains. Their host edges have endpoints in disjoint classes, so those pairs are nonadjacent in $\mathcal{L}(G)$. Thus the constructed forest is an elimination forest of height $|D| + \max_{i\in[q]} |E(G[V_i])|$, including when D is empty. □

Deleting vertices of $\mathcal{L}(G)$ deletes the corresponding host edge copies. The remaining components have disjoint sets of incident host vertices, yielding a host cut covered by the deleted edges.

For $0 < \rho < 1$, a *ρ-balanced separator* of an N-vertex graph H is a set $S \subseteq V(H)$ such that every component of $H - S$ has at most ρN vertices. We write $b_\rho(H)$ for its minimum size. A *balanced separator* is a 1/2-balanced separator, and we abbreviate $b_{1/2}(H)$ to $b(H)$.

Lemma 4.4 (Balanced separators from tree decompositions [WAPL15, Lemma 5.3])**.** *Every finite nonempty graph H satisfies $b(H) \le \mathrm{tw}(H) + 1$.*

Lemma 4.5 (From host cuts to line-graph separators)**.** *Let G be an unweighted host multigraph with $m > 0$ edges, counting parallel edges separately. Suppose that, for some $\beta > 0$, every set A with $\mu_G(A) \in [1/4, 3/4]$ satisfies*

$$|\delta_G(A)| \ge \beta m. \tag{25}$$

Then $b_\rho(\mathcal{L}(G)) \ge \min\{(3/4 - \rho)m, \beta m\}$ for every $1/2 \le \rho < 3/4$.

Proof. Fix a ρ-balanced separator S of $\mathcal{L}(G)$, and write F_S for the corresponding multiset of host edges, so $|F_S| = |S|$. If $|S| \ge (3/4 - \rho)m$, the claim is immediate. Assume henceforth that $|S| < (3/4 - \rho)m$.

Grouping the surviving host edges. The line graph $\mathcal{L}(G) - S$ has $m - |S| > (1/4 + \rho)m$ vertices, with each component of order at most ρm. Choose a union Z of components with $m/4 \le |Z| \le \rho m$. If a component has size in this interval, take it. Otherwise, every component has fewer than $m/4$ vertices. Add components until their total first reaches $m/4$. The total is then below $m/2 \le \rho m$.

Let X be the host edges corresponding to vertices in Z, and let Y be the other surviving host edges. Then $|X| = |Z| \ge m/4$ and $|Y| = m - |S| - |X| > m/4$. If a host vertex were incident to edges in both multisets, their line-graph vertices would be adjacent, with one in Z and one outside Z. This contradicts the choice of Z as a union of components.

Forming the host cut. Let $A, B \subseteq V(G)$ be the sets of endpoints of edges in X, Y, respectively. The preceding paragraph gives $A \cap B = \varnothing$. Every edge of X contributes two degree units to $\mathrm{vol}_G(A)$, including a loop. The same holds for Y and B. Thus $\mathrm{vol}_G(A) \ge 2|X| \ge m/2$ and $\mathrm{vol}_G(B) \ge 2|Y| \ge m/2$. Since the total host volume is $2m$, $\mathrm{vol}_G(A) \le 2m - \mathrm{vol}_G(B) \le 3m/2$. Since $\mu_G(A) = \mathrm{vol}_G(A)/(2m)$, we obtain $\mu_G(A) \in [1/4, 3/4]$. Every surviving edge lies wholly in A or

wholly in $B \subseteq V(G) \setminus A$, so every edge crossing A belongs to F_S. The cut assumption therefore yields $|S| = |F_S| \geq |\delta_G(A)| \overset{(25)}{\geq} \beta m$, proving the bound. □

Lemma 4.6. *For every finite graph H, $b(H) \leq \mathrm{td}(H)$. Moreover, every elimination forest F of an N-vertex graph H satisfies* $\mathrm{totd}(F) \geq (N/2)b(H)$.

Proof. Fix an elimination forest F of H. If every root subtree has at most $N/2$ vertices, then $b(H) = 0$ and both claims follow. Otherwise, delete the root of the unique subtree of size greater than $N/2$, and continue into a child subtree of size greater than $N/2$ while one exists. Such a child is unique, since two disjoint subtrees of that size cannot fit among N vertices.

The deleted vertices form an ancestor path P. Every remaining subtree has at most $N/2$ vertices, and no graph edge joins distinct remaining subtrees. Thus P is balanced and $b(H) \leq |P| \leq h(F)$. Minimizing height proves the first claim. The original subtree rooted at the last vertex of P has more than $N/2$ vertices, each at depth at least $|P| \geq b(H)$. Their depths alone give $\mathrm{totd}(F) \geq (N/2)b(H)$. □

Corollary 4.7 (NO treedepth bound)**.** *Under the cut assumption of Lemma 4.5,* $\mathrm{td}(\mathcal{L}(G)) \geq \min\{m/4, \beta m\}$.

Proof. Take $\rho = 1/2$. Then

$$\mathrm{td}(\mathcal{L}(G)) \overset{\text{Lem. 4.6}}{\geq} b(\mathcal{L}(G)) \overset{\text{Lem. 4.5}}{\geq} \min\{m/4, \beta m\}.$$

□

Let G_{w} be a weighted partition instance from Theorem A.11, G^* its rounded unweighted host from Lemma 4.1, and $H = \mathcal{L}(G^*)$ its simple line graph. The construction and its guarantees are

$$G_{\mathrm{w}} \xrightarrow{\text{Lemma 4.1}} G^* \xrightarrow{\text{Lemmas 4.3 and 4.5}} H.$$

The approximation gap below uses these estimates; the later reduction in Section 6.5 bounds the degree of H.

4.3 The approximation gap

Dense treedepth gap. Using the convenient YES coefficient 4ε, the ratio of the NO and YES coefficients is $c/(8\sqrt{\varepsilon})$, which grows without bound as ε decreases. Fix $C \geq 1$ and choose a fixed rational $\varepsilon > 0$ small enough that Lemma 4.1 applies, $c\sqrt{\varepsilon} \leq 1/4$, and $\sqrt{\varepsilon} < c/(8C)$. Since ε is fixed, $q = \lceil 2/\varepsilon \rceil$ is a constant. Apply that lemma with this q to obtain an unweighted host G^* with m edges, and output $\mathcal{L}(G^*)$.

In the YES case, let D be the multiset of host edges joining different partition classes. The host bounds are

$$\begin{aligned} |D| &\overset{\text{Lem. 4.2}}{\leq} m \max_{i \in [q]} \Phi_{G^*}(V_i) \overset{\text{Lem. 4.1}}{\leq} 3\varepsilon m, \\ |E(G^*[V_i])| &\overset{\text{Lem. 4.2}}{\leq} \mathrm{vol}_{G^*}(V_i)/2 \overset{\text{Lem. 4.1}}{\leq} (1+\varepsilon)m/q. \end{aligned} \tag{26}$$

Consequently

$$\begin{aligned} \mathrm{td}(\mathcal{L}(G^*)) &\overset{\text{Lem. 4.3}}{\leq} |D| + \max_{i \in [q]} |E(G^*[V_i])| \\ &\overset{(26)}{\leq} 3\varepsilon m + (1+\varepsilon)m/q \leq 3\varepsilon m + (1+\varepsilon)\varepsilon m/2 \\ &= (7/2 + \varepsilon/2)\varepsilon m < 4\varepsilon m, \end{aligned} \tag{27}$$

where $q = \lceil 2/\varepsilon \rceil$ gives $1/q \le \varepsilon/2$, and $\varepsilon < 1$ gives $7/2+\varepsilon/2 < 4$. In the NO case, $(c/2)\sqrt{\varepsilon} \le 1/8 < 1/4$, so Corollary 4.7 gives

$$\mathrm{td}(\mathcal{L}(G^*)) \overset{\text{Lem. 4.1, Cor. 4.7}}{\geq} (c/2)\sqrt{\varepsilon}\, m > 4C\varepsilon m,$$

by the choice of ε. The NO bound therefore exceeds the YES bound by a factor greater than C. Indeed, a C-approximation returning an elimination forest has height at most $4C\varepsilon m$ in the YES case, whereas every valid forest has larger height in the NO case. Comparing the returned height with this rational threshold distinguishes the source promise. The threshold uses only the fixed parameters and the observable output order m; all construction sizes are polynomial for these fixed parameters. □

5 NP-hardness of approximation

We prove NP-hardness below 18/17 for treedepth, below 12/11 for treewidth, and below 63/62 for total elimination depth. All three reductions use the normalized bounded-occurrence parity source of Theorem 5.1. Treedepth and treewidth share one tripartite construction. A second tripartite encoding gives the total-depth gap in Section 5.3.

5.1 Treedepth

We encode parity equations in a tripartite graph, then convert its vertex-cover number into a dense treedepth gap. Corollary 6.22 gives the bounded-degree hardness consequence after sparsification.

The truth blocks and assignment witnesses follow Bonnet, Neuen, and Sokołowski [BNS25, Section 4 and Lemma 10], building on Bonnet's treewidth construction [Bon25, Section 3.1]. The anchor graph is the one in [BNS25, Lemma 6]. We modify the witness gadgets for parity equations and use the anchor construction through the corollary below. Appendix A.4 records which ingredients are inherited, and Appendix B.3 gives the corollary's derivation.

5.1.1 The source gap and normalization

We use Håstad's Max-3-LIN-2 gap [Hås97], in the distinct-variable, bounded-occurrence form stated by Hazan, Safra, and Schwartz [HSS03, Theorem 10]. Appendix A.5 adds right-hand-side balance and divisibility of occurrence counts to obtain the following statement.

Theorem 5.1 (Normalized bounded-occurrence Max-3-LIN-2 gap)**.** *For every fixed $0 < \epsilon < 1/4$, there is a constant $b_\epsilon \ge 6$ depending only on ϵ such that the following promise problem is NP-hard. The input is an unweighted system F with finite Boolean variable set $\mathcal{X}(F)$ and $m \ge 1$ equations, indexed by $j \in \{1, \dots, m\}$, of the form $x + y + z = a \pmod 2$, where $a \in \{0, 1\}$ and $x, y, z \in \mathcal{X}(F)$ are distinct. Equations may be repeated. For $x \in \mathcal{X}(F)$, write d_x for the number of equations containing x. Write a_j for the right-hand side of equation j, and $\mathrm{OPT}(F)$ for the maximum number simultaneously satisfied, counting repetitions. The instances satisfy*

$$\begin{gathered} d_x \in 6\mathbb{Z}_{>0}, \qquad d_x \le b_\epsilon, \qquad |\{d_x : x \in \mathcal{X}(F)\}| \le 2, \\ \sum_{x \in \mathcal{X}(F)} d_x = 3m, \\ |\{j \in \{1, \dots, m\} : a_j = 0\}| = |\{j \in \{1, \dots, m\} : a_j = 1\}| = m/2. \end{gathered} \tag{28}$$

YES: $\mathrm{OPT}(F) \ge (1 - \epsilon)m$.

NO: $\mathrm{OPT}(F) \le (1/2+\epsilon)m$.

For the normalization, let F_0 be the initial source system, with m_0 equations. Let F_0' be a copy on fresh variables with every right-hand side complemented, and put $F_1 = F_0 \dot\cup F_0'$. The two operations are

$$F_0 \xrightarrow{\text{Lemma A.12}} F_1 \xrightarrow{\text{Lemma A.13}} F.$$

The second arrow repeats every equation six times. Consequently

$$m \overset{\text{Lems. A.12, A.13}}{=} 12m_0,$$
$$\mathrm{OPT}(F) \overset{\text{Lems. A.12, A.13}}{=} 12\,\mathrm{OPT}(F_0).$$

Thus the satisfied fraction is unchanged, and the maximum occurrence count is multiplied by six. We retain the correspondence between each variable of F_0 and its copy, and between the repeated equations in the two halves. This recorded pairing will provide a balanced optimal independent set in the treewidth reduction. The restriction to at most two occurrence counts is inherited from the source theorem; the treedepth and treewidth reductions use only the other normalization conditions. We henceforth assume (28). Divisibility makes the truth-block sizes integral, and right-hand-side balance makes the three graph parts equal.

5.1.2 A tripartite graph with equal parts

We construct a tripartite graph G_F with independent parts $A \dot\cup B \dot\cup C$. Its vertices have two roles: vertices in *truth blocks* encode values of individual variables, and *assignment witnesses* represent satisfying assignments to the three variables of an equation. All blocks are pairwise disjoint, and every witness is a new vertex. For every $x \in \mathcal{X}(F)$, create independent truth blocks $T_x^0 \subseteq A$ and $T_x^1 \subseteq B$, with

$$|T_x^0| = |T_x^1| = 5d_x/6 \in \mathbb{Z}_{>0}.$$

Integrality follows from (28). Put all edges between these two blocks. Selecting T_x^a in an independent set will encode the assignment $x = a$. The total size of one block per variable is

$$\frac{5}{6}\sum_{x\in\mathcal{X}(F)} d_x \overset{(28)}{=} \frac{5m}{2}. \tag{29}$$

For each equation $j \in \{1,\dots,m\}$, create four witnesses, one for each of its satisfying assignments. If its right-hand side is zero, call the witness for assignment 000 special and place it in A. If its right-hand side is one, call the witness for assignment 111 special and place it in B. Place the other three witnesses in C. Join the special witness to the other three witnesses of its equation, forming a star. We call these four vertices and their star edges the *equation gadget*. Its star permits at most three witnesses in an independent set, even when no truth block is selected.

For every witness representing a satisfying assignment σ and every variable x of its equation, join it to the entire truth block $T_x^{1-\sigma(x)}$. Add no other edges, so witnesses from distinct equations are nonadjacent.

Every listed edge respects the tripartition. A special 000 witness lies in A and its truth-block neighbors all lie in B. A special 111 witness lies in B and its truth-block neighbors all lie in A. The three remaining witnesses lie in C, and their truth-block neighbors lie in $A \cup B$. Star edges join the special witness to C. Thus G_F is simple, unweighted, and tripartite.

The truth vertices number $5m$, and there are $4m$ witnesses, so G_F has $9m$ vertices. Equation (28) gives $m/2$ special witnesses in each of A and B, and three witnesses per equation in C. Hence

$$|A| = |B| = |C| \overset{(29),(28)}{=} 3m. \tag{30}$$

The source graph G_F already has bounded maximum degree. For $v \in T_x^a$, its opposite truth block and witness neighbors give

$$\deg_{G_F}(v) = \frac{5d_x}{6} + 2d_x = \frac{17d_x}{6} \overset{(28)}{\leq} 3b_\epsilon.$$

For a witness w of an equation on x, y, z, its three truth-block neighborhoods and at most three star neighbors give

$$\begin{aligned}\deg_{G_F}(w) &\leq \frac{5}{6}(d_x + d_y + d_z) + 3 \\ &\overset{(28)}{\leq} \frac{5}{2}b_\epsilon + 3 \overset{\text{Thm. 5.1}}{\leq} 3b_\epsilon,\end{aligned}$$

where the last inequality uses $b_\epsilon \geq 6$.

5.1.3 The independent-set optimum

For an independent set I containing every truth block that it meets, the selected block T_x^a fixes $x = a$. Variables with neither block selected remain unassigned. Complete this partial assignment by choosing each unassigned variable independently and uniformly from $\{0, 1\}$. Denote the resulting random assignment by σ. For $j \in \{1, \ldots, m\}$, let r_j count its unassigned variables and let w_j count its witness vertices in I. Define

$$p_j = \Pr[\text{equation } j \text{ is satisfied by } \sigma].$$

Lemma 5.2 (Witnesses under a partial assignment)**.** *With this notation, $w_j - p_j \leq 5r_j/6$ for every $j \in \{1, \ldots, m\}$.*

Proof. Every selected witness agrees with the fixed variables, since otherwise it has a neighbor in a selected truth block. If $r_j = 0$, the equation is either satisfied with one consistent witness or unsatisfied with none, so $w_j \leq p_j$. If $r_j \geq 1$, flipping one unassigned variable pairs every satisfying completion with an unsatisfying one. Distinctness of the three variables ensures that this flip changes the parity. Hence half of the 2^{r_j} completions satisfy the equation. Thus $p_j = 1/2$ and there are $2^{r_j - 1}$ consistent satisfying witnesses. The star permits at most three, giving

$$w_j \leq \min\{2^{r_j-1}, 3\} = r_j, \qquad p_j = \frac{1}{2} \qquad (r_j = 1, 2, 3). \tag{31}$$

Therefore

$$w_j - p_j \overset{(31)}{\leq} r_j - \frac{1}{2} \overset{r_j \leq 3}{\leq} \frac{5}{6}r_j \qquad (r_j = 1, 2, 3).$$

□

Lemma 5.3. *The graph G_F satisfies*

$$\Delta(G_F) \leq 3b_\epsilon, \qquad \alpha(G_F) = 5m/2 + \mathrm{OPT}(F), \qquad \mathrm{vc}(G_F) = 13m/2 - \mathrm{OPT}(F).$$

Proof. Start with any independent set I, and fill every truth block that it meets. This preserves independence because a block has no internal edges and its vertices have identical neighborhoods. It suffices to bound the filled set, which we continue to call I. The two blocks of a variable are joined completely, so at most one is selected. Let $U \subseteq \mathcal{X}(F)$ be the set of unassigned variables. Relative to one truth block per variable, the missing truth vertices number $\frac{5}{6}\sum_{x\in U} d_x$. For the random completion above, counting those variables' occurrences and applying linearity of expectation give

$$\sum_{j=1}^{m} r_j = \sum_{x\in U} d_x, \qquad \sum_{j=1}^{m} p_j \le \mathrm{OPT}(F). \tag{32}$$

The second bound holds because every completion satisfies at most $\mathrm{OPT}(F)$ equations. Lemma 5.2 therefore gives

$$\begin{aligned} |I| &\overset{(29),(32)}{=} \frac{5m}{2} + \sum_{j=1}^{m}\left(w_j - \frac{5}{6}r_j\right) \\ &\overset{\text{Lem. 5.2}}{\le} \frac{5m}{2} + \sum_{j=1}^{m} p_j \overset{(32)}{\le} \frac{5m}{2} + \mathrm{OPT}(F). \end{aligned}$$

For the reverse bound, take an assignment satisfying $\mathrm{OPT}(F)$ equations. Select its truth blocks, of total size $\frac{5m}{2}$ by (29), and, for each satisfied equation, its unique witness with that assignment. Such a witness has no neighbor in the selected truth blocks. Only one witness is selected per equation, so no star edge has both endpoints selected. There are no edges between distinct equation gadgets. The resulting independent set has size $\frac{5m}{2} + \mathrm{OPT}(F)$.

The matching bounds prove

$$\alpha(G_F) = \frac{5m}{2} + \mathrm{OPT}(F). \tag{33}$$

The graph has $9m$ vertices, so

$$\mathrm{vc}(G_F) \overset{(1)}{=} 9m - \alpha(G_F) \overset{(33)}{=} \frac{13m}{2} - \mathrm{OPT}(F).$$

□

5.1.4 The anchor graph

Let G be tripartite with specified independent parts A, B, C, and let $\ell \ge 1$ be an integer. Add pairwise disjoint sets K_A, K_B, K_C of size ℓ and distinct vertices z_A, z_B, z_C, with all added vertices distinct and outside $V(G)$. The graph $H_\ell(G)$ has vertex set $V(G) \cup K_A \cup K_B \cup K_C \cup \{z_A, z_B, z_C\}$ and edge set

$$E(H_\ell(G)) = E(G) \cup \bigcup_{X\in\{A,B,C\}} \left(\binom{K_X}{2} \cup \{\{u,v\} : u \in K_X,\ v \in X \cup \{z_X\}\} \right).$$

Thus K_X is a clique joined completely to $X \cup \{z_X\}$, and there are no other new edges. We call K_X the *anchor clique* of part X. This is the anchor graph of [BNS25, Lemma 6] and has $|V(G)| + 3\ell + 3$ vertices.

Corollary 5.4 (Anchors as long as the largest part)**.** *If G is tripartite with nonempty specified independent parts A, B, C and $\ell \ge \max\{|A|, |B|, |C|\}$, then $\mathrm{td}(H_\ell(G)) = \ell + \mathrm{vc}(G) + 1$.*

We derive this from the anchor construction in Appendix B.3, including the short-anchor analysis. Bonnet, Neuen, and Sokołowski require $\ell \ge \mathrm{vc}(G)$ for their identity [BNS25, Lemma 6]. Our corollary only requires ℓ to be at least the largest part size, so it applies with $\ell = 3m$ even though $\mathrm{vc}(G_F) \overset{\text{Lem. 5.3}}{\ge} 11m/2$.

5.1.5 The treedepth identity and the hardness gap

Proposition 5.5. *For every normalized source system F, the graph $H = H_{3m}(G_F)$ has $18m + 3$ vertices and satisfies*

$$\mathrm{td}(H) = \frac{19m}{2} - \mathrm{OPT}(F) + 1. \tag{34}$$

Proof. Each source part has size $3m$ by (30), so Corollary 5.4 and Lemma 5.3 give

$$\mathrm{td}(H) \overset{\text{Cor. 5.4}}{=} 3m + \mathrm{vc}(G_F) + 1 \overset{\text{Lem. 5.3}}{=} \frac{19m}{2} - \mathrm{OPT}(F) + 1.$$

The graph has $9m + 9m + 3 = 18m + 3$ vertices and $O(m^2)$ edges. Repeated equations have distinct witness gadgets, so no parallel edges arise. The source normalization changes size by a constant factor, giving a polynomial construction. □

The treedepth reduction follows the chain

$$F \xrightarrow{\text{Lemma 5.3}} G_F \xrightarrow{\text{Proposition 5.5}} H_{3m}(G_F).$$

The first arrow encodes $\mathrm{OPT}(F)$ in the vertex-cover number, and the second converts that number into treedepth. The source gap then gives the two bounds below; Corollary 6.22 later transfers them to bounded-degree graphs.

Dense treedepth gap. The leading NO and YES coefficients below approach 9 and 17/2 as the source error decreases, giving ratio 18/17. Fix $1 \le r < 18/17$ and put $\delta_0 = 9 - 17r/2 > 0$. Choose a fixed rational $0 < \epsilon < \min\{1/4, \delta_0/(2(1+r))\}$. Then $9 - \epsilon - r(17/2 + \epsilon) = \delta_0 - (1+r)\epsilon > \delta_0/2$. Put $\gamma = 9 - \epsilon - r(17/2 + \epsilon) > 0$ and fix $t = \lfloor (r-1)/\gamma \rfloor + 1$. Replace F by t disjoint copies on disjoint variable sets, and let m denote their total equation count. Since the original count is at least one,

$$\gamma m > r - 1. \tag{35}$$

Both the equation count and the optimum are multiplied by the fixed integer t, which depends only on r, ϵ. Thus replication is polynomial and preserves the source promise and (28). By Proposition 5.5 and the source promise, the two cases satisfy

$$\begin{aligned}
\text{YES: } \mathrm{td}(H) &\overset{(34)}{=} 19m/2 - \mathrm{OPT}(F) + 1 \\
&\overset{\text{Thm. 5.1}}{\le} 19m/2 - (1-\epsilon)m + 1 = (17/2 + \epsilon)m + 1, \\
\text{NO: } \mathrm{td}(H) &\overset{(34)}{=} 19m/2 - \mathrm{OPT}(F) + 1 \\
&\overset{\text{Thm. 5.1}}{\ge} 19m/2 - (1/2+\epsilon)m + 1 = (9-\epsilon)m + 1.
\end{aligned}$$

The NO lower bound minus r times the YES upper bound is $\gamma m + 1 - r \overset{(35)}{>} 0$, proving the gap. □

5.2 Treewidth

We apply Lemma 3.17 to the parity witness graph from Section 5.1. For a fixed integer $R \ge 1$, take R disjoint copies without changing their source edges, then complete the three global parts to cliques, obtaining $K_R(G_F)$. This is the replicated adaptation of Bonnet's co-tripartite completion [Bon25, Section 3.1], described and proved in Section 3.4. Processing copies separately in each decomposition path controls the additional bag size independently of R.

Dense treewidth gap. Fix $1 \le r < 12/11$ and put $\delta_0 = 6 - 11r/2 > 0$. Choose a fixed rational $0 < \epsilon < \min\{1/4, \delta_0/(2(1+r))\}$, so that $6 - \epsilon - r(11/2+\epsilon) > \delta_0/2$. Put $\delta = 6 - \epsilon - r(11/2+\epsilon) > 0$, and set the fixed integer $R = \lfloor 3r/\delta \rfloor + 1$, so that

$$R\delta > 3r. \tag{36}$$

These choices preserve the source separation and control the additional bag size in the replicated decomposition. Use Theorem 5.1 with error ϵ to obtain a system F with $m \ge 1$ equations and occurrence bound b_ϵ. Apply the witness construction of Section 5.1. By (30) and Lemma 5.3, the resulting simple unweighted graph G_F satisfies

$$|V(G_F)| = 9m, \qquad |A| = |B| = |C| = 3m, \qquad \Delta(G_F) \le 3b_\epsilon,$$

and $\mathrm{vc}(G_F) \overset{\text{Lem. 5.3}}{=} 13m/2 - \mathrm{OPT}(F)$. The two graph transformations are

$$F \xrightarrow{\text{Lemma 5.3}} G_F \xrightarrow{\text{Lemma 3.17}} K_R(G_F).$$

Since $\mathrm{OPT}(F) \le m$, every instance satisfies

$$\mathrm{vc}(G_F) \overset{\text{Lem. 5.3}}{=} \frac{13m}{2} - \mathrm{OPT}(F) \ge \frac{13m}{2} - m = \frac{11m}{2} > 3m.$$

The source promise gives the two bounds

$$\begin{aligned}
\text{YES:}\quad \mathrm{vc}(G_F) &\overset{\text{Lem. 5.3}}{=} \frac{13m}{2} - \mathrm{OPT}(F) \\
&\overset{\text{Thm. 5.1}}{\le} \frac{13m}{2} - (1-\epsilon)m = \left(\frac{11}{2} + \epsilon\right) m, \\
\text{NO:}\quad \mathrm{vc}(G_F) &\overset{\text{Lem. 5.3}}{=} \frac{13m}{2} - \mathrm{OPT}(F) \\
&\overset{\text{Thm. 5.1}}{\ge} \frac{13m}{2} - \left(\frac{1}{2} + \epsilon\right) m = (6-\epsilon)m.
\end{aligned} \tag{37}$$

Applying Lemma 3.17 gives

$$\begin{aligned}
\text{YES:}\quad \mathrm{tw}(K_R(G_F)) &\overset{\text{Lem. 3.17}}{\le} R\,\mathrm{vc}(G_F) + 3m - 1 \\
&\overset{(37)}{\le} R(11/2+\epsilon)m + 3m - 1, \\
\text{NO:}\quad \mathrm{tw}(K_R(G_F)) &\overset{\text{Lem. 3.17}}{\ge} R\,\mathrm{vc}(G_F) - 1 \\
&\overset{(37)}{\ge} R(6-\epsilon)m - 1.
\end{aligned}$$

For every $m \ge 1$, the NO bound minus r times the YES bound is

$$(R\delta - 3r)m + (r-1) \overset{(36)}{>} 0,$$

by the choice of R and $r \ge 1$. Thus the bounds are separated by factor r even with the additive constants -1. The output has $9Rm$ vertices, and its three global cliques have equal size $3Rm$. □

5.3 Total elimination depth

Auxiliary pairs encode the normalized parity source of Theorem 5.1 in a tripartite graph. Completing its parts to cliques follows Bonnet's construction [Bon25, Section 3.1]. Three added independent sets then transfer the number of satisfied equations to total depth, giving the limiting gap $189/186 = 63/62$.

5.3.1 The three-path objective

Let G be a simple tripartite graph with specified nonempty independent parts $A \dot\cup B \dot\cup C$, and put $N = |V(G)|$. Complete each part to a clique by setting

$$\widehat{G} = \left(V(G),\ E(G) \cup \bigcup_{X \in \{A,B,C\}} \binom{X}{2} \right).$$

For an integer $M \geq 1$, form $H_M^\Sigma(G)$ by adding disjoint independent sets T_A, T_B, T_C, with

$$|T_X| = M, \qquad N_{H_M^\Sigma(G)}(t) = X \qquad (X \in \{A,B,C\},\ t \in T_X).$$

The resulting graph is simple and unweighted.

In an elimination forest F_G of $\widehat{G}$, the vertices of each clique part lie on one ancestor path. Write y_X for the deepest vertex of part X, with depths starting at one, and define

$$W(F_G) = \sum_{X \in \{A,B,C\}} (\mathrm{depth}_{F_G}(y_X) + 1), \qquad W(G) = \min_{F_G} W(F_G).$$

Here $\mathrm{totd}(F_G)$ denotes the sum of all vertex depths in the forest. The following normalization and path identities are proved in Appendix B.4.

Lemma 5.6 (Normalization of added vertices)**.** *Every elimination forest of $H_M^\Sigma(G)$ can be transformed without increasing total depth so that all vertices of each T_X are leaf children of the deepest vertex of X. Consequently,*

$$\mathrm{totd}(H_M^\Sigma(G)) = \min_{F_G} \big(\mathrm{totd}(F_G) + MW(F_G)\big), \tag{38}$$

and

$$MW(G) \leq \mathrm{totd}(H_M^\Sigma(G)) \leq MW(G) + \frac{N(N+1)}{2}. \tag{39}$$

For $X \in \{A,B,C\}$, let Y, Z be the other two parts. Let $S_0 \subseteq V(G)$ range over covers of all X–$(Y \cup Z)$ edges, and let $S_1 \subseteq (Y \cup Z) \setminus S_0$ be a minimum vertex cover of the remaining YZ graph. Define

$$\mathrm{cost}_X(G) = \min_{S_0} \left(2|S_0| + \mathrm{vc}\big(G[(Y \cup Z) \setminus S_0]\big)\right). \tag{40}$$

All cover graphs refer to G, before clique completion. In the associated forest, S_0 lies on a path common to all three parts, S_1 lies on a path shared by Y, Z, and the remaining vertices of each part lie on its own path.

Lemma 5.7 (Cost of the three paths)**.** *For every tripartite graph G with nonempty specified parts,*

$$W(G) = N + 3 + \min\{\mathrm{cost}_A(G), \mathrm{cost}_B(G), \mathrm{cost}_C(G)\}. \tag{41}$$

5.3.2 The tripartite parity construction

Fix $0 < \epsilon < 1/4$ and let F be an instance from Theorem 5.1, with variable set $\mathcal{X}(F)$, $m \geq 1$ equations, and satisfying (28). Thus d_x is the number of equations containing variable x, $d_x \in 6\mathbb{Z}_{>0}$, and $\sum_{x\in\mathcal{X}(F)} d_x = 3m$. The source occurrence bound is $d_x \leq b_\epsilon$. Construct $G = G_F^\Sigma$ with independent parts A, B, C as follows. All blocks and all auxiliary vertices mentioned below are pairwise disjoint. An *auxiliary pair* consists of fresh vertices $b \in B$, $c \in C$ and the edge bc. Attaching an endpoint to a block means adding every edge between that endpoint and the block.

Literal blocks. For every $x \in \mathcal{X}(F)$, put two blocks A_x^0, A_x^1 in A, with $|A_x^0| = |A_x^1| = 9d_x/2 \in \mathbb{Z}_{>0}$.

Opposite pairs. For every $x \in \mathcal{X}(F)$, create $2d_x$ auxiliary pairs. For d_x of them, attach b to A_x^0 and c to A_x^1; for the other d_x, attach b to A_x^1 and c to A_x^0.

Equation pairs. In each equation $e \in \{1, \dots, m\}$, order its distinct variables as x_1, x_2, x_3 and write a_e for its right-hand side. For every satisfying local assignment $\sigma = (\sigma_1, \sigma_2, \sigma_3) \in \{0,1\}^3$ with $\sigma_1 + \sigma_2 + \sigma_3 = a_e \pmod 2$, and every $j \in \{1,2,3\}$, create one auxiliary pair. Attach b to all three blocks $A_{x_i}^{1-\sigma_i}$, $i \in \{1,2,3\}$, and attach c to $A_{x_j}^{\sigma_j}$. This gives twelve pairs per equation.

Padding pairs. Add $18m$ auxiliary pairs having no neighbors in A.

Add no other edges to G. In particular, its BC graph is a matching. There are $6m$ opposite pairs, $12m$ equation pairs, and $18m$ padding pairs, so

$$|A| = 27m, \qquad |B| = |C| = 36m, \qquad N = 99m. \tag{42}$$

The two vertex types satisfy

$$\begin{aligned} \deg_G(v) &= 2d_x + 8d_x = 10d_x \quad && (v \in A_x^0 \cup A_x^1), \\ \deg_G(v) &\leq 3(9b_\epsilon/2) + 1 && (v \in B \cup C). \end{aligned}$$

The second bound accounts for at most three literal-block neighborhoods and the paired endpoint. Since $b_\epsilon \geq 6$,

$$\Delta(G_F^\Sigma) \leq 14b_\epsilon. \tag{43}$$

This is the degree before completing the parts and adding T_A, T_B, T_C.

In the reduction flow, transformations appear below the arrows and references above them:

$$\begin{aligned} \text{parity gap} &\xrightarrow[\text{normalization}]{\text{[HSS03, Theorem 10]; Appendix A.5}} F, \\ F &\xrightarrow[\text{auxiliary pairs}]{\text{Proposition 5.8}} G_F^\Sigma, \\ G_F^\Sigma &\xrightarrow[\text{clique completion}]{\text{[Bon25, Section 3.1]}} \widehat{G}_F^\Sigma, \\ \widehat{G}_F^\Sigma &\xrightarrow[\text{add } T_A,T_B,T_C]{\text{Lemma 5.6}} H_M^\Sigma(G_F^\Sigma). \end{aligned}$$

Proposition 5.8 (The encoded parity optimum)**.** *The tripartite graph above satisfies*

$$W(G_F^\Sigma) = 192m - 6\,\mathrm{OPT}(F) + 3. \tag{44}$$

In particular, $W(G_F^\Sigma) \geq 186m + 3$.

Proof. We compute the A-first cost and lower-bound the two competing costs in (41).

The A-first cover. By Lemma B.9, each literal block can be taken wholly in S_0 or wholly outside it. Omitting a literal block forces all its auxiliary neighbors into S_0. Retain exactly these forced endpoints: removing any unforced auxiliary endpoint from S_0 saves two and increases the residual cover cost by at most one. The residual BC graph remains a matching, so each pair contributes one when neither endpoint is forced, two when one is forced, and four when both are forced. Relative to an unforced pair, the extra costs are therefore $0, 1, 3$.

With every literal block in S_0, the cost is

$$2|A| + 36m \overset{(42)}{=} 90m. \tag{45}$$

For variable x, omitting one block saves $9d_x$ and incurs opposite-pair extra cost $2d_x$, giving net saving $7d_x$. Omitting both saves $18d_x$ and incurs extra cost $6d_x$, giving net saving $12d_x$. Allocate these net savings equally among the d_x equations containing x.

For an equation e, let s be the number of its variables with exactly one omitted block and let b be the number with both blocks omitted. A variable with exactly one omitted block A_x^v is fixed to value v. When $b = 0$, let n_e be the number of the four satisfying local assignments compatible with these fixed values. Write Δ_e for the allocated variable savings minus the extra costs of the twelve equation pairs. Then

$$\Delta_e = \begin{cases} 3s - 12 + (3+s)n_e, & b = 0, \\ 3s + 4b - 12, & b > 0. \end{cases} \tag{46}$$

To verify the first formula, a compatible assignment forces precisely s of its three c endpoints and no b endpoint. An incompatible assignment forces all three b endpoints and adds two further units for every forced c endpoint. Let $\mathcal{S}_e$ be the four satisfying local assignments, and let f_σ count the fixed variables matching $\sigma \in \mathcal{S}_e$. Each coordinate takes each value in exactly two members of $\mathcal{S}_e$, so $\sum_{\sigma \in \mathcal{S}_e} f_\sigma = 2s$. Each of the n_e compatible assignments has $f_\sigma = s$. Thus the total extra equation cost, separating compatible and incompatible assignments, is

$$E_e = n_e s + 3(4 - n_e) + 2(2s - n_e s) = 12 + 4s - (3+s)n_e.$$

Subtracting E_e from $7s$ proves the first formula. When $b > 0$, all twelve b endpoints are forced, and the number of forced c endpoints is $2s + 4b$. The extra cost is then $12 + 4s + 8b$, subtracted from $7s + 12b$. For $b = 0$ and $s \le 2$, the unfixed variables give $n_e = 2^{2-s}$. Substitution in (46) and the bound $s + b \le 3$ therefore give

$$\begin{aligned} &\Delta_e = -s(3-s)/2 \le 0 && (b = 0,\ s \le 2), \\ &\Delta_e = 3s + 4b - 12 \overset{s+b\le 3}{\le} b - 3 \le 0 && (b > 0), \\ &\Delta_e = 6n_e - 3 \in \{-3, 3\} && (b = 0,\ s = 3). \end{aligned} \tag{47}$$

In particular, an equation with an unfixed variable has saving at most zero. Let $U \subseteq \mathcal{X}(F)$ consist of variables having neither or both blocks omitted. Sample one vector z_U uniformly from $\{0,1\}^U$, with independent coordinates, and keep the fixed values outside U. Each unfinished equation contains a variable of U; conditioning on its other two distinct variables leaves that variable fair, so the equation is satisfied with probability $1/2$. Its new expected saving is therefore $(3-3)/2 = 0$, while fully fixed equations keep their saving. Writing Δ_e^{old} and Δ_e^{new} for the respective savings, linearity gives

$$\mathbb{E}\left[\sum_{e=1}^{m} \Delta_e^{\text{new}}\right] \overset{\text{linearity}}{=} \sum_{e=1}^{m} \mathbb{E}[\Delta_e^{\text{new}}] \overset{(47)}{\ge} \sum_{e=1}^{m} \Delta_e^{\text{old}}.$$

No independence between equations is needed. Some outcome attains at least its expectation. A full assignment satisfying k equations has saving $3k - 3(m-k) = 6k - 3m$. Conversely, any assignment is realized by omitting the block matching each assigned value. If $S_{\max}$ denotes the maximum total saving over the normalized block choices, this proves

$$S_{\max} \overset{(47)}{=} 6\,\mathrm{OPT}(F) - 3m. \tag{48}$$

Therefore

$$\begin{aligned} \mathrm{cost}_A(G) &\overset{\text{Lem. B.9, (45)}}{=} 90m - S_{\max} \\ &\overset{(48)}{=} 90m - (6\,\mathrm{OPT}(F) - 3m) = 93m - 6\,\mathrm{OPT}(F). \end{aligned} \tag{49}$$

The competing covers. Consider a B-first or C-first cover. Normalize each literal block using Lemma B.9. Its state is T if it is in S_0, U if it is in S_1, and V if it is in neither. For a block of variable x, these states cost $9d_x, 9d_x/2, 0$, respectively.

An *ordinary pair* has its b endpoint attached to one literal block, called its tail, and its c endpoint attached to one literal block, called its head. For fixed block states, its minimum auxiliary cost in a B-first cover is

$$2 + \mathbf{1}[\text{tail} \neq T,\ \text{head} = V]. \tag{50}$$

The middle edge forces at least one endpoint into S_0, at cost two. If the tail is T, choosing $c \in S_0$ attains this cost. Otherwise $b \in S_0$ is forced. The head attachment is then already covered when the head is T or U; when it is V, the cheapest choice is to add c to S_1, costing one more. In a C-first cover the same formula holds with tail and head exchanged. Here $\mathbf{1}[\mathcal{E}]$ is one if condition $\mathcal{E}$ holds and zero otherwise.

For an equation pair, its b endpoint has three block neighbors. In the B-first case its extra cost above two is $\mathbf{1}[\text{head} = V] \max_{1\le i\le 3} \mathbf{1}[\text{tail}_i \neq T]$. In the C-first case it is $\mathbf{1}[\text{head} \neq T] \max_{1\le i\le 3} \mathbf{1}[\text{tail}_i = V]$. Each maximum dominates the average of its three entries; in particular,

$$\begin{aligned} &2 + \mathbf{1}[\text{head} = V] \max_{1\le i\le 3} \mathbf{1}[\text{tail}_i \neq T] \\ &\overset{\text{max}\ge\text{average}}{\ge} \frac{1}{3}\sum_{i=1}^{3} (2 + \mathbf{1}[\text{head} = V,\ \text{tail}_i \neq T]). \end{aligned}$$

For this cost analysis, represent each equation pair by three weighted directed ordinary pairs, one from each tail block to the head block. The baseline two is averaged along with the extra cost. Let $\mathcal{P}_e$ be the resulting multiset of pairs and $\mathcal{P}_o$ the multiset of opposite pairs, directed from their tail to their head. Set $\mathcal{P} = \mathcal{P}_o \,\dot\cup\, \mathcal{P}_e$ and assign

$$\omega(p) = \begin{cases} 1, & p \in \mathcal{P}_o, \\ 1/3, & p \in \mathcal{P}_e. \end{cases} \qquad \sum_{p\in\mathcal{P}} \omega(p) = 6m + \frac{1}{3}(36m) = 18m.$$

These weights describe the cost analysis; the constructed graph remains unweighted. Write $\mathrm{tail}(p)$ and $\mathrm{head}(p)$ for the two literal blocks of p. Let $\mathcal{B} = \{A_x^v : x \in \mathcal{X}(F),\ v \in \{0,1\}\}$. For a fixed block-state map $z : \mathcal{B} \to \{T, U, V\}$, its total literal-block cost is

$$L(z) = \sum_{x\in\mathcal{X}(F)} \sum_{v\in\{0,1\}} d_x \left(9\mathbf{1}[z(A_x^v) = T] + \frac{9}{2}\mathbf{1}[z(A_x^v) = U] \right).$$

Put

$$I_p(z) = \mathbf{1}[z(\mathrm{tail}(p)) \neq T,\ z(\mathrm{head}(p)) = V] \qquad (p \in \mathcal{P}).$$

If $C_{\mathrm{np}}(z)$ is the literal-block and non-padding-pair cost of a B-first cover with these states, the preceding inequality gives

$$\begin{aligned} C_{\mathrm{np}}(z) &\overset{\max\geq\text{average}}{\geq} L(z) + \sum_{p\in\mathcal{P}} \omega(p)\big(2 + I_p(z)\big) \\ &= 36m + L(z) + \sum_{p\in\mathcal{P}} \omega(p) I_p(z). \end{aligned} \tag{51}$$

For every $x \in \mathcal{X}(F)$ and $v \in \{0,1\}$,

$$\begin{aligned} \sum_{p\in\mathcal{P}:\,\mathrm{head}(p)=A_x^v} \omega(p) &= d_x + 2d_x = 3d_x, \\ \sum_{p\in\mathcal{P}:\,\mathrm{tail}(p)=A_x^v} \omega(p) &= d_x + 2d_x = 3d_x. \end{aligned} \tag{52}$$

Indeed, opposite pairs contribute d_x in each direction. In each equation containing x, the literal is a tail in two satisfying assignments and all three positions, each with weight $1/3$, and a head in two assignments at its own position, each with total weight one.

Every directed pair satisfies the pointwise inequality

$$1 \leq I_p(z) + \mathbf{1}[z(\mathrm{head}(p)) \neq V] + \mathbf{1}[z(\mathrm{tail}(p)) = T].$$

Indeed, if neither block indicator pays the unit, the head is V and the tail is not T, so $I_p(z) = 1$. Multiply by $\omega(p)$ and sum, using $\sum_{p\in\mathcal{P}} \omega(p) = 18m$. The equal weighted in- and out-degrees give the certificate

$$\begin{aligned} 18m &\overset{(52)}{\leq} \sum_{p\in\mathcal{P}} \omega(p) I_p(z) \\ &\quad + \sum_{x\in\mathcal{X}(F)} \sum_{v\in\{0,1\}} 3d_x\big(2\mathbf{1}[z(A_x^v) = T] + \mathbf{1}[z(A_x^v) = U]\big) \\ &\overset{6d_x\leq 9d_x,\ 3d_x\leq 9d_x/2}{\leq} \sum_{p\in\mathcal{P}} \omega(p) I_p(z) + L(z). \end{aligned} \tag{53}$$

For a C-first cover reverse all ordinary-pair directions; the same bounds apply by (52). For either competing cover, this certificate and the weighted baseline prove

$$C_{\mathrm{np}}(z) \overset{(51),\,(53)}{\geq} 36m + 18m = 54m. \tag{54}$$

Each padding pair costs at least two, since one of its endpoints must be in S_0. Therefore

$$\mathrm{cost}_B(G), \mathrm{cost}_C(G) \overset{(54)}{\geq} 3(18m) + 2(18m) = 90m. \tag{55}$$

A uniformly random assignment satisfies each parity equation with probability $1/2$, so

$$\frac{m}{2} \leq \mathrm{OPT}(F) \leq m. \tag{56}$$

It follows that

$$\begin{aligned} \mathrm{cost}_A(G) &\overset{(49)}{=} 93m - 6\,\mathrm{OPT}(F) \overset{(56)}{\leq} 90m \\ &\overset{(55)}{\leq} \min\{\mathrm{cost}_B(G), \mathrm{cost}_C(G)\}. \end{aligned} \tag{57}$$

Thus the A-first cost attains the minimum in the path identity, giving

$$\begin{aligned} W(G) &\overset{(41),(57)}{=} N + 3 + \mathrm{cost}_A(G) \\ &\overset{(42),(49)}{=} 99m + 3 + 93m - 6\,\mathrm{OPT}(F) \\ &= 192m - 6\,\mathrm{OPT}(F) + 3 \overset{(56)}{\geq} 186m + 3. \end{aligned}$$

□

5.3.3 The approximation gap

Theorem 5.9 (Dense total-depth gap)**.** *For every fixed $1 \leq r < 63/62$, it is NP-hard to approximate total elimination depth within factor r on connected simple unweighted graphs. The reduction starts from a bounded-occurrence parity source, with a fixed error $\epsilon = \epsilon(r) > 0$ chosen in the proof. Its tripartite graph before clique completion satisfies $\Delta(G_F^\Sigma) \leq 14b_\epsilon$.*

Proof. Fix $1 \leq r < 63/62 = 189/186$. Choose a fixed rational

$$0 < \epsilon < \min\left\{\frac{1}{4}, \frac{189 - 186r}{6(1+r)}\right\}.$$

Then

$$\gamma := (189 - 6\epsilon) - r(186 + 6\epsilon) > 0.$$

Apply Theorem 5.1 with this error, obtaining a system with $m_0 \geq 1$ equations. Fix an integer $K > 6(r-1)/\gamma$, and take K disjoint copies with fresh variable names. Their total equation count $m = Km_0$ satisfies

$$3(r-1) < \gamma m/2. \tag{58}$$

This preserves its optimum fraction, the occurrence bound, and (28). Construct $G = G_F^\Sigma$ and put $N = 99m$. Fix an integer $C_r \geq 9900r/\gamma$ and set

$$M = C_r m.$$

Both K and C_r depend only on r, ϵ. Since $m \geq 1$, $N(N+1) = 9801m^2 + 99m \leq 9900m^2$, so these choices give

$$N(N+1) \leq 9900m^2, \qquad M \geq \frac{rN(N+1)}{\gamma m}. \tag{59}$$

For the fixed parameters, $M = \Theta(m)$. Proposition 5.8 and Lemma 5.6 give

$$\begin{aligned} \text{YES:}\quad W(G) &\overset{(44)}{=} 192m - 6\,\mathrm{OPT}(F) + 3 \\ &\overset{\text{Thm. 5.1}}{\leq} (186 + 6\epsilon)m + 3, \\ \text{NO:}\quad W(G) &\overset{(44)}{=} 192m - 6\,\mathrm{OPT}(F) + 3 \\ &\overset{\text{Thm. 5.1}}{\geq} (189 - 6\epsilon)m + 3. \end{aligned} \tag{60}$$

Consequently,

$$\begin{aligned} \text{YES:}\quad \mathrm{totd}(H_M^\Sigma(G)) &\overset{(39),(60)}{\leq} M\big((186 + 6\epsilon)m + 3\big) + N(N+1)/2, \\ \text{NO:}\quad \mathrm{totd}(H_M^\Sigma(G)) &\overset{(39),(60)}{\geq} M\big((189 - 6\epsilon)m + 3\big). \end{aligned}$$

The NO lower bound minus r times the YES upper bound is at least

$$\begin{aligned} & M(\gamma m - 3(r-1)) - rN(N+1)/2 \\ & \overset{(58)}{>} M\gamma m/2 - rN(N+1)/2 \\ & \overset{(59)}{\geq} 0. \end{aligned}$$

The output has $99m+3M = O(m)$ vertices and $O(m^2)$ edges, and is constructed deterministically in polynomial time. Its completed core is connected, because each part is a nonempty clique, auxiliary pairs join B to C, and literal-block edges join A to the other parts. Every added vertex has a neighbor in its corresponding part, so the whole output is connected. An r-approximation would distinguish the source YES and NO cases. □

6 Hardness on graphs of bounded degree

6.1 The degree-reduction approach

Let H be one of the dense graphs constructed above, with $N = |V(H)|$. Its edge set consists of $E_0 \subseteq \binom{V(H)}{2}$, which we keep during sparsification, and clique and biclique factors indexed by finite disjoint sets $\mathcal{C}$ and $\mathcal{B}$. Each clique factor $f \in \mathcal{C}$ has a nonempty support $U_f \subseteq V(H)$ and contributes all pairs in $\binom{U_f}{2}$. Each biclique factor has disjoint nonempty sides $A_f, B_f \subseteq V(H)$, support $U_f = A_f \dot\cup B_f$, and contributes all edges uv with $u \in A_f$ and $v \in B_f$. Thus

$$E(H) = E_0 \cup \bigcup_{f\in\mathcal{C}} \binom{U_f}{2} \cup \bigcup_{f\in\mathcal{B}} \{uv : u \in A_f,\ v \in B_f\}. \tag{61}$$

For each factor choose an integer threshold $t_f \geq 1$. Two disjoint vertex sets are *anticomplete* if no edge joins them. For an integer $t \geq 1$, a (t,t) *hole* is a pair of anticomplete sets of t vertices each. In a bipartite graph with specified sides, the two sets lie on opposite sides. A clique replacement on U_f has no hole at threshold t_f if every two disjoint t_f-subsets of U_f have at least one edge between them. A biclique replacement on $A_f \dot\cup B_f$ has no hole at threshold t_f if every t_f-subset of A_f and every t_f-subset of B_f have at least one edge between them.

Different factors may share vertices or edges. A factor specifies edges to be included, not an induced subgraph, so other factors or E_0 may add further edges on its support.

Replace each factor by a subgraph of its specified edge set on the same support. Let J be the union of these replacement edges with E_0. Then J is a spanning subgraph of H. The construction has two steps.

1. *Preserve the YES bound by deleting edges.* Every elimination forest or tree decomposition of H is also valid for J. Thus treedepth, treewidth, and total elimination depth do not increase.

2. *Recover a dense solution from a sparse one.* Each replacement satisfies its no-hole condition at threshold t_f. For clique factors the subsets are disjoint, and for biclique factors they lie on opposite sides. The completion lemmas show that, after deleting a small set D of exceptional vertices, an arbitrary forest or decomposition of J has every edge of $H - D$ represented by comparable endpoints or by vertices in a common bag. For forests, restore D as a chain above all existing roots. For tree decompositions, add D to every bag. The height or width increases by at most $|D|$, and the total depth by at most $N|D|$.

These steps give a sparse instance and bound the additive loss incurred when recovering a dense solution. The following lemma states when such a loss preserves the approximation gap.

Lemma 6.1 (Preservation of approximation gaps)**.** *Let a minimization promise problem be hard to distinguish between* $\mathrm{OPT}(I) \le Y(N)$ *in the YES case and* $\mathrm{OPT}(I) \ge r'Y(N)$ *in the NO case, where* $r' > 1$, $Y(N) > 0$, *and* N *is the source size. Assume the bounds below admit a polynomial-time computable separating threshold whenever they are strictly separated; in particular this holds for the explicit rational bounds used here. Suppose a deterministic polynomial-time reduction maps each source instance* I *to a target minimization instance* I'*. Assume that for every fixed pair* $\epsilon_{\mathrm{YES}}, \epsilon_{\mathrm{NO}} > 0$ *we can choose the reduction's fixed parameters and a size threshold* N_0 *so that, on promised instances with* $N \ge N_0$*,*

$$\mathrm{OPT}(I') \le (1+\epsilon_{\mathrm{YES}})Y(N) \quad (\textit{YES}), \qquad \mathrm{OPT}(I') \ge (r'-\epsilon_{\mathrm{NO}})Y(N) \quad (\textit{NO}).$$

Also assume that the source promise remains hard after any fixed size threshold. Then every fixed $1 \le r < r'$ *remains hard after the reduction.*

Proof. Choose positive ϵ_{YES} and ϵ_{NO} so that $\epsilon_{\mathrm{NO}} + r\epsilon_{\mathrm{YES}} < r' - r$. Then $r(1+\epsilon_{\mathrm{YES}}) < r' - \epsilon_{\mathrm{NO}}$, so an r-approximation returning a feasible solution has value at most $r(1+\epsilon_{\mathrm{YES}})Y(N)$ in the YES case and at least $(r'-\epsilon_{\mathrm{NO}})Y(N)$ in the NO case. Comparing its value with the supplied computable threshold distinguishes the cases on sufficiently large instances. □

Keep the total loss below the hardness gap. Choose $t_f = \lceil \zeta |U_f| \rceil$, so that $t_f - 1 < \zeta |U_f|$ for each clique or biclique factor. If every vertex lies in at most a fixed number of factor supports, then $\sum_{f \in \mathcal{C} \cup \mathcal{B}} |U_f| = O(N)$. Hence $|D| \le \varepsilon N$ for any prescribed fixed $\varepsilon > 0$, by choosing the fixed positive ζ sufficiently small. The resulting losses are at most εN for treedepth and treewidth, and εN^2 for total depth. Given a target factor r below the dense threshold, choose a dense factor $r' > r$. The YES bound Y has the same linear or quadratic scale as the corresponding loss, so the threshold fractions can make the loss at most δY for any fixed $\delta > 0$. Taking $\delta < (r'-r)/(r+1)$ lets Lemma 6.1 preserve factor r.

For the maximum degree to be bounded, we also require bounded degree in E_0. If a biclique has a side smaller than its threshold, its no-hole condition is vacuous and we may use the empty replacement. Otherwise a threshold proportional to its total support size bounds the ratio of its side sizes. This ratio bounds the number of auxiliary vertices mapped to each prescribed vertex. The expander construction then gives the required no-hole condition with a constant degree contribution from each factor. Since each vertex lies in a bounded number of factor supports, its total degree is bounded. These constants may depend on the desired approximation ratio, but not on N. For treewidth, each side of every biclique must additionally be partitionable into a bounded number of sets, each contained in a clique-factor support. The corresponding clique-factor bags anchor the side sets in the biclique completion lemma. For the UGC applications, the bounded-degree source promise of Theorem 3.1 bounds the degree of the retained source edges E_0.

6.2 Completing sparse factors

The no-hole conditions above are inherited by deleting vertices. Recall that forest restriction preserves the ancestor order between surviving vertices and never increases their depths.

The following lemmas identify a small exceptional set whose deletion allows a sparse factor to be completed. For a clique, the surviving vertices must lie on one ancestor path or in one bag. For a biclique, it is enough that every surviving cross-pair is comparable or appears together in a bag.

6.2.1 Clique factors

Lemma 6.2 (Clique completion in an elimination forest)**.** *Let J be a finite simple graph, let $U \subseteq V(J)$, and let $t \geq 1$ be an integer. Suppose $J[U]$ has no two disjoint anticomplete sets of size t. Every elimination forest F of J has an ancestor path P of F such that $|U \setminus V(P)| \leq 2(t-1)$.*

Proof. If $|U| < t$, then $|U| \leq t-1 \leq 2(t-1)$, so the empty path satisfies the claim. Now assume $|U| \geq t$. Traverse each rooted tree in depth-first order. Introduce a vertex when first reached, traverse all its child subtrees, and forget it after those subtrees have been traversed. Throughout the traversal maintain two subsets of U: L is the set of vertices already forgotten, and R is the set of vertices not yet introduced. Initially $L = \varnothing$ and $R = U$. When a vertex of U is introduced, remove it from R. When it is forgotten, add it to L. The vertices in $U \setminus (L \cup R)$ have been introduced but not forgotten, so they lie on the current ancestor path. We traverse the rooted trees one after another, with this path empty between trees.

At every point, no edge of J joins L to R. Indeed, if $x \in L$ and $y \in R$, then y cannot be an ancestor of x, since ancestors are introduced first. Also, x cannot be an ancestor of y, since the traversal finishes the subtree of x before forgetting x. Thus x and y are incomparable, and the elimination-forest property rules out the edge xy.

Consider the instant immediately after the tth vertex of U is forgotten. Then $|L| = t$. If $|R| \geq t$, then L and R contain disjoint anticomplete subsets of size t, contrary to the hypothesis. Hence $|R| \leq t-1$. Immediately before this forgetting event, $|L| = t-1$ and R is unchanged. Let P be the current ancestor path then. The sets L, R, and $U \setminus (L \cup R)$ partition U, and every vertex in the third set lies on P. Thus

$$U \setminus V(P) \subseteq L \cup R, \qquad |L| = t-1, \qquad |R| \leq t-1. \tag{62}$$

Consequently,

$$|U \setminus V(P)| \overset{(62)}{\leq} |L| + |R| \overset{(62)}{\leq} (t-1) + (t-1) = 2(t-1).$$

□

Lemma 6.3 (Clique completion in a tree decomposition)**.** *Let J be a finite simple graph, let $U \subseteq V(J)$, and let $t \geq 1$ be an integer. Suppose $J[U]$ has no two disjoint anticomplete sets of size t. Every tree decomposition of J has a node u whose bag B_u satisfies $|U \setminus B_u| \leq 3(t-1)$.*

Proof. Let $(T, (B_u)_{u \in V(T)})$ be the supplied decomposition. If $|U| < t$, choose any node u of T. Then $|U \setminus B_u| \leq |U| \leq t-1 \leq 3(t-1)$. Otherwise consider an edge of the decomposition tree. Its deletion leaves two sides. Call a side heavy if it contains T_v for at least t vertices $v \in U$. If both sides were heavy, choose a set $X \subseteq U$ of exactly t vertices with T_v contained in one side, and a set $Y \subseteq U$ of exactly t vertices with T_v contained in the other side. The sets are disjoint, since no nonempty T_v can lie on both sides, so they contain $2t$ vertices in total. For $x \in X$ and $y \in Y$, we have $T_x \cap T_y = \varnothing$, so no bag contains both x and y. Thus X and Y are anticomplete, contrary to the hypothesis. Orient the edge toward its heavy side if one exists, and arbitrarily if neither is heavy. Following outgoing edges cannot revisit a node, since the underlying tree has no cycles. The walk therefore ends at a node with no outgoing edge. Call such a node a *sink*, and let u be one.

For each component C of $T - u$, at most $t-1$ vertices $v \in U$ have $T_v \subseteq V(C)$. Otherwise, the edge from u into C would be oriented away from the sink u. Let $W = U \setminus B_u$. Every vertex $v \in W$ has T_v contained in one component of $T - u$. Write $W_C = \{v \in W : T_v \subseteq V(C)\}$. Thus $|W_C| \leq t-1$ for each component C. If $t = 1$, this gives $W = \varnothing$. For $t \geq 2$, suppose $|W| \geq 3t-2$. Select components one at a time until the sum of their $|W_C|$ first reaches t, and let W_1 be the union

of their W_C sets. The last component added contributes at most $t-1$ vertices, so, together with the assumed lower bound on $|W|$,

$$t \le |W_1| \le 2t-2, \qquad |W| \ge 3t-2. \tag{63}$$

Let $W_2 = W \setminus W_1$. Then $|W_2| \overset{(63)}{\ge} (3t-2)-(2t-2) = t$. For $x \in W_1$ and $y \in W_2$, the sets T_x, T_y lie in different components of $T-u$ and are therefore disjoint, so no bag contains both x and y. Hence W_1, W_2 are anticomplete in J, contrary to the hypothesis. Thus $|W| \le 3t-3$. Since $W = U \setminus B_u$ and $3t-3 = 3(t-1)$, this is the required bound. □

6.2.2 Biclique factors

We use König's theorem [Die17, Theorem 2.1.1]: in a bipartite graph, the maximum matching size equals the minimum vertex-cover size.

Both biclique completion lemmas use the following interval argument.

Lemma 6.4 (Interval completion)**.** *Let J be a finite simple graph, let $A, B \subseteq V(J)$ be disjoint, and let $t \ge 1$ be an integer. Assign each $v \in A \cup B$ a nonempty interval I_v in a common finite linear order. Suppose every edge of J between A and B has intersecting endpoint intervals, and the bipartite graph between A and B has no (t,t) hole. There is a set $D \subseteq A \cup B$ with $|D| \le 4(t-1)$ such that $I_u \cap I_v \ne \varnothing$ for all $u \in A \setminus D$ and $v \in B \setminus D$. If every disjoint cross-pair has its A-interval before its B-interval, the bound is $|D| \le 2(t-1)$.*

Proof. Let K be the bipartite graph on $A \dot\cup B$ whose edges are the pairs with disjoint intervals. Partition its edges according to which endpoint interval comes first. Suppose a matching contains $m_0 \ge 2t-1$ edges $u_i v_i$ with $u_i \in A$, $v_i \in B$, and the u_i-interval first. Order these edges by the last point of I_{u_i}. For $i \le t$ and $j \ge m_0 - t + 1$, we have $i \le j$ and

$$\max I_{u_i} \le \max I_{u_j} < \min I_{v_j}.$$

Thus $\{u_1, \dots, u_t\}$ and $\{v_{m_0-t+1}, \dots, v_{m_0}\}$ are anticomplete t-sets in J: their vertices are distinct because the edges form a matching, and their intervals are disjoint. This contradicts the no-hole condition. Interchanging A, B gives the same bound for the other orientation. Every matching in K therefore has at most $4t-4$ edges, or at most $2t-2$ when only the first orientation occurs. König's theorem gives a vertex cover D of K with the corresponding size bound. No surviving cross-pair is an edge of K, proving the claim. □

This deletion set can be computed in polynomial time. Enumerate the disjoint-interval pairs to construct K, and find a maximum bipartite matching. Starting at the unmatched vertices of A, traverse unmatched edges toward B and matched edges toward A. If Z is the set reached, then $(A \setminus Z) \cup (B \cap Z)$ is a minimum vertex cover [Die17, Section 2.1] and supplies D.

Lemma 6.5 (Biclique completion in an elimination forest)**.** *Let F be an elimination forest of a finite simple graph J, let $A, B \subseteq V(J)$ be disjoint, and let $t \ge 1$ be an integer. Suppose the bipartite graph of J between A and B has no (t,t) hole. There is a set $D \subseteq A \cup B$ such that $|D| \le 4(t-1)$ and each pair in $(A \setminus D) \times (B \setminus D)$ is comparable in F.*

Proof. Traverse the rooted trees consecutively in depth-first order, introducing each vertex when first reached and forgetting it after its child subtrees. Assign distinct times to these events, and let $I_v = [t_{\mathrm{in}}(v), t_{\mathrm{out}}(v)]$ be the interval from the introduction to the forgetting of v. These intervals are nested for comparable vertices and disjoint for incomparable ones. Every edge of J has comparable endpoints, so Lemma 6.4 gives $D \subseteq A \cup B$ with $|D| \le 4(t-1)$ and intersecting intervals for every surviving cross-pair. Such pairs are therefore comparable in F. □

For treewidth, we first complete the clique factors and use their bags to anchor the two sides of each biclique. We use the following elementary path-projection observation. Let S_1, S_2 be connected vertex sets in a tree, with $x \in S_1$ and $y \in S_2$, and let P be the path from x to y. If $z \in S_1 \cap S_2$, let p be the unique vertex of P closest to z. Both the x–z and y–z paths contain p, so connectedness gives $p \in S_1 \cap S_2 \cap V(P)$. The reverse implication is immediate. Thus S_1, S_2 intersect if and only if their intersections with P intersect. Their path traces will be prefixes and suffixes, so the case with a fixed ordering in Lemma 6.4 applies.

When vertices are deleted from a graph, we restrict its tree decomposition by removing them from every bag.

Lemma 6.6 (Biclique completion between two bags)**.** *Let J be a finite simple graph and let $t \geq 1$ be an integer. Fix a tree decomposition $(T, (B_u)_{u \in V(T)})$ of J. Suppose $A, B \subseteq V(J)$ are disjoint, $A \subseteq B_x$ and $B \subseteq B_y$ for some $x, y \in V(T)$, and the bipartite graph of J between A and B has no (t,t) hole. There is a set $D \subseteq A \cup B$ with $|D| \leq 2(t-1)$ such that for every $u \in A \setminus D$ and $v \in B \setminus D$, some $z \in V(T)$ satisfies $\{u, v\} \subseteq B_z \setminus D$.*

Proof. If $x = y$, take $D = \varnothing$. Otherwise let P be the unique path from x to y in T, and list its vertices as $z_0 = x, z_1, \ldots, z_\ell = y$ in their order along the path. Give $V(P)$ the linear order induced by this list. For $u \in A$, we have $x \in T_u$, and T_u is connected. Thus $T_u \cap V(P) = \{z_0, \ldots, z_a\}$ for some $0 \leq a \leq \ell$, the vertex set of a nonempty prefix of P. Similarly, for $v \in B$, $T_v \cap V(P) = \{z_c, \ldots, z_\ell\}$ for some $0 \leq c \leq \ell$, the vertex set of a suffix ending at y. By the observation above, T_u, T_v intersect if and only if these path intervals intersect.

Every edge of J between A and B has intersecting traces and hence intersecting path intervals. A disjoint prefix and suffix always have the prefix first. Lemma 6.4 therefore gives $D \subseteq A \cup B$ with $|D| \leq 2(t-1)$ such that every surviving cross-pair has intersecting path intervals. Choose a node z in this intersection; then $\{u, v\} \subseteq B_z \setminus D$, as required. □

6.2.3 Comparison bounds

Definition 6.7 (Graphs and replacement conditions)**.** *Let H be a finite simple graph whose edges are specified by* (61)*. For each $f \in \mathcal{C} \cup \mathcal{B}$, fix an integer $t_f \geq 1$. Let $J \subseteq H$ be a spanning subgraph with $E_0 \subseteq E(J)$. Assume the following conditions.*

1. *For each $f \in \mathcal{C}$, every two disjoint sets $X, Y \subseteq U_f$ with $|X| = |Y| = t_f$ have an edge of J between them.*
2. *For each $f \in \mathcal{B}$, every $X \subseteq A_f$ and $Y \subseteq B_f$ with $|X| = |Y| = t_f$ have an edge of J between them.*

Each completion lemma deletes a small set so that a solution for J also covers one dense factor. Take the union D of these sets. Further deletions preserve each completion, so restricting the solution gives one for $H - D$. Restoring D above forest roots or in every bag increases height or width by at most $|D|$, and total depth by at most $N|D|$, where $N = |V(H)|$.

For treewidth, we first choose a deletion set $D_{\mathcal{C}}$ for the clique factors, so each $U_f \setminus D_{\mathcal{C}}$ lies in a bag. If every biclique side lies in a clique support, these bags contain $A_f \setminus D_{\mathcal{C}}$ and $B_f \setminus D_{\mathcal{C}}$, as required by the biclique completion lemma. Summing the deletion bounds gives

$$\mathrm{exc}_{\mathrm{td}} = 2\sum_{f \in \mathcal{C}} (t_f - 1) + 4\sum_{f \in \mathcal{B}} (t_f - 1), \qquad \mathrm{exc}_{\mathrm{tw}} = 3\sum_{f \in \mathcal{C}} (t_f - 1) + 2\sum_{f \in \mathcal{B}} (t_f - 1). \tag{64}$$

Lemma 6.8 (Treedepth comparison)**.** *Let H, J and the thresholds t_f satisfy Definition 6.7, and let* $\mathrm{exc}_{\mathrm{td}}$ *be defined by* (64)*. Every elimination forest F of J can be completed to an elimination forest $\widehat{F}$ of H with $h(\widehat{F}) \le h(F) + \mathrm{exc}_{\mathrm{td}}$. In particular,* $\mathrm{td}(J) \le \mathrm{td}(H) \le \mathrm{td}(J) + \mathrm{exc}_{\mathrm{td}}$.

Proof. Fix an elimination forest F of J. For each $f \in \mathcal{C}$, choose the path P_f from Lemma 6.2 and put $D_f = U_f \setminus V(P_f)$, so $|D_f| \le 2(t_f - 1)$. For each $f \in \mathcal{B}$, let $D_f \subseteq A_f \cup B_f$ be the set from Lemma 6.5, with $|D_f| \le 4(t_f - 1)$. Set $D = \bigcup_{f \in \mathcal{C} \cup \mathcal{B}} D_f$. Thus

$$\begin{aligned} |D| &\le \sum_{f \in \mathcal{C} \cup \mathcal{B}} |D_f| \\ &\overset{\text{Lems. 6.2, 6.5}}{\le} 2 \sum_{f \in \mathcal{C}} (t_f - 1) + 4 \sum_{f \in \mathcal{B}} (t_f - 1) \\ &\overset{(64)}{=} \mathrm{exc}_{\mathrm{td}} \,. \end{aligned} \tag{65}$$

Restrict F to a forest F' on $V(H) \setminus D$. The surviving vertices of each clique factor lie on P_f, and the surviving cross-pairs of each biclique are comparable by the choice of D_f. Retained edges are covered because $E_0 \subseteq E(J)$. Restriction preserves these ancestor relations, so (61) makes F' an elimination forest of $H - D$ with $h(F') \le h(F)$. By Lemma 3.8, restoring D gives an elimination forest $\widehat{F}$ of H with

$$\begin{aligned} h(\widehat{F}) &\overset{\text{Lem. 3.8}}{\le} h(F) + |D| \\ &\overset{(65)}{\le} h(F) + \mathrm{exc}_{\mathrm{td}} \,. \end{aligned}$$

Choosing F of minimum height proves $\mathrm{td}(H) \le \mathrm{td}(J) + \mathrm{exc}_{\mathrm{td}}$. The other comparison follows from $J \subseteq H$. □

Lemma 6.9 (Total-depth comparison)**.** *Let H, J and the thresholds t_f satisfy Definition 6.7. Put $N = |V(H)|$, and let* $\mathrm{exc}_{\mathrm{td}}$ *be defined by* (64)*. Every elimination forest F of J can be completed to an elimination forest $\widehat{F}$ of H with* $\mathrm{totd}(\widehat{F}) \le \mathrm{totd}(F) + N \,\mathrm{exc}_{\mathrm{td}}$*. In particular,* $\mathrm{totd}(J) \le \mathrm{totd}(H) \le \mathrm{totd}(J) + N \,\mathrm{exc}_{\mathrm{td}}$.

Proof. Fix an elimination forest F of J, and use the exceptional set D and restricted forest F' from the proof of Lemma 6.8. Put $d = |D| \le \mathrm{exc}_{\mathrm{td}}$. Restriction does not increase total depth, so $\mathrm{totd}(F') \le \mathrm{totd}(F)$. Restoring D above all roots adds d to the depths of the $N - d$ surviving vertices and gives each of the d new vertices depth at most d. Consequently,

$$\mathrm{totd}(\widehat{F}) \overset{\text{Lem. 3.8}}{\le} \mathrm{totd}(F) + d(N - d) + d^2 = \mathrm{totd}(F) + Nd \le \mathrm{totd}(F) + N \,\mathrm{exc}_{\mathrm{td}} \,.$$

Minimizing over F gives the upper comparison, and the lower comparison follows from $J \subseteq H$. □

The following comparison lemma assumes that each side of a biclique lies within one clique-factor support. When a side is covered by several such supports, Theorem 6.19 splits the biclique into sub-bicliques to meet this assumption.

Lemma 6.10 (Treewidth comparison)**.** *Let H, J and the thresholds t_f satisfy Definition 6.7, and let* $\mathrm{exc}_{\mathrm{tw}}$ *be defined by* (64)*. Suppose that for every $f \in \mathcal{B}$ there are $g, h \in \mathcal{C}$ with $A_f \subseteq U_g$ and $B_f \subseteq U_h$. Every tree decomposition of J can be completed to one of H whose width increases by at most* $\mathrm{exc}_{\mathrm{tw}}$*. In particular,* $\mathrm{tw}(J) \le \mathrm{tw}(H) \le \mathrm{tw}(J) + \mathrm{exc}_{\mathrm{tw}}$.

Proof. Fix a tree decomposition $(T, (B_u)_{u\in V(T)})$ of J. For each $f \in \mathcal{C}$, choose u_f by Lemma 6.3 and put $D_f = U_f \setminus B_{u_f}$, so $|D_f| \le 3(t_f - 1)$. Set $D_{\mathcal{C}} = \bigcup_{f\in\mathcal{C}} D_f$ and restrict the bags to $B_u \setminus D_{\mathcal{C}}$. Then $U_f \setminus D_{\mathcal{C}} \subseteq B_{u_f} \setminus D_{\mathcal{C}}$ for each $f \in \mathcal{C}$.

For $f \in \mathcal{B}$, choose $g, h \in \mathcal{C}$ with $A_f \subseteq U_g$ and $B_f \subseteq U_h$. The restricted sides $A_f \setminus D_{\mathcal{C}}$ and $B_f \setminus D_{\mathcal{C}}$ lie in $B_{u_g} \setminus D_{\mathcal{C}}$ and $B_{u_h} \setminus D_{\mathcal{C}}$, respectively. They still satisfy the no-hole condition at threshold t_f, since any t_f-subsets of them were also subsets of the original sides. Lemma 6.6 therefore gives $D_f \subseteq (A_f \cup B_f) \setminus D_{\mathcal{C}}$ with $|D_f| \le 2(t_f - 1)$. Put $D = D_{\mathcal{C}} \cup \bigcup_{f\in\mathcal{B}} D_f$. Then

$$\begin{aligned} |D| &\le \sum_{f\in\mathcal{C}\cup\mathcal{B}} |D_f| \\ &\overset{\text{Lems. 6.3, 6.6}}{\le} 3\sum_{f\in\mathcal{C}}(t_f - 1) + 2\sum_{f\in\mathcal{B}}(t_f - 1) \qquad (66) \\ &\overset{(64)}{=} \mathrm{exc}_{\mathrm{tw}}\,. \end{aligned}$$

In the bags $B_u \setminus D$, each surviving clique factor lies in $B_{u_f} \setminus D$, each surviving biclique edge is covered by the choice of D_f, and retained edges remain covered because $E_0 \subseteq E(J)$. Equation (61) therefore gives a decomposition of $H - D$: every surviving vertex occurs at the same nodes as before, so its occurrences remain nonempty and connected.

Define $\widehat{B}_u = (B_u \setminus D) \cup D$ for every $u \in V(T)$. This retains all edges of $H - D$, covers every edge incident with D, and makes each vertex of D occur throughout T. Thus $(T, (\widehat{B}_u)_{u\in V(T)})$ is a decomposition of H. Since $|\widehat{B}_u| \le |B_u| + |D|$ for every u, its width increases by at most $|D| \overset{(66)}{\le} \mathrm{exc}_{\mathrm{tw}}$. Choosing a minimum-width decomposition proves the upper comparison. The lower comparison follows from $J \subseteq H$. □

All these completions are computable in polynomial time in the graph, factor description, and supplied forest or decomposition encoding. The forest paths are obtained by the stated depth-first sweeps. For the clique bags, count the occurrence subtrees contained on each side of every decomposition-tree edge, orient the edges, and find a sink. Path traversals give the interval endpoints for anchored bicliques; the matching and vertex-cover procedure above gives their deletion sets. Finally take the union of the deletion sets and perform the stated restriction and restoration. The running time includes the length of the supplied decomposition, which may contain redundant bags.

6.3 Sparse replacements at arbitrary sizes

We now construct bounded-degree replacements satisfying the no-hole conditions used above, with $t_f - 1$ a small fixed fraction of the factor size. The expander mixing lemma turns an eigenvalue bound into an edge between two large sets. Fixed-length walk powers lower the required set size, and grouping auxiliary vertices gives the prescribed clique supports and biclique sides. Lemma 6.16 records the resulting bounds.

6.3.1 Edge expansion and adjacency matrices

We begin with edge expansion, which requires a set to send many edges to its complement. For our replacements, we must also force an edge between two prescribed large sets, even when they do not cover the graph. An eigenvalue bound will control these edge counts through the mixing lemma. The definitions and standard spectral facts below follow [HLW06, Section 2].

Let $\mathcal{E}$ be a finite undirected multigraph on $M \ge 1$ vertices, allowing loops and parallel edges. A *step* traverses one edge occurrence in a specified direction. A nonloop edge gives one step in each

direction, and a loop gives one step that stays at its vertex. Parallel edge occurrences give distinct steps. In this subsection, d-regular means that every vertex has d possible outgoing steps, counting loop steps once.

For vertex sets $S, T \subseteq V(\mathcal{E})$, let $e_{\mathcal{E}}(S,T)$ be the number of steps that start in S and end in T, counted with multiplicity. The sets may overlap. If $S = T$, each nonloop internal edge contributes two steps and each loop contributes one. If S, T are disjoint, each edge between them contributes one step from S to T. In particular, $|\delta_{\mathcal{E}}(S)| = e_{\mathcal{E}}(S, V(\mathcal{E}) \setminus S)$.

Definition 6.11 (Edge expanders)**.** *For an integer $d \geq 1$ and a real $c > 0$, a d-regular graph $\mathcal{E}$ is a* (d,c)-edge expander *if*

$$|\delta_{\mathcal{E}}(S)| \geq c|S| \quad \textit{whenever } 0 < |S| \leq M/2.$$

A sequence of graphs whose orders tend to infinity is a bounded-degree expander family *if the same constants d, c work for every graph in the sequence.*

The boundary condition says that separating S from its complement requires deleting at least $c|S|$ edge occurrences. Bounded degree keeps the total number of edges linear in the number of vertices. To bound the edges between two specified sets, we express their number using the adjacency matrix and then bound its eigenvalues.

Now assume that $\mathcal{E}$ is d-regular. Define its *adjacency matrix* $A_{\mathcal{E}}$ by

$$(A_{\mathcal{E}})_{uv} = e_{\mathcal{E}}(\{u\}, \{v\}) \qquad (u, v \in V(\mathcal{E})).$$

Reversing steps gives $(A_{\mathcal{E}})_{uv} = (A_{\mathcal{E}})_{vu}$, and regularity gives $\sum_{v \in V(\mathcal{E})} (A_{\mathcal{E}})_{uv} = d$ for every $u \in V(\mathcal{E})$. Write $\mathbf{1}$ for the all-ones vector and T for transpose. Then

$$A_{\mathcal{E}}^{\mathsf{T}} = A_{\mathcal{E}}, \qquad A_{\mathcal{E}}\mathbf{1} = d\mathbf{1}, \tag{67}$$

A real vector $x = (x_v)_{v \in V(\mathcal{E})}$ assigns a real number to each vertex, and matrix multiplication gives $(A_{\mathcal{E}}x)_u = \sum_{v \in V(\mathcal{E})} (A_{\mathcal{E}})_{uv} x_v$ for $u \in V(\mathcal{E})$. Thus this coordinate is the sum of the values at the endpoints of the steps leaving u.

For $S \subseteq V(\mathcal{E})$, its *indicator vector* $\mathbf{1}_S$ has coordinate one on S and zero elsewhere. For real vectors x, y indexed by $V(\mathcal{E})$, $x^{\mathsf{T}}y = \sum_{v \in V(\mathcal{E})} x_v y_v$. The Euclidean length of x is $\|x\|_2 = (\sum_{v \in V(\mathcal{E})} x_v^2)^{1/2}$. Two vectors are *orthogonal* if $x^{\mathsf{T}}y = 0$. In particular, x is orthogonal to $\mathbf{1}$ if and only if $\sum_{v \in V(\mathcal{E})} x_v = 0$. The step count can now be written as $e_{\mathcal{E}}(S,T) = \mathbf{1}_S^{\mathsf{T}} A_{\mathcal{E}} \mathbf{1}_T$.

A nonzero vector x is an *eigenvector* of $A_{\mathcal{E}}$ with *eigenvalue* μ if $A_{\mathcal{E}}x = \mu x$. The spectral theorem for real symmetric matrices states that there is a basis of eigenvectors that have length one and are pairwise orthogonal, and every vector is a unique linear combination of these basis vectors [HLW06, Section 2]. We may take the first basis vector to be $\mathbf{1}/\sqrt{M}$, with eigenvalue d. All remaining basis vectors then have coordinate sum zero. Moreover, multiplication by $A_{\mathcal{E}}$ preserves zero sums:

$$\mathbf{1}^{\mathsf{T}} A_{\mathcal{E}} x \overset{(67)}{=} (A_{\mathcal{E}}\mathbf{1})^{\mathsf{T}} x \overset{(67)}{=} d\mathbf{1}^{\mathsf{T}} x = 0 \quad \text{if } \mathbf{1}^{\mathsf{T}} x = 0.$$

Subtracting the average coordinate from every coordinate of a vector produces a zero-sum vector. Thus zero-sum vectors describe deviations from a constant assignment. The eigenvalue bound below controls these deviations and will give the error bound in the mixing lemma.

Definition 6.12 (Spectral expanders)**.** *Write $\mathbf{1}^{\perp} = \{x \in \mathbb{R}^M : \mathbf{1}^{\mathsf{T}} x = 0\}$ for the zero-sum subspace. Since $A_{\mathcal{E}}$ maps this subspace into itself, the graph has* nonconstant eigenvalue bound $\lambda \geq 0$ *if every eigenvalue of the restriction of $A_{\mathcal{E}}$ to $\mathbf{1}^{\perp}$ has absolute value at most λ. We call it a* spectral expander *when $\lambda < d$. A sequence of such graphs with orders tending to infinity is a* bounded-degree spectral expander family *if d is fixed and λ/d has a common upper bound below one, independent of M.*

The word nonconstant excludes the chosen all-ones eigenvector, not any further eigenvectors with the same eigenvalue d. The absolute-value bound controls both positive and negative eigenvalues, which will be needed when we take walk powers.

Lemma 6.13 (Spectral edge expansion [HLW06, Theorem 2.4])**.** *Let $\mathcal{E}$ be a d-regular graph on M vertices with nonconstant eigenvalue bound $\lambda < d$. Then*

$$|\delta_{\mathcal{E}}(S)| \geq \frac{d-\lambda}{2}|S| \qquad \text{whenever } 0 < |S| \leq M/2.$$

By Lemma 6.13, such a spectral family is an expander family under the preceding definition.

6.3.2 The mixing lemma and walk powers

Our goal is to show that the edge count between two large sets is positive. If the $d|S|$ steps leaving S had endpoints uniformly distributed over the M vertices, then $d|S||T|/M$ would end in T. The expander mixing lemma bounds the difference between this quantity and the actual step count. We use its standard form, including when S, T overlap.

Lemma 6.14 (Expander mixing lemma [HLW06, Lemma 2.5])**.** *Let $\mathcal{E}$ be a d-regular graph on M vertices with nonconstant eigenvalue bound λ. For all $S, T \subseteq V(\mathcal{E})$,*

$$\left| e_{\mathcal{E}}(S,T) - \frac{d|S||T|}{M} \right| \leq \lambda\sqrt{|S||T|}.$$

To guarantee a step between S and T, it is enough that the main term exceed the error term. For nonempty S, T, the lemma gives

$$e_{\mathcal{E}}(S,T) \overset{\text{Lem. 6.14}}{\geq} \frac{d|S||T|}{M} - \lambda\sqrt{|S||T|} = d\sqrt{|S||T|}\left(\frac{\sqrt{|S||T|}}{M} - \frac{\lambda}{d}\right).$$

Thus $e_{\mathcal{E}}(S,T) > 0$ if $\sqrt{|S||T|}/M > \lambda/d$. In particular, for any integer $t \geq 1$, sets of at least t vertices are joined whenever $t > (\lambda/d)M$. Thus λ/d determines the fraction of vertices above which holes are impossible. A smaller ratio gives a smaller threshold and hence a smaller completion loss.

To reduce this ratio, we use a graph whose steps represent sequences of a fixed number of steps in the original graph. We define this construction next and derive its eigenvalue bound.

For an integer $h \geq 0$, a *walk of length h* consists of vertices $x_0, x_1, \ldots, x_h$ and a chosen step from x_{i-1} to x_i for each $1 \leq i \leq h$. Vertices and edges may be repeated. Different choices among parallel steps give different walks, even if their vertex sequences agree. Reversing the order and directions of its steps gives a walk of the same length in the opposite direction.

For an integer $h \geq 1$, the *hth walk power* of $\mathcal{E}$ has the same vertex set and one step from u to v for each length-h walk from u to v in $\mathcal{E}$. Its adjacency matrix is $A_{\mathcal{E}}^h$: matrix multiplication counts walks by length. This matrix is symmetric, so the walk power is undirected. There are d choices at each step, giving degree d^h. If $A_{\mathcal{E}}x = \mu x$, then $A_{\mathcal{E}}^h x = \mu^h x$, so the nonconstant eigenvalue bound is λ^h [HLW06, Section 2]. The ratio of this bound to the degree is $\lambda^h/d^h = (\lambda/d)^h$. When $\lambda/d < 1$, a fixed sufficiently large h makes this ratio smaller than any prescribed positive constant. The new degree d^h still does not depend on the number of vertices. Thus h counts steps in an auxiliary walk, whereas t counts vertices in the sets that a replacement must join.

We use the following explicit starting family.

Theorem 6.15 (Margulis–Gabber–Galil expanders [HLW06, Construction 8.1 and Theorem 8.2])**.** *For every integer $\nu \geq 1$, put $\mathbb{Z}_\nu = \{0, \ldots, \nu - 1\}$, with arithmetic modulo ν. On vertex set $\mathbb{Z}_\nu^2$, define $\mathcal{G}_\nu$ by giving each vertex (x, y) the eight outgoing steps with endpoints*

$$(x \pm 2y, y), \quad (x \pm (2y+1), y), \quad (x, y \pm 2x), \quad (x, y \pm (2x+1)).$$

The eight choices remain distinct steps when endpoints coincide, including loop steps when the endpoint is (x, y). They define an eight-regular undirected multigraph under our step convention, since the transformations with opposite signs are inverses. Its nonconstant eigenvalues have absolute value at most $5\sqrt{2}$. Put $\lambda_0 = 5\sqrt{2}/8 < 1$. For every integer $h \geq 1$, the hth walk power $\mathcal{G}_\nu^{[h]}$ has degree 8^h and normalized transition operator

$$P_{\nu,h} = 8^{-h} A_{\mathcal{G}_\nu}^h, \qquad \|P_{\nu,h} x\|_2 \leq \lambda_0^h \|x\|_2 \quad (x \in \mathbf{1}^\perp).$$

Thus its normalized operator norm on the zero-sum subspace is at most λ_0^h; when $\nu = 1$ that subspace is zero. The walk-power assertion follows from the preceding calculation. For fixed h, enumerating its $\nu^2 8^h$ steps constructs the multigraph deterministically in time polynomial in ν.

These graphs have square orders ν^2. To obtain replacements on arbitrary prescribed sets, we group auxiliary vertices and map each group to one prescribed vertex, using separate maps for the two biclique sides. The resulting lemma is proved in Appendix B.5.

Lemma 6.16 (Mapping to prescribed vertex sets)**.** *For every fixed $\zeta > 0$ and $\kappa \geq 1$, there is a constant $d_{\zeta,\kappa}$ such that the following connected simple graphs, each of maximum degree at most $d_{\zeta,\kappa}$, can be constructed deterministically in polynomial time.*

1. *For any specified nonempty finite set U, a graph on U in which every two disjoint sets $X, Y \subseteq U$ with $|X|, |Y| \geq \lceil \zeta |U| \rceil$ have an edge between them.*
2. *For any specified disjoint nonempty finite sets A, B with $\max\{|A|, |B|\} / \min\{|A|, |B|\} \leq \kappa$, a bipartite graph on $A \dot\cup B$ with parts A, B in which every two sets $X \subseteq A$, $Y \subseteq B$ with $|X|, |Y| \geq \lceil \zeta(|A| + |B|) \rceil$ have an edge between them.*

6.4 General sparsification theorems

We choose thresholds proportional to factor sizes. Bounding the number of factors containing each vertex bounds both the total completion loss and the degree after replacement, provided the retained edges also have bounded degree.

Use the edge decomposition in (61) and the replacement graph J from Section 6. The input supplies E_0, the factor index sets, and each factor's support or sides. Put $N = |V(H)|$. Fix constants $d_0 \geq 0$, $d_1 \geq 1$, and assume:

1. $\Delta((V(H), E_0)) \leq d_0$.
2. $|\{f \in \mathcal{C} \cup \mathcal{B} : v \in U_f\}| \leq d_1$ for every $v \in V(H)$.

Counting incidences between vertices and factor supports gives

$$\sum_{f \in \mathcal{C} \cup \mathcal{B}} |U_f| \overset{\text{Hyp. 2}}{\leq} d_1 N. \tag{68}$$

For a fixed real $\zeta > 0$, the integer $t_f = \lceil \zeta |U_f| \rceil \geq 1$ is the number of vertices in each set that the replacement of factor f must join. Write R_f for that replacement graph, and let $\mathcal{R} = (R_f)_{f \in \mathcal{C} \cup \mathcal{B}}$ be the collection of all replacements. The construction proceeds as follows:

$$H \xrightarrow{\text{Lemma 6.17}} (E_0, \mathcal{R}) \xrightarrow{\text{Lemma 6.17}} J.$$

Each R_f is built on its prescribed support; the output keeps E_0 and takes their union:

$$V(J) = V(H), \qquad E(J) = E_0 \cup \bigcup_{f \in \mathcal{C} \cup \mathcal{B}} E(R_f).$$

The following lemma bounds the degree of this union and the sum of the threshold losses.

Lemma 6.17 (Common sparse factor replacements)**.** *Let H have the edge description* (61)*, and let fixed constants $d_0 \geq 0$, $d_1 \geq 1$ satisfy conditions 1–2. Fix $\zeta > 0$ and set $t_f = \lceil \zeta |U_f| \rceil$ for every factor. There is a deterministic polynomial-time construction of a spanning subgraph $J \subseteq H$ with $E_0 \subseteq E(J)$ that replaces each factor by a graph satisfying its no-hole condition at threshold t_f, and has maximum degree at most $d_0 + d_1 c_\zeta$, where c_ζ depends only on ζ. Moreover, $\sum_{f \in \mathcal{C} \cup \mathcal{B}} (t_f - 1) < \zeta d_1 N$ whenever at least one factor is present.*

Proof. Choose a fixed rational q with $0 < q \leq \zeta$; take $q = \zeta$ when ζ is rational. Set $\kappa_q = \max\{1, 1/q - 1\}$ and $c_\zeta = d_{q,\kappa_q}$ from Lemma 6.16. Use that lemma for each clique factor. For a biclique with side sizes a, b, put $u = \lceil q(a+b) \rceil$. If $\min\{a, b\} < u$, use the empty graph on its support: no admissible pair of u-sets exists. Otherwise $\min\{a, b\} \geq u \geq q(a + b)$, so

$$\frac{\max\{a, b\}}{\min\{a, b\}} = \frac{a + b}{\min\{a, b\}} - 1 \leq \frac{1}{q} - 1 \leq \kappa_q.$$

Lemma 6.16 therefore applies to this factor. Each replacement has degree at most c_ζ and no hole at $\lceil q|U_f| \rceil \leq t_f$; every pair of admissible t_f-sets contains a pair of these smaller sets. This proves the desired condition for every fixed real $\zeta > 0$, while the algorithm uses only the hardcoded rational q. The same dichotomy covers all q: if $q > 1/2$ the nonempty branch is impossible, and if $q = 1/2$ it requires $a = b$.

Take the union of the replacements with E_0. The replacements of vacuous biclique factors need not be connected. The completion lemmas require only the no-hole conditions; when connected factor replacements are needed separately, use Lemma 6.16 with its side-ratio assumption. For every $v \in V(H)$, the degree in the union satisfies

$$\begin{aligned} \deg_J(v) &\overset{\text{Lem. 6.16}}{\leq} \deg_{(V(H), E_0)}(v) + |\{f \in \mathcal{C} \cup \mathcal{B} : v \in U_f\}| c_\zeta \\ &\overset{\text{Hyps. 1, 2}}{\leq} d_0 + d_1 c_\zeta. \end{aligned}$$

Finally, the ceiling defining t_f gives

$$t_f - 1 < \zeta |U_f| \qquad (f \in \mathcal{C} \cup \mathcal{B}). \tag{69}$$

If at least one factor is present, then

$$\begin{aligned} \sum_{f \in \mathcal{C} \cup \mathcal{B}} (t_f - 1) &\overset{(69)}{<} \zeta \sum_{f \in \mathcal{C} \cup \mathcal{B}} |U_f| \\ &\overset{(68)}{\leq} \zeta d_1 N. \end{aligned}$$

Each factor has nonempty support, so there are at most $d_1 N$ factors. For a fixed maximum walk length h, the construction enumerates $O(8^h \sum_{f\in\mathcal{C}\cup\mathcal{B}} |U_f|) = O(8^h d_1 N)$ auxiliary steps. Sorting the resulting endpoint pairs and merging repetitions uses $O(8^h d_1 N \log(2+8^h d_1 N))$ comparisons. Vertex labels have $O(\log(2+N))$ bits, so this is a deterministic polynomial-time construction for the fixed parameters. □

Theorem 6.18 (Treedepth sparsification)**.** *Let H have the edge description* (61)*, and let fixed constants $d_0 \geq 0$, $d_1 \geq 1$ satisfy conditions 1–2. For every fixed $\varepsilon > 0$ one can construct deterministically in polynomial time a spanning subgraph $J \subseteq H$ with $E_0 \subseteq E(J)$ whose maximum degree depends only on d_0, d_1, ε and such that $\mathrm{td}(H) - \varepsilon N \leq \mathrm{td}(J) \leq \mathrm{td}(H)$. Every elimination forest of J can be transformed into one of H whose height increases by at most εN.*

Proof. Apply Lemma 6.17 with $\zeta = \varepsilon/(4d_1)$. Its degree bound depends only on the stated constants. The exception bound in Lemma 6.8 satisfies

$$\begin{aligned} \mathrm{exc}_{\mathrm{td}} &\overset{(64)}{=} 2\sum_{f\in\mathcal{C}}(t_f-1) + 4\sum_{f\in\mathcal{B}}(t_f-1) \\ &\leq 4\sum_{f\in\mathcal{C}\cup\mathcal{B}}(t_f-1) \overset{\text{Lem. 6.17}}{\leq} 4\zeta d_1 N = \varepsilon N. \end{aligned} \tag{70}$$

Lemma 6.8 now gives both conclusions. □

Theorem 6.19 (Treewidth sparsification)**.** *Let H have the edge description* (61)*, and let fixed constants $d_0 \geq 0$, $d_1 \geq 1$ satisfy conditions 1–2. Fix an integer $s \geq 1$. Suppose the supplied factor description also includes, for every $f \in \mathcal{B}$, integers $1 \leq \ell_f, m_f \leq s$ and partitions $A_f = A_{f,1} \dot\cup \cdots \dot\cup A_{f,\ell_f}$ and $B_f = B_{f,1} \dot\cup \cdots \dot\cup B_{f,m_f}$ into nonempty sets, together with indices $g_{f,i}, h_{f,j} \in \mathcal{C}$ such that $A_{f,i} \subseteq U_{g_{f,i}}$ and $B_{f,j} \subseteq U_{h_{f,j}}$ for every $i \in [\ell_f]$ and $j \in [m_f]$. For every fixed $\varepsilon > 0$, one can construct deterministically in polynomial time a spanning subgraph $J \subseteq H$ with $E_0 \subseteq E(J)$ whose maximum degree depends only on d_0, d_1, s, ε and such that $\mathrm{tw}(H) - \varepsilon N \leq \mathrm{tw}(J) \leq \mathrm{tw}(H)$. Every tree decomposition of J can be transformed into one of H whose width increases by at most εN.*

Proof. Apply Lemma 6.17 to the original factors with $\zeta = \varepsilon/(d_1 \max\{3, 2s^2\})$. It constructs J with maximum degree depending only on the stated constants and thresholds $t_f = \lceil \zeta |U_f| \rceil$.

The supplied partitions satisfy

$$\ell_f m_f \leq s^2 \qquad (f \in \mathcal{B}). \tag{71}$$

Using their supplied indices, their product pieces form the disjoint union

$$A_f \times B_f = \dot\bigcup_{i\in[\ell_f],\, j\in[m_f]} (A_{f,i} \times B_{f,j}).$$

Describe the same graph H by keeping E_0 and its clique factors and replacing the biclique indexed by f with the $\ell_f m_f$ bicliques indexed by (f,i,j), on sides $A_{f,i}, B_{f,j}$. Give each of these bicliques threshold t_f. Each has its sides contained in the clique supports $U_{g_{f,i}}$ and $U_{h_{f,j}}$.

The same graph J satisfies this refined factor description. The clique conditions hold by Lemma 6.17. Any t_f-subsets of $A_{f,i}$ and $B_{f,j}$ are also subsets of A_f and B_f, so the original replacement joins them. If either new side has fewer than t_f vertices, this condition is vacuous.

Using the smaller-side alternative in each refined biclique, the completion in Lemma 6.10 increases the decomposition width by at most

$$3\sum_{f\in\mathcal{C}}(t_f-1)+\sum_{f\in\mathcal{B}}\sum_{i\in[\ell_f]}\sum_{j\in[m_f]}\min\{2(t_f-1),|A_{f,i}|,|B_{f,j}|\}.$$

Bounding every minimum by $2(t_f-1)$ gives the uniform estimate

$$\begin{aligned}
&3\sum_{f\in\mathcal{C}}(t_f-1)+2\sum_{f\in\mathcal{B}}\ell_f m_f(t_f-1)\\
&\overset{(71)}{\le} 3\sum_{f\in\mathcal{C}}(t_f-1)+2s^2\sum_{f\in\mathcal{B}}(t_f-1)\\
&\le \max\{3,2s^2\}\sum_{f\in\mathcal{C}\cup\mathcal{B}}(t_f-1)\\
&\overset{\text{Lem. 6.17}}{\le} \max\{3,2s^2\}\zeta d_1 N=\varepsilon N.
\end{aligned}$$

The comparison lemma also gives the stated parameter inequalities. □

Theorem 6.20 (Total-depth sparsification)**.** *Let H have the edge description* (61)*, and let fixed constants $d_0\ge 0$, $d_1\ge 1$ satisfy conditions 1–2. For every fixed $\varepsilon>0$ one can construct deterministically in polynomial time a spanning subgraph $J\subseteq H$ with $E_0\subseteq E(J)$ whose maximum degree depends only on d_0,d_1,ε and such that $\mathrm{totd}(H)-\varepsilon N^2\le\mathrm{totd}(J)\le\mathrm{totd}(H)$. Every elimination forest F of J can be transformed into an elimination forest $\widehat{F}$ of H satisfying $\mathrm{totd}(\widehat{F})\le\mathrm{totd}(F)+\varepsilon N^2$.*

Proof. Use the same factor replacements as in Theorem 6.18, with $\zeta=\varepsilon/(4d_1)$. Equation (70) gives $\mathrm{exc}_{\mathrm{td}}\le\varepsilon N$, and Lemma 6.9 gives the completion and both parameter bounds, with total-depth increase at most $N\,\mathrm{exc}_{\mathrm{td}}\overset{(70)}{\le}\varepsilon N^2$. □

If a construction has a fixed number d of designated vertices that are not needed for sparsification, deleting them lowers treedepth and treewidth by at most d. Restoring them above all existing roots increases total depth by at most dN, where N is the original graph order. These costs are $O(1)$ for treedepth and treewidth, whose promise bounds are linear in N, and $O(N)$ for total depth, whose bounds are quadratic. They are therefore $o(Y)$, so Lemma 6.1 absorbs them on sufficiently large instances. We use this to omit the three vertices z_A,z_B,z_C from the NP treedepth construction.

6.5 Hardness consequences

Fix a target factor r below a dense hardness threshold, and choose a dense factor $r'>r$. Write Y for its YES upper bound. It is linear in N for treedepth and treewidth and quadratic for total depth. The sparsification losses τN and τN^2 can therefore be made at most δY by choosing τ small. Taking $\delta<(r'-r)/(r+1)$, Lemma 6.1 transfers the factor-r hardness. We verify the factor conditions below, using the following observation for the SSE construction. In these applications we choose connected replacements using Lemma 6.16. The biclique side-size ratios, when bicliques occur, are bounded as checked below. Although the general sparsification theorems do not require balance, these connected replacements preserve the connected components of the dense graph: each replaced factor still connects its entire support.

The transfer steps are summarized below. Here H is a dense graph, H' its sparse replacement, R its regular extension, and R_ℓ replaces each edge of R by a path of fixed even length ℓ. The first

row applies to all three objectives. The second applies to treedepth and treewidth, and to total depth under SSE.

$$H \xrightarrow{\text{Theorems 6.18–6.20}} H' \xrightarrow{\text{Proposition 6.27}} R,$$

$$R \xrightarrow{\text{Corollary 6.29}} R_\ell.$$

Lemma 6.21 (Clique factors of a line graph)**.** *Let G be an unweighted host multigraph. Identify each edge occurrence $e \in E(G)$ with its vertex in $\mathcal{L}(G)$, and put $U_v = \{e \in E(G) : v \text{ is an endpoint of } e\}$ for $v \in V(G)$. The nonempty sets U_v give clique factors for $\mathcal{L}(G)$, and each line-graph vertex belongs to at most two factors.*

Proof. Two line-graph vertices are adjacent if and only if their host edges share an endpoint, so $E(\mathcal{L}(G)) = \bigcup_{v \in V(G)} \binom{U_v}{2}$. An edge has at most two endpoints. A loop contributes one vertex to the factor at its endpoint. □

We may require arbitrarily large sources: disjoint copies preserve the satisfiable fraction of a parity system and the independence fraction and maximum degree of an independent-set source. A relative independent-set approximation on the copies transfers to the original by choosing the copy with the largest selected set. For SSE, parallel copies of every host edge preserve normalized volumes and normalized cut sizes and multiply the line-graph order.

In all SSE cases, use Lemma 4.1 with source parameter η and $q = \lceil 2/\eta \rceil$, as in Section 4.3. Write $N = |E(G^*)|$, counting multiplicity, and $\beta = (c/2)\sqrt{\eta} \le 1/4$. The line graph has YES height bound $4\eta N$ and NO balanced-separator bound βN, the latter by Lemma 4.5 with $\rho = 1/2$. Lemma 6.21 gives $d_1 = 2$, $d_0 = 0$, and no biclique factors, regardless of the host degree. Sparsification replaces these host-incidence cliques directly; the sparse output need not itself be a line graph. Since β/η grows without bound as η decreases, any fixed target gap can be obtained before sparsification.

Corollary 6.22 (Bounded-degree treedepth hardness)**.** *For each fixed $r \ge 1$, approximating treedepth within factor r on simple unweighted graphs of bounded maximum degree is NP-hard if $r < 18/17$, Unique Games-hard if $r < 4/3$, and SSE-hard for every fixed r. The degree bound depends only on the excluded factor. All reductions are deterministic.*

Proof. Apply Theorem 6.18 and Lemma 6.1 in each case.

NP-hardness. Retain every edge of G_F, whose maximum degree is at most $3b_\epsilon$ by Lemma 5.3. After omitting z_A, z_B, z_C, replace the three anchor cliques and the three anchor–part bicliques. Each vertex lies in at most two factor supports, and each biclique has both sides of size $3m$. Thus $d_1 = 2$, $d_0 = 3b_\epsilon$, and $\kappa = 1$. Deleting z_A, z_B, z_C lowers treedepth by at most 3, by Lemma 3.8. The resulting graph has order $18m$, so sparsification lowers treedepth by at most $18\tau m$. The resulting YES bound is $(17/2+\epsilon)m + 1$, and the NO lower bound is $(9-\epsilon)m - 2 - 18\tau m$. Fix a target factor $r < 18/17$ and let $\gamma = 9 - \epsilon - r(17/2+\epsilon) > 0$, using the choice of ϵ in Section 5.1.5. Choose

$$\tau < \gamma/36, \qquad m > 2(2+r)/\gamma, \tag{72}$$

taking enough source copies for the second condition. Then

$$(9-\epsilon)m - 2 - 18\tau m - r\big((17/2+\epsilon)m + 1\big) \overset{(72)}{\ge} \frac{\gamma m}{2} - (2+r) \overset{(72)}{>} 0,$$

so the gap survives both losses.

UGC hardness. Fix a target factor $r < 4/3$, and choose r' with $r < r' < 4/3$. Take q, η from the dense gap calculation in Section 3.2.3 for factor r', and apply Theorem 3.1 with this η in $S_q(G)$. Replace the q layer cliques by bounded-degree expanders and retain all cross-layer source edges. These retained edges have maximum degree at most $(q-1)d_\eta$, so $d_1 = 1$ and $d_0 = (q-1)d_\eta$. Write Y for the dense YES upper bound. A layer clique gives

$$n \le \mathrm{td}(S_q(G)) \le Y \qquad \text{in the YES case.} \tag{73}$$

The sparsification loss is therefore at most $\tau q n \overset{(73)}{\le} \tau q Y$. Choose $\tau < (r'-r)/(q(r+1))$. Lemma 6.1 then transfers the dense factor-r' gap to factor r.

SSE hardness. Use the common line-graph factors above. Lemma 4.6 converts the separator bound into NO treedepth bound βN. For a target factor r, choose η small enough that $\gamma := \beta - 4r\eta > 0$, which is possible since $\beta = (c/2)\sqrt{\eta}$. Choose

$$\tau < \gamma. \tag{74}$$

The sparse NO bound exceeds r times the YES bound by at least $(\gamma - \tau)N \overset{(74)}{>} 0$. □

The NP-hardness assertion answers the bounded-degree conjecture of Schirrmacher, Siebertz, and Vigny [SSV25, Conjecture 1]. Weighted treedepth hardness at maximum degree ten is proved in [DSSV25].

Corollary 6.23 (Bounded-degree treewidth hardness)**.** *For each fixed $r \ge 1$, approximating treewidth within factor r on simple unweighted graphs of bounded maximum degree is NP-hard if $r < 12/11$, Unique Games-hard if $r < 4/3$, and SSE-hard for every fixed r. The degree bound depends only on the excluded factor. All reductions are deterministic.*

Proof. Apply Theorem 6.19 and Lemma 6.1.

NP-hardness. In $K_R(G_F)$, replace the three global cliques and retain every edge of each source copy. Lemma 5.3 bounds the retained degree by $d_0 = 3b_\epsilon$. Each vertex lies in one clique factor, so $d_1 = 1$. There are no biclique factors. The clique replacements use $\kappa = 1$ in Lemma 6.16. Fix a target factor $r < 12/11$. The dense gap calculation in Section 5.2 chooses ϵ, R so that

$$\gamma = R(6-\epsilon) - r\big(R(11/2+\epsilon)+3\big) > 0. \tag{75}$$

The dense YES and NO bounds are respectively $R(11/2+\epsilon)m + 3m - 1$ and $R(6-\epsilon)m - 1$, using Lemma 3.17. Since the output order is $9Rm$, sparsification preserves the YES upper bound and lowers the NO bound by at most $9\tau Rm$. Choose

$$\tau < \gamma/(18R). \tag{76}$$

The sparse NO lower bound minus r times the sparse YES upper bound is

$$(\gamma - 9\tau R)m + (r-1) \overset{(76)}{>} \frac{\gamma m}{2} + r - 1 \overset{(75)}{>} 0.$$

UGC hardness. Fix a target factor $r < 4/3$, and choose r' with $r < r' < 4/3$. Take ζ, R from the dense gap calculation in Section 3.4.2 for factor r', and use Theorem 3.1 with $\eta = \zeta/2$, as in Theorem 3.19. This is admissible since $0 < \eta < 1/24 < 1/10$. Apply it in $K_R(P_3(G))$. Replace only

the three global cliques by bounded-degree expanders and retain every product edge in each source copy. These source edges already have bounded degree, so $d_1 = 1$ and $d_0 = 2d_\eta = D_{3,\zeta}$, with no biclique factors. Lemma 3.18 gives YES cover bound $(3/2 + 3\zeta/2)n \le (3/2 + 3\zeta)n$ and NO cover bound $(2 - \zeta)n \ge (2 - 3\zeta)n$ for $P_3(G)$, as required by Section 3.4.2. The dense graph has global cliques of size Rn, so its treewidth is at least $Rn - 1$. Write Y for the dense YES upper bound, so

$$Y \ge Rn - 1. \tag{77}$$

Since the graph has order $3Rn$, the sparsification loss, when $Rn \ge 2$, is at most

$$3\tau Rn \le 6\tau(Rn - 1) \overset{(77)}{\le} 6\tau Y.$$

Choose $\tau < (r' - r)/(6(r + 1))$. Lemma 6.1 then transfers the dense factor-r' gap to factor r.

SSE hardness. Use the common line-graph factors. Lemma 4.4 gives $b(H) \le \mathrm{tw}(H) + 1$ for the dense line graph. Thus the sparse NO width is at least $\beta N - 1 - \tau N$. The YES line-graph construction has treedepth at most $4\eta N$. An elimination forest of height h gives a tree decomposition of width at most $h - 1$: connect its roots and use its ancestor paths as bags. A fixed vertex occurs on its descendant subtree, and every edge is covered at its deeper endpoint. Hence the dense YES treewidth is at most $4\eta N$. By Lemma 2.1, deleting edges to obtain the sparse graph cannot increase this bound. Fix η so that $\gamma := \beta - 4r\eta > 0$, which is possible because $\beta = (c/2)\sqrt{\eta}$. Choose

$$\tau < \gamma/2, \qquad N > 2/\gamma, \tag{78}$$

taking sufficiently many parallel copies for the second condition. Then

$$\beta N - 1 - \tau N \overset{(78)}{>} (\beta - \gamma)N = 4r\eta N,$$

so the sparse NO width exceeds r times the YES bound. □

The SSE treewidth hardness at every fixed factor on unrestricted graphs is due to Wu, Austrin, Pitassi, and Liu [WAPL15, Theorem 1.1]. The bounded-degree statement here follows by applying our treewidth sparsification theorem to the line-graph construction.

For the total-depth transfer below, we use the following bound for lifting an approximate solution through a reduction.

Lemma 6.24 (Lifting approximate solutions)**.** *Suppose each solution of a minimization instance I' with cost z can be converted in polynomial time into a solution of I with cost at most $z/a + \eta L(I)$, where $a > 0$, $\eta \ge 0$, and*

$$0 \le L(I) \le \mathrm{OPT}(I). \tag{79}$$

If

$$\mathrm{OPT}(I') \le a\,\mathrm{OPT}(I) + \eta a L(I), \tag{80}$$

then an r-approximation for I' gives an $(r + (r + 1)\eta)$-approximation for I. If instead

$$\mathrm{OPT}(I') \le a\,\mathrm{OPT}(I), \tag{81}$$

the factor is $r + \eta$.

Proof. An r-approximate solution has $z \le r\,\mathrm{OPT}(I')$, so its conversion has cost at most

$$\begin{aligned}\frac{z}{a} + \eta L(I) &\le \frac{r\,\mathrm{OPT}(I')}{a} + \eta L(I) \\ &\overset{(80)}{\le} r\,\mathrm{OPT}(I) + (r+1)\eta L(I) \\ &\overset{(79)}{\le} \big(r + (r+1)\eta\big)\,\mathrm{OPT}(I).\end{aligned}$$

Under (81), the conversion cost satisfies

$$\frac{z}{a} + \eta L(I) \overset{(81)}{\le} r\,\mathrm{OPT}(I) + \eta L(I) \overset{(79)}{\le} (r+\eta)\,\mathrm{OPT}(I).$$

□

Corollary 6.25 (Bounded-degree total-depth hardness)**.** *Approximating total depth within every fixed factor $1 \le r < 63/62$ is NP-hard on simple unweighted graphs of bounded maximum degree. On such graphs, total depth is also Unique Games-hard to approximate within every fixed factor $1 \le r < 8/7$, and SSE-hard to approximate within every fixed factor $r \ge 1$. The degree bound depends only on the excluded factor. All reductions are deterministic.*

Proof. Apply Theorem 6.20. It loses at most τN^2 and completes sparse forests at the same additive cost.

NP-hardness. Fix $1 \le r < r' < 63/62$. Use the dense family $H = H_M^{\Sigma}(G_F^{\Sigma})$ from Theorem 5.9 for factor r', where F is a parity system with m equations and occurrence bound b_ϵ from Theorem 5.1. The three part cliques and the three bicliques $X \times T_X$ give $d_1 = 2$. Retain every edge of G_F^{Σ}; its maximum degree is at most $14b_\epsilon$ by (43), so $d_0 = 14b_\epsilon$. The part sizes are $27m, 36m, 36m$ by (42), while each T_X has size $M = \Theta(m)$. Thus all biclique side-size ratios are bounded by a constant depending only on r'.

Write $N = 99m + 3M$ for the output order and Y for the dense YES upper bound. Proposition 5.8 and (39) give

$$\mathrm{totd}(H) \overset{(39),\,(44)}{\ge} M(186m+3) \ge Mm. \tag{82}$$

Since $M = \Theta(m)$ and $N = \Theta(m)$, fix $c_* > 0$ such that

$$Mm \ge c_* N^2, \qquad Y \ge Mm. \tag{83}$$

The second inequality follows because Y is an upper bound on the dense YES optimum. Choose

$$0 < \tau < c_* \min\{r' - r, 1/2\}. \tag{84}$$

Sparsification preserves the YES upper bound Y and reduces the NO lower bound by at most

$$\begin{aligned}\tau N^2 &\overset{(83)}{\le} (\tau/c_*)Mm \\ &\overset{(84)}{<} (r'-r)Mm \overset{(83)}{\le} (r'-r)Y.\end{aligned}$$

Hence it preserves the factor-r gap. Moreover, writing H' for the sparse graph gives

$$\begin{aligned}\mathrm{totd}(H') &\overset{\text{Thm. 6.20; (82)}}{\ge} Mm - \tau N^2 \\ &\overset{(83),\,(84)}{\ge} Mm/2.\end{aligned} \tag{85}$$

This is $\Omega(N^2)$ by (83), as needed for the regular-graph transfer below.

UGC hardness. Fix a target factor $r < 8/7$, and choose r' with $r < r' < 8/7$. Use the parameters q, η from the dense gap calculation in Section 3.3.4 for factor r', starting from Theorem 3.1. Replace the q layer cliques by bounded-degree expanders and retain every cross-layer source edge. Thus $d_1 = 1$ and $d_0 = (q-1)d_\eta$. Each layer is a clique, so its n vertices lie on one ancestor path and have depths at least $1, 2, \ldots, n$. Summing over all q layers gives the lower bound

$$\begin{aligned} L &:= q\,\frac{n(n+1)}{2} \le \mathrm{totd}(S_q(G)), \\ L &\ge \frac{qn^2}{2} = \frac{N^2}{2q}, \quad N = qn. \end{aligned} \tag{86}$$

Write Y for the dense YES upper bound. In the YES case,

$$L \overset{(86)}{\le} \mathrm{totd}(S_q(G)) \le Y. \tag{87}$$

Thus the sparsification loss satisfies

$$\tau N^2 \overset{(86)}{\le} 2q\tau L \overset{(87)}{\le} 2q\tau Y.$$

Choose $\tau < (r' - r)/(2q(r+1))$. Lemma 6.1 then transfers the dense factor-r' gap to factor r.

SSE hardness. The YES line-graph construction has height at most $4\eta N$ and has N vertices, so its total depth is at most $N \cdot 4\eta N = 4\eta N^2$. Lemma 4.6 gives NO total depth at least $\beta N^2/2$. For a target factor r, choose η small enough that $\gamma := \beta/2 - 4r\eta > 0$, which is possible since $\beta = (c/2)\sqrt{\eta}$. Choose

$$\tau < \gamma. \tag{88}$$

After sparsification, the NO lower bound minus r times the YES upper bound is $(\gamma - \tau)N^2 \overset{(88)}{>} 0$. □

6.6 Regular graphs and subdivisions

Both constructions below recover a forest by contracting connected vertex sets. The following observation records the common argument.

Lemma 6.26 (Projection through connected parts)**.** *Let G, H be finite simple graphs, and let $(X_v)_{v \in V(G)}$ partition $V(H)$ into nonempty connected sets. Suppose every edge $uv \in E(G)$ has an edge of H between X_u and X_v. From any elimination forest F of H, one can compute in polynomial time an elimination forest $\overline{F}$ of G such that*

$$\mathrm{depth}_{\overline{F}}(v) \le \mathrm{depth}_F(x) \quad (v \in V(G),\ x \in X_v).$$

Consequently $h(\overline{F}) \le h(F)$ and $\mathrm{totd}(F) \ge \sum_{v \in V(G)} |X_v| \,\mathrm{depth}_{\overline{F}}(v)$.

Proof. For each v, choose the shallowest vertex r_v of X_v. Lemma 2.2 makes it unique and an ancestor of every vertex in X_v. If xy witnesses an edge $uv \in E(G)$, with $x \in X_u$ and $y \in X_v$, then x, y are comparable in F. Both r_u, r_v are ancestors of their deeper endpoint, so they too are comparable. Restrict F to the representatives and rename r_v as v; this gives $\overline{F}$. Restriction and the ancestor property give, for every $x \in X_v$,

$$\mathrm{depth}_{\overline{F}}(v) \le \mathrm{depth}_F(r_v) \overset{\text{Lem. 2.2}}{\le} \mathrm{depth}_F(x).$$

Taking the maximum and summing over the partition prove the two consequences. Computing depths, choosing representatives, and restricting the forest are polynomial-time operations. No connectedness of G or H is required; for empty graphs the empty partition and forest suffice. □

Proposition 6.27. *For every finite simple graph G of order n and fixed odd $d \geq 3$ with $d > \Delta(G)$, one can construct in polynomial time a simple d-regular graph R containing G as an induced subgraph, with $(d+2)n \leq |V(R)| \leq (d+3)n$ and*

$$\begin{aligned} \mathrm{td}(G) \leq \mathrm{td}(R) &\leq \mathrm{td}(G) + d + 2, \\ \mathrm{tw}(G) \leq \mathrm{tw}(R) &\leq \max\{\mathrm{tw}(G), d+2\}, \\ (d+2)\,\mathrm{totd}(G) \leq \mathrm{totd}(R) &\leq (d+3)\,\mathrm{totd}(G) + \tfrac{1}{2}(d+2)(d+3)n. \end{aligned}$$

Proof. For each $v \in V(G)$, set $t = d - \deg_G(v)$, so $1 \leq t \leq d$. Take graphs Q_v whose vertex sets are pairwise disjoint and disjoint from $V(G)$. Choose $P_v \subseteq V(Q_v)$ with $|P_v| = t$, and join v to every vertex of P_v. Define Q_v as follows.

1. If t is even, take K_{d+1} and remove a perfect matching on P_v.
2. If $t \geq 3$ is odd, take K_{d+2} and remove a cycle with vertex set P_v and a perfect matching on $V(Q_v) \setminus P_v$. The latter set has even size $d + 2 - t$.
3. If $t = 1$, write $P_v = \{p\}$, take K_{d+2}, and choose distinct $a, b \in V(Q_v) \setminus \{p\}$. Remove pa, pb and a perfect matching on $V(Q_v) \setminus \{p, a, b\}$.

Define the graph R by

$$\begin{aligned} V(R) &= V(G) \,\dot\cup \dot{\bigcup_{v \in V(G)}} V(Q_v), \\ E(R) &= E(G) \cup \bigcup_{v \in V(G)} E(Q_v) \cup \bigcup_{v \in V(G)} \{\{v, p\} : p \in P_v\}. \end{aligned}$$

The three cases give

$$\deg_{Q_v}(p) = d - 1 \quad (p \in P_v), \qquad \deg_{Q_v}(x) = d \quad (x \in V(Q_v) \setminus P_v).$$

Hence $\deg_R(v) = \deg_G(v) + |P_v| = d$ for every $v \in V(G)$, and $\deg_R(x) = d$ for every $x \in V(Q_v)$. Each Q_v has minimum degree $d - 1$, and its order satisfies

$$d + 1 \leq |V(Q_v)| \leq d + 2 \qquad (v \in V(G)). \tag{89}$$

Every connected component therefore has at least d vertices, so Q_v is connected because $2d > d + 2$ for $d \geq 3$. The resulting graph R is simple.

Appending $V(Q_v)$ as a chain below v increases the height by at most $d + 2$. If the chain has $|V(Q_v)|$ vertices, their total depth is at most

$$|V(Q_v)|\,\mathrm{depth}_F(v) + \frac{|V(Q_v)|(|V(Q_v)|+1)}{2} \overset{(89)}{\leq} (d+2)\,\mathrm{depth}_F(v) + \frac{(d+2)(d+3)}{2}.$$

Summing over v and including the original vertices gives the stated total-depth upper bound. Attach a bag $\{v\} \cup V(Q_v)$ to an existing bag containing v in a tree decomposition of G. Doing this for every v proves the treewidth bound.

For the total-depth lower bound, partition $V(R)$ into connected sets $X_v = \{v\} \cup V(Q_v)$. Each edge uv of G itself witnesses an edge between X_u and X_v. Thus any elimination forest F of R projects by Lemma 6.26 to a forest $\overline{F}$ of G with

$$\mathrm{depth}_F(x) \overset{\text{Lem. 6.26}}{\geq} \mathrm{depth}_{\overline{F}}(v) \qquad (v \in V(G),\ x \in X_v). \tag{90}$$

Consequently,

$$\begin{aligned}\operatorname{totd}(F) &\overset{(90)}{\geq} \sum_{v\in V(G)} |X_v| \operatorname{depth}_{\overline{F}}(v)\\ &\overset{(89)}{\geq} (d+2)\operatorname{totd}(\overline{F}) \geq (d+2)\operatorname{totd}(G).\end{aligned}$$

This argument also gives a polynomial-time projection of forests. □

Corollary 6.28. *The hardness ranges in Corollaries 6.22, 6.23, and 6.25 also hold on d-regular graphs for a suitable fixed odd degree d depending only on the excluded factor.*

Proof. Fix a target factor r and a larger factor r' in the same hardness range. For NP-hardness of total depth, take $1 \leq r < r' < 63/62$. Choose an odd d larger than the maximum degree of the bounded-degree source family. We may increase d further, then require a sufficiently large source order n, using the enlargements described in Section 6.5.

For treedepth and treewidth, Proposition 6.27 changes the promise bounds by at most the fixed constant $d+2$. Their YES bounds have linear scale, so this is $o(Y)$ as the source order grows. Lemma 6.1 preserves the factor-r gap.

A chain on all n vertices is always an elimination forest, so

$$\operatorname{totd}(G) \leq \frac{n(n+1)}{2}. \tag{91}$$

For total depth, Proposition 6.27 gives

$$\begin{aligned}0 &\overset{\text{Prop. 6.27}}{\leq} \frac{\operatorname{totd}(R)}{d+2} - \operatorname{totd}(G)\\ &\overset{\text{Prop. 6.27}}{\leq} \frac{\operatorname{totd}(G)}{d+2} + \frac{d+3}{2}n\\ &\overset{(91)}{\leq} \frac{n(n+1)}{2(d+2)} + \frac{d+3}{2}n.\end{aligned} \tag{92}$$

For the UGC and SSE total-depth families, let $Y = \Theta(n^2)$ be the YES upper bound. For any fixed $\epsilon > 0$, choose d large enough that the first term in the last bound is at most $\epsilon Y/2$ for all sufficiently large n, then choose n large enough that the second term is at most $\epsilon Y/2$. The normalized error is then at most ϵY, and the NO bound does not decrease. Choosing $\epsilon < (r'-r)/(r+1)$, Lemma 6.1 preserves the factor-r gap.

For the NP total-depth family, let $G = H'$ be the sparse instance from the preceding proof, and write $s = |V(G)| = 99m + 3M$, with m and M as defined there. That proof gives $\operatorname{OPT}(G) = \operatorname{totd}(H') \overset{(85)}{\geq} Mm/2 =: L$. Since $M = \Theta(m)$, we have $s = \Theta(m)$ and $L = \Omega(s^2)$. Choose $\epsilon > 0$ such that

$$r(1+\epsilon) < r'. \tag{93}$$

Use (92) with $n = s$, and fix $c > 0$ such that

$$L \geq cs^2 \tag{94}$$

throughout this family. For $s \geq 1$, the first term is at most $s^2/(d+2) \overset{(94)}{\leq} L/(c(d+2))$. Choose an odd d, also larger than the maximum degree of G, and then a source order s such that

$$\frac{1}{c(d+2)} \leq \frac{\epsilon}{2}, \qquad s \geq \frac{d+3}{c\epsilon}. \tag{95}$$

For these choices, each term in the last bound of (92) is at most $\epsilon L/2$, using (94) for the second. Therefore

$$\frac{\mathrm{totd}(R)}{d+2} - \mathrm{totd}(G) \overset{(92),\,(94),\,(95)}{\leq} \epsilon L, \tag{96}$$

and hence

$$\mathrm{OPT}(R) \overset{(96)}{\leq} (d+2)\,\mathrm{OPT}(G) + (d+2)\epsilon L.$$

The projection in Proposition 6.27 converts every forest of R of cost z into one of G of cost at most $z/(d+2)$. In particular, if $z \leq r\,\mathrm{OPT}(R)$, the projected forest $\overline{F}$ satisfies

$$\begin{aligned}\mathrm{totd}(\overline{F}) &\overset{\text{Prop. 6.27}}{\leq} z/(d+2) \leq r\,\mathrm{OPT}(R)/(d+2)\\ &\overset{(96)}{\leq} r\,\mathrm{OPT}(G) + r\epsilon L \leq r(1+\epsilon)\,\mathrm{OPT}(G).\end{aligned}$$

The last inequality uses $L \leq \mathrm{OPT}(G)$. Thus an r-approximation on regular graphs yields factor $r(1+\epsilon) \overset{(93)}{<} r'$ on the source family. □

Corollary 6.29. *For every fixed $g \geq 3$, the treedepth and treewidth hardness ranges above hold on bounded-degree graphs that are bipartite, 2-degenerate, and have girth at least g. Here 2-degenerate means that every nonempty subgraph has a vertex of degree at most two, and girth is the length of a shortest cycle (infinite for an acyclic graph). On the same class, total depth is SSE-hard to approximate within every fixed factor.*

Proof. Replace every edge of G by a path of fixed even length $\ell \geq \max\{2, g/3\}$, obtaining G_ℓ. All original vertices lie in one side of a bipartition, since every replacement path has even length. Each internal vertex has degree two, and deleting all internal vertices leaves the original vertices isolated, so G_ℓ is 2-degenerate. Original vertices retain their degrees, so $\Delta(G_\ell) \leq \max\{\Delta(G), 2\}$. Every cycle uses at least three replacement paths and therefore has length at least $3\ell \geq g$. By Lemma 2.1, contracting each replacement path gives $\mathrm{tw}(G) \overset{\text{Lem. 2.1}}{\leq} \mathrm{tw}(G_\ell)$ and $\mathrm{td}(G) \overset{\text{Lem. 2.1}}{\leq} \mathrm{td}(G_\ell)$. Conversely, for each path $u = x_0, x_1, \ldots, x_\ell = v$, attach the bags $\{u, v, x_1\}, \{v, x_1, x_2\}, \ldots, \{v, x_{\ell-2}, x_{\ell-1}\}$ along a path to an old bag containing u, v. These bags cover the replacement path and preserve the connectedness condition for each vertex, so $\mathrm{tw}(G_\ell) \leq \max\{\mathrm{tw}(G), 2\}$. If $\mathrm{tw}(G) \geq 2$, the two bounds give equality. If $\mathrm{tw}(G) \leq 1$, then G is a forest. Subdivision preserves acyclicity and whether an edge exists, so it preserves treewidth zero or one as well. Thus treewidth is preserved in every case. A path on s vertices has an elimination tree of height at most $\lceil \log_2(s+1) \rceil$: choose a middle vertex as root and recurse on the two remaining subpaths. Thus the $\ell - 1$ internal vertices on each replaced edge have such a tree of height $h = \lceil \log_2 \ell \rceil$. Attach this tree below the deeper of the original edge's endpoints in an elimination forest of G. The original endpoints are comparable, so the edge to the shallower endpoint is represented as well. This gives

$$\mathrm{td}(G_\ell) \leq \mathrm{td}(G) + h, \qquad h = \lceil \log_2 \ell \rceil. \tag{97}$$

Lemma 6.1 absorbs the fixed height increase in (97), since the YES bounds grow linearly with the source order. For total depth, start with a d-regular instance G on n vertices from Corollary 6.28. The identity $dn = 2|E(G)|$ and oddness of d imply that n is even. Add a perfect matching on the vertex set to make every degree even, using auxiliary matching edges that may be parallel to original edges. Each connected component of the augmented multigraph has even degrees and therefore has an Euler tour, a closed walk traversing every edge once. Orient its edges along this tour, then

delete the matching edges. Each vertex has outdegree $(d+1)/2$ before deletion, so its outdegree in the original graph is at least $(d-1)/2$.

Assign the internal vertices of each replacement path to the tail of its oriented edge. For each vertex v, let X_v consist of v and the internal vertices assigned to it. The sets X_v partition $V(G_\ell)$. Each is connected, and

$$s = 1 + (\ell-1)(d-1)/2, \qquad |X_v| \geq s \quad (v \in V(G)). \tag{98}$$

Each oriented edge uv has a replacement path whose last edge joins X_u to X_v. Lemma 6.26 therefore projects any elimination forest F of G_ℓ to an elimination forest $\overline{F}$ of G, and gives

$$\begin{aligned} \operatorname{totd}(F) &\overset{\text{Lem. 6.26}}{\geq} \sum_{v \in V(G)} |X_v| \operatorname{depth}_{\overline{F}}(v) \\ &\overset{(98)}{\geq} s \operatorname{totd}(\overline{F}) \geq s \operatorname{totd}(G). \end{aligned} \tag{99}$$

For the upper bound, use the same midpoint-recursive trees below the deeper endpoints in an optimal elimination forest F of G, and call the resulting forest of G_ℓ $\widehat{F}$. Each of the $\ell - 1$ new vertices x on a replacement path uv satisfies

$$\operatorname{depth}_{\widehat{F}}(x) \leq \operatorname{depth}_F(\text{deeper endpoint of } uv) + h. \tag{100}$$

Each vertex is the deeper endpoint of at most d edges, so summing the original and new depths gives

$$\begin{aligned} s \operatorname{totd}(G) &\overset{(99)}{\leq} \operatorname{totd}(G_\ell) \\ &\overset{(100)}{\leq} \operatorname{totd}(G) + (\ell-1) \sum_{uv \in E(G)} \operatorname{depth}_F(\text{deeper endpoint of } uv) + h(\ell-1)|E(G)| \\ &\leq [1 + d(\ell-1)] \operatorname{totd}(G) + h(\ell-1)|E(G)|. \end{aligned}$$

Since $d \geq 3$, the definition of s gives

$$3s - [1 + d(\ell-1)] = 2 + \frac{(d-3)(\ell-1)}{2} \geq 0.$$

Using $|E(G)| = dn/2$ and putting $C = h(\ell-1)d/2$, we obtain

$$s \operatorname{totd}(G) \leq \operatorname{totd}(G_\ell) \leq 3s \operatorname{totd}(G) + Cn.$$

First fix a target factor r and any fixed $r' > r$, and choose a regular SSE source family with NO bound Z and YES upper bound Y satisfying $Z > 3r'Y$. This fixes the odd degree d. Fix the required even subdivision length ℓ, and hence h, s, C, before enlarging the source. Choose $c_Y > 0$ such that $Y \geq c_Y n^2$ throughout this family, and take $n > rC/[3s(r'-r)c_Y]$. These choices give

$$Z > 3r'Y, \qquad 3s(r'-r)Y > rCn. \tag{101}$$

The subdivided instance has YES bound at most $3sY + Cn$ and NO bound at least sZ. Therefore the NO lower bound minus r times the YES upper bound is at least

$$\begin{aligned} sZ - r(3sY + Cn) &\overset{(101)}{>} 3s(r'-r)Y - rCn \\ &\overset{(101)}{>} 0 \end{aligned}$$

Such orders are obtained from parallel host-edge copies as described in Section 6.5. □

Equation (104) also transfers the total-depth hardness conclusions of both corollaries to trivially perfect completion.

7 Applications

We first transfer the total-depth gaps to trivially perfect completion, then specify the vertex-query search conventions and their hardness transfers. The YES deletion sets and NO separator bounds also give hardness for deletion and labeling objectives. Established automata and hypergraph reductions transfer the parameter gaps, while the linear treewidth gaps yield exponential lower bounds for junction-tree table size.

Several application routes are summarized below. The separator promise means the line-graph YES deletion and NO separator bounds from Section 4; the objective definitions and the individual transfers follow in the indicated sections.

$$\begin{aligned} \text{total depth} &\xrightarrow{\text{Lemma 7.1}} \text{trivially perfect completion},\\ \text{line-graph separator promise} &\xrightarrow{\text{Section 7.3}} \text{deletion and label sizes},\\ \text{treewidth} &\xrightarrow{\text{Sections 7.4 and 7.5}} \text{hypergraph widths and table size}. \end{aligned}$$

7.1 Trivially perfect completion by number of added edges

The total-depth transfer must account for the input's original edges.

The *closure* of a rooted forest joins each vertex to all its proper ancestors. A graph is *trivially perfect* if and only if it is such a closure [HBB$^+$21, Section 2]. Let $f(J)$ be the minimum number of edges added to J to obtain a trivially perfect graph, and recall that $\mathrm{totd}(J)$ is its minimum total elimination depth.

Lemma 7.1 (Completion identity [HBB$^+$21, Theorem 8])**.** *For every graph J,*

$$f(J) = \mathrm{totd}(J) - |V(J)| - |E(J)|. \tag{102}$$

For any $C \geq 1$, a factor-C completion of cost z satisfies

$$f(J) \leq z \leq C f(J). \tag{103}$$

Adding the known offset $|V(J)| + |E(J)|$ cannot worsen the multiplicative guarantee: the offset is added once to the returned cost, while a factor-C comparison allows C times the offset. Thus

$$\begin{aligned} \mathrm{totd}(J) &\overset{(102)}{=} f(J) + |V(J)| + |E(J)| \\ &\overset{(103)}{\leq} z + |V(J)| + |E(J)| \\ &\overset{(103)}{\leq} C f(J) + |V(J)| + |E(J)| \\ &\leq C\big(f(J) + |V(J)| + |E(J)|\big) \overset{(102)}{=} C\,\mathrm{totd}(J), \end{aligned} \tag{104}$$

where the final inequality uses $C \geq 1$. Thus Corollary 6.25 also gives NP-hardness within every fixed factor below 63/62 and UGC hardness below 8/7 for trivially perfect completion on graphs of bounded maximum degree. From a supplied trivially perfect supergraph K, construct a generating forest in polynomial time by choosing a universal vertex as the root of each connected component, deleting it, and recursing on the remaining components. Such a vertex exists by the closure characterization, which also holds for induced subgraphs by restricting their generating forests. The resulting forest has closure K and total depth $|E(K)| + |V(K)|$, so the adjusted cost also supplies the forest needed by the total-depth approximation transfer.

Theorem 7.2. *For every fixed $C \geq 1$, approximating the minimum number of edges added to obtain a trivially perfect graph within factor C is SSE-hard on graphs of maximum degree bounded by a constant depending only on C.*

Proof. Apply (104) to the SSE hardness of total depth in Corollary 6.25. □

Deciding whether at most k edges suffice to complete a graph to a trivially perfect graph is NP-hard. Chain Completion asks for the fewest edges added across a fixed bipartition to make the neighborhoods on one side linearly ordered by inclusion. Its NP-hardness [Yan81] and the polynomial reduction to trivially perfect completion [BCK+16, Lemma 5.4] give this decision hardness. Bliznets, Cygan, Komosa, Mach, and Pilipczuk show that, under the Exponential Time Hypothesis (ETH), for some fixed $c \geq 1$ there is no $2^{O(k^{1/4}/\log^c k)} n^{O(1)}$-time algorithm for this completion problem, where k is the number of added edges and n is the input order [BCK+16, Theorem 1.1]. The SSE hardness of *chordal* Minimum Fill-In at every fixed factor, whose target graphs have no induced cycle of length at least four, is already known [WAPL15, Theorem 1.2]. Restricting the target class to trivially perfect graphs changes the optimum, so that theorem alone does not prove the statement here.

7.2 Vertex-query search

On a connected graph G, a vertex query either finds the target or identifies the component containing it after the queried vertex is removed. Under the confirming convention, the target must itself be queried, even if it is the only remaining candidate. With unit query costs, the minimum sum of query counts over all targets is $\mathrm{totd}(G)$; with a uniform target distribution, the minimum expected cost is $\mathrm{totd}(G)/|V(G)|$.

Some search models instead stop without a query once the target is uniquely identified. If $I(G)$ is the minimum sum of query counts under this identification convention, then

$$I(G) \leq \mathrm{totd}(G) \leq I(G) + |V(G)|. \tag{105}$$

A confirming strategy can stop early, giving the first inequality. Conversely, add a final query of the identified target on each outcome where it has not yet been queried. This costs at most one extra query per target and gives the second inequality. Thus $\mathrm{totd}(G)/|V(G)|$ and $I(G)/|V(G)|$ differ by at most one, but need not be equal: on the three-vertex path their values are $5/3$ and 1, respectively.

The identification approximation of [Szy26a] also gives an $O(\sqrt{\log n})$-approximation for the uniform confirming objective. Add a final confirming query where necessary. On a connected graph with at least two vertices every strategy makes at least one query, so the extra expected cost of at most one is absorbed by a constant factor. The one-vertex case is solved directly.

Our NP and UGC total-depth reductions have connected outputs. The UGC graph $S_q(G)$ is connected whenever G has an edge. The edgeless source is always a YES instance and can be mapped to K_1. In the NP construction, opposite pairs and clause pairs join A to B and C, and every auxiliary pair has an edge between B and C (Section 5.3). The bounded-degree replacements retain these joining edges and use connected sparse replacement graphs from Lemma 6.16. Thus the same constant-factor hardness holds for average-case confirming search with uniform weights and unit query costs, also at bounded maximum degree.

It also holds under the identification convention. Indeed, fix a target factor r strictly below the relevant threshold and first choose a source gap with a larger ratio $r' > r$. Its YES total-depth upper bound Y is $\Theta(N^2)$ in the output order N, and its NO lower bound Z satisfies $Z > r'Y$. Equation (105) gives identification bounds Y and $Z - N$, whose gap satisfies

$$(Z - N) - rY > (r' - r)Y - N > 0$$

for sufficiently large source instances. The source padding used in the reductions ensures this size condition. Dividing by N preserves the ratio for average cost. A returned strategy is converted to a forest by adding the terminal queries and recursively using its query as the root of the current candidate component; queries outside that component reveal no information and may be deleted. These operations are polynomial-time for an explicitly represented strategy.

7.3 Deletion and labeling objectives

Vertex integrity. For $S \subseteq V(J)$, let $\mathcal{C}(J-S)$ be the set of connected components of $J-S$, and define

$$c_J(S) = \max\big(\{|V(C)| : C \in \mathcal{C}(J-S)\} \cup \{0\}\big),$$
$$\mathrm{vi}(J) = \min_{S \subseteq V(J)} \big(|S| + c_J(S)\big).$$

The elementary inequality $\mathrm{td}(J) \le \mathrm{vi}(J)$ follows by putting S on a chain and its components on separate chains below it. In the layered graph, deleting all copies of a minimum source vertex cover leaves q separate cliques on $\alpha(G)$ vertices. Thus the same YES construction gives $\mathrm{vi}(S_q(G)) \overset{\text{Lem. 3.2}}{\le} q(n-\alpha(G)) + \alpha(G)$, while Theorem 3.9 supplies the NO lower bound. The parameter choices in Section 3.2.3 therefore also give UGC hardness for vertex integrity below $4/3$. For SSE hardness, let $J = \mathcal{L}(G^*)$ be the output in the dense construction of Section 4.3, put $N = |V(J)|$, and let D consist of the output vertices corresponding to crossing host edges. Every component of $J-D$ lies within the vertices representing one class's internal edges. Thus

$$\begin{aligned} \mathrm{vi}(J) &\le |D| + c_J(D) \\ &\overset{(26)}{\le} 3\varepsilon N + (1+\varepsilon)N/q \\ &\le \left(3 + \frac{1+\varepsilon}{2}\right)\varepsilon N \le 4\varepsilon N, \end{aligned} \tag{106}$$

using $q = \lceil 2/\varepsilon \rceil$ and $\varepsilon \le 1$. In the NO case, the separator lemma with $\rho = 2/3$ gives

$$b_{2/3}(J) \overset{\text{Lem. 4.5}}{\ge} \beta N, \tag{107}$$

where $b_{2/3}(J)$ is the minimum size of a set whose deletion leaves every component of order at most $2N/3$, and $\beta = (c/2)\sqrt{\varepsilon}$. Choose ε small enough that the source construction in Section 4.3 applies, $\beta \le 1/12$, and $\varepsilon \le 1$. For any $S \subseteq V(J)$, either S is a $2/3$-balanced separator, giving $|S| \ge b_{2/3}(J) \overset{(107)}{\ge} \beta N$, or $c_J(S) > 2N/3 \ge \beta N$. In both cases $|S| + c_J(S) \ge \beta N$, so

$$\mathrm{vi}(J) \ge \beta N. \tag{108}$$

The ratio of the NO lower bound to the YES upper bound grows as $1/\sqrt{\varepsilon}$, so it exceeds any fixed factor for sufficiently small ε. Hence vertex integrity has no constant-factor approximation under SSEH.

The *fracture number* is $\mathrm{fr}(J) = \min_{S \subseteq V(J)} \max\{|S|, c_J(S)\}$, and *starwidth* $\mathrm{sw}(J)$ is the minimum width of a tree decomposition whose underlying tree is a star.

Lemma 7.3 (Elementary integrity comparisons)**.** *For every nonempty graph J,*

$$\mathrm{fr}(J) \le \mathrm{vi}(J) \le 2\,\mathrm{fr}(J), \qquad \mathrm{sw}(J) + 1 \le \mathrm{vi}(J) \le 2(\mathrm{sw}(J)+1).$$

Proof. For every $S \subseteq V(J)$,

$$\max\{|S|, c_J(S)\} \le |S| + c_J(S) \le 2\max\{|S|, c_J(S)\}.$$

Minimizing gives the fracture-number comparisons. A star decomposition with center bag S and one leaf bag $S \cup C$ for each component C of $J - S$ has width at most $|S| + c_J(S) - 1$. Minimizing over S gives $\mathrm{sw}(J) + 1 \le \mathrm{vi}(J)$.

Conversely, let a star decomposition of J have width $k = \mathrm{sw}(J)$, and let S be its center bag. By connectedness of the bags containing a vertex, each vertex outside S occurs in only one leaf bag. Edge coverage therefore prevents edges between vertices in different leaf bags outside S. Every component of $J - S$ therefore lies in one leaf bag. The center bag and each leaf bag have at most $k + 1$ vertices, so

$$|S| \le k + 1, \qquad c_J(S) \le k + 1. \tag{109}$$

Hence $\mathrm{vi}(J) \le |S| + c_J(S) \overset{(109)}{\le} 2(k+1)$. □

For the same SSE instances, the comparisons give

$$\begin{aligned}
\text{YES:}\quad \mathrm{fr}(J) &\overset{\text{Lem. 7.3}}{\le} \mathrm{vi}(J) \overset{(106)}{\le} 4\varepsilon N,\\
\mathrm{sw}(J) + 1 &\overset{\text{Lem. 7.3}}{\le} \mathrm{vi}(J) \overset{(106)}{\le} 4\varepsilon N,\\
\text{NO:}\quad \mathrm{fr}(J) &\overset{\text{Lem. 7.3}}{\ge} \mathrm{vi}(J)/2 \overset{(108)}{\ge} \beta N/2,\\
\mathrm{sw}(J) &\overset{\text{Lem. 7.3}}{\ge} \mathrm{vi}(J)/2 - 1 \overset{(108)}{\ge} \beta N/2 - 1.
\end{aligned}$$

Since $\beta/(8\varepsilon) = c/(16\sqrt{\varepsilon})$ grows without bound as ε decreases, choose ε for the desired factor. For an integer $t \ge 1$, let G_t^* replace each edge occurrence of G^* by t parallel copies. For every $v \in V(G^*)$ and $A \subseteq V(G^*)$,

$$\begin{gathered}
|E(G_t^*)| = t|E(G^*)|, \qquad \deg_{G_t^*}(v) = t\deg_{G^*}(v),\\
\mathrm{vol}_{G_t^*}(A) = t\,\mathrm{vol}_{G^*}(A), \qquad |\delta_{G_t^*}(A)| = t|\delta_{G^*}(A)|,\\
|V(\mathcal{L}(G_t^*))| = tN.
\end{gathered}$$

Thus the normalized source promises are unchanged and the output order grows without bound. The additive 1 is negligible compared with the linear YES bound $4\varepsilon N$ and is absorbed by Lemma 6.1. Gima et al. give an $O(\log \mathrm{OPT})$ approximation for vertex integrity [GHK+24, Theorem 5.1]. Earlier NP-hardness includes vertex integrity on connected line graphs [GHK+24, Corollary 4.11] and starwidth [vE17].

General and average customizable labels. A customizable hub labeling assigns $L(v) \subseteq V(J)$ so that

$$V(P) \cap L(s) \cap L(t) \neq \varnothing \qquad \text{for every } s\text{–}t \text{ path } P,$$

including the zero-length path at a vertex [BS22]. The labeling is *hierarchical* if there is a total order $v_1, \dots, v_{|V(J)|}$ of $V(J)$ such that $L(v_i) \subseteq \{v_i, \dots, v_{|V(J)|}\}$ for every i. Without this requirement we call it general. Average, maximum, and total label size mean, respectively, $\sum_{v \in V(J)} |L(v)|/|V(J)|$, $\max_{v \in V(J)} |L(v)|$, and $\sum_{v \in V(J)} |L(v)|$. Blum and Storandt prove

$$\frac{\sum_{v \in V(J)} |L(v)|}{|V(J)|} \overset{\text{[BS22, Cor. 10, full version]}}{\ge} \frac{b_{2/3}(J)}{4}. \tag{110}$$

Choose a fixed $\varepsilon > 0$ small enough that the source construction in Section 4.3 applies and $(c/2)\sqrt{\varepsilon} \le 1/12$ and $\sqrt{\varepsilon} < c/(32C)$, where $C \ge 1$ is the desired factor. For the resulting graph $J = \mathcal{L}(G^*)$, order the YES forest's vertices in postorder, so every vertex precedes its ancestors. For this forest F, define

$$L(v) = \{u \in V(J) : u \text{ is an ancestor of } v \text{ in } F\}, \qquad v \in V(J),$$

where a vertex is its own ancestor. This is a hierarchical labeling. On any s–t path, its shallowest vertex z is an ancestor of every path vertex by Lemma 2.2. Thus $z \in L(s) \cap L(t)$, so these are valid customizable labels. The YES height bound gives

$$\max_{v \in V(J)} |L(v)| \overset{(27)}{\le} 4\varepsilon |V(J)|.$$

In the NO case,

$$\frac{\sum_{v \in V(J)} |L(v)|}{|V(J)|} \overset{(110)}{\ge} \frac{b_{2/3}(J)}{4} \overset{\text{Lem. 4.5}}{\ge} \frac{c}{8}\sqrt{\varepsilon}\,|V(J)|.$$

Consequently every constant approximation factor is SSE-hard for average or maximum general label size and average hierarchical label size.

For hierarchical labels, the total-depth gaps give stronger conclusions under the other two hardness assumptions as well. The following identity also supplies a polynomial-time decoding of labels into a forest.

Lemma 7.4 (Hierarchical labels and total depth)**.** *For every nonempty graph J, the minimum total size of a hierarchical customizable hub labeling is* $\mathrm{totd}(J)$*. Its minimum average size is* $\mathrm{totd}(J)/|V(J)|$*.*

Proof. Ancestor sets in an elimination forest are valid hierarchical labels, as proved above, and their total size is the forest's total depth. This proves the upper bound.

Conversely, let L be a hierarchical labeling and fix a compatible total order. In each connected component choose its highest-ranked vertex as the root, delete it, and recurse on the remaining components. The resulting forest F is an elimination forest: every graph edge either meets the chosen root or remains within one recursive component. Suppose a is an ancestor of v in F. The component in which a was chosen contains an a–v path P, all of whose vertices have rank at most that of a. The cover condition gives a vertex in $L(a) \cap L(v) \cap V(P)$. Since hierarchy permits only vertices of rank at least that of a in $L(a)$, this vertex must be a. Writing $\mathrm{Anc}_F(v)$ for the ancestors of v including v itself, we have proved

$$\mathrm{Anc}_F(v) \subseteq L(v) \qquad (v \in V(J)). \tag{111}$$

Consequently,

$$\mathrm{totd}(J) \overset{\text{definition of totd}}{\le} \mathrm{totd}(F) = \sum_{v \in V(J)} \mathrm{depth}_F(v) \overset{(111)}{\le} \sum_{v \in V(J)} |L(v)|.$$

This proves the reverse bound. The recursion is polynomial-time. If only the labels are supplied, recover a compatible order by topologically sorting the constraints $v < u$ for $u \in L(v) \setminus \{v\}$; hierarchy guarantees that these constraints are acyclic. Dividing the total-size identity by $|V(J)|$ proves the average-size identity. □

For a customizable contraction hierarchy (CCH), an elimination order adds edges between the remaining neighbors of each eliminated vertex. Orient the resulting edges toward the later vertex in that order. The upward search space of v is the set of vertices reachable from v along these oriented edges, including v itself. Blum and Storandt show that these search spaces form a hierarchical customizable hub labeling. Their Corollary 5 identifies the corresponding minimum average label size [BS22, Corollary 5, full version]. Their correspondence is constructive: for a fixed order, the upward search spaces equal the canonical hierarchical labels, which are contained in every hierarchical labeling respecting that order [BS22, Lemmas 2 and 4, full version]. Consequently the minimum total upward vertex search-space size is also $\mathrm{totd}(J)$, and its minimum average is $\mathrm{totd}(J)/|V(J)|$. An order gives all upward search spaces by polynomial-time graph searches; the preceding lemma then decodes them into an elimination forest.

Corollary 7.5. *Minimum total or average hierarchical customizable-label size, and minimum total or average CCH upward vertex search-space size, are NP-hard to approximate within every fixed factor $1 \le r < 63/62$, Unique Games-hard within every fixed factor $1 \le r < 8/7$, and SSE-hard within every fixed factor $r \ge 1$. These conclusions hold on graphs of bounded maximum degree and on regular graphs, with the degree depending only on the excluded factor. The SSE conclusions also hold on bounded-degree bipartite 2-degenerate graphs of any prescribed fixed girth lower bound.*

Proof. Apply Lemma 7.4 and the CCH correspondence to Corollaries 6.25, 6.28, and 6.29. Multiplication by the known graph order preserves each approximation factor. The polynomial decodings above also preserve these factors. □

7.4 Transfers through established reductions

The following consequences use existing correspondences. We state the transformations and their losses to specify which parameter gaps they preserve.

Automata and restricted cycle rank. The *star height* of a regular language is the minimum nesting depth of Kleene stars in a regular expression for the language. A partial deterministic finite automaton (DFA) has an initial state, accepting states, and at most one transition from each state per letter, but may omit transitions. Its language consists of words labeling paths from the initial state to an accepting state. It is *bideterministic* if it has one accepting state and remains deterministic when its transitions are reversed and its initial and accepting states are exchanged. It is *trim* if every state lies on such a path. A binary DFA uses a two-letter alphabet. For a digraph D, the *cycle rank* $\mathrm{cr}(D)$ is zero if D is acyclic, is $1 + \min_{v \in V(D)} \mathrm{cr}(D - v)$ if D is strongly connected and has a directed cycle, and otherwise is the maximum cycle rank of its strongly connected components, the maximal sets whose vertices can reach one another by directed paths [Gru08, Sections 2–3].

Gruber's cycle-rank/star-height correspondence and polynomial binary encoding [Gru08, Theorems 20 and 23] transfer our three treedepth gaps to star height for languages represented by trim binary bideterministic partial DFAs. We first add a universal vertex a to G to ensure connectivity and cancel the unit difference between treedepth and symmetric cycle rank. Define the symmetric digraph D by

$$V(D) = V(G) \cup \{a\}, \qquad E(D) = \{(u, v), (v, u) : uv \in E(G + a)\}.$$

Use state set $V(D)$ and alphabet $E(D)$, with transitions $u \xrightarrow{(u,v)} v$ for $(u, v) \in E(D)$, and take a as the initial and sole accepting state. The resulting partial DFA accepts the label sequences of directed walks from a back to a. It is trim because the digraph is strongly connected, and it is

bideterministic because each letter labels only one arc. Every trim bideterministic DFA is minimal: reverse determinism and the sole accepting state imply that a word can be accepted from at most one state. Trimness supplies an accepting suffix from each state, so any two states are distinguishable by such a suffix. Putting a above an optimal forest of G gives one direction of

$$\mathrm{td}(G + a) = \mathrm{td}(G) + 1. \tag{112}$$

For the reverse direction, in any elimination tree of $G + a$ every vertex is comparable with a, because a is universal. Every root-to-leaf path therefore contains a: a branch leaving its ancestor path above a would contain an incomparable vertex. Restricting the tree to G removes a from each such path and lowers its height by at least one. Gruber's symmetric-digraph correspondence now gives

$$\mathrm{cr}(D) \overset{\text{[Gru08, proof of Thm. 12]}}{=} \mathrm{td}(G + a) - 1 \overset{(112)}{=} \mathrm{td}(G).$$

Gruber's binary encoding constructs in polynomial time a bideterministic DFA for the encoded language, preserves its star height, and retains the original initial and sole accepting state [Gru08, proof of Theorem 23]. Delete states that are not on an initial-to-accepting path; this polynomial trimming preserves the language and bideterminism. The resulting trim DFA is minimal by the preceding argument and has its initial state as its sole accepting state. Thus every state is reachable from the initial state and can return to it, so the transition digraph is strongly connected. Bideterminism over two letters bounds its indegree and outdegree by two. McNaughton's theorem states that a bideterministic language's star height equals the cycle rank of the transition digraph of its minimal trim DFA [Gru08, Theorem 20]. Hence the encoded language's star height is the cycle rank of this bounded-degree digraph. Gruber proves cycle-rank NP-hardness even for digraphs of maximum outdegree two and maximum total degree four [Gru08, Corollary 24]. Our consequence is the transfer of the present approximation gaps.

Edge ranking. An *edge ranking* assigns positive integer labels to edges so that every path containing two distinct edges of the same label also contains an edge with a larger label. Its minimum possible largest label is the edge-ranking number $\mathrm{er}(B)$. Let G^* be the host multigraph in Section 4, and put $H = \mathcal{L}(G^*)$. The SSE reduction proves hardness of $\mathrm{td}(H)$ at every fixed factor. For a simple graph B, the standard identity

$$\mathrm{er}(B) \overset{\text{[BDJ+94]}}{=} \mathrm{td}(\mathcal{L}(B)) \tag{113}$$

transfers treedepth hardness of its line graph to edge ranking. We turn G^* into a simple bipartite graph before applying this identity. Replace every loop occurrence by an edge to a fresh pendant vertex, keeping parallel edge occurrences distinct. This preserves the line graph: each loop is replaced by one edge incident with its original endpoint, and the new pendant vertices have no other incident edges. Then subdivide each edge occurrence once, using a fresh vertex x_e. Call the resulting graph B. Each new edge joins an old vertex to x_e, so it is not a loop. Different occurrences use different subdivision vertices, so their new edges cannot be parallel. Thus B is simple. It is bipartite, with the old and pendant vertices on one side and the subdivision vertices on the other.

For each edge occurrence $e = uv$ before subdivision, the two new edges are ux_e and x_ev. Their vertices in $\mathcal{L}(B)$ are adjacent. Contracting these pairs, one pair for every edge occurrence, recovers H. Thus H is a minor of $\mathcal{L}(B)$. Conversely, replace each vertex of an elimination forest of H by a two-vertex chain, attaching its child subtrees below the lower copy. Every edge of $\mathcal{L}(B)$ joins two copies of adjacent vertices of H, or the two copies of one vertex. The expanded forest is therefore valid and has at most twice the original height, proving

$$\mathrm{td}(\mathcal{L}(B)) \leq 2\,\mathrm{td}(H). \tag{114}$$

Consequently,

$$\mathrm{td}(H) \overset{\text{Lem. 2.1}}{\leq} \mathrm{td}(\mathcal{L}(B)) \overset{(113)}{=} \mathrm{er}(B) \overset{(114)}{\leq} 2\,\mathrm{td}(H).$$

Applying the SSE treedepth hardness at factor $2C$ gives hardness for edge ranking on simple bipartite graphs at every fixed factor $C \geq 1$. Lam and Yue proved that computing an optimal edge ranking is NP-hard at ICS 1996 [LY96].

Hypergraph widths and binary queries. A hypergraph H has a vertex set $V(H)$ and a collection $E(H)$ of subsets called hyperedges; its *rank* is the largest hyperedge size. A tree decomposition consists of a tree T and a bag $B_t \subseteq V(H)$ for each node t. Every vertex must occur in at least one bag, the nodes whose bags contain any fixed vertex must form a connected subtree of T, and each hyperedge must be contained in a bag. An edge cover of a bag B_t is a set of hyperedges $\lambda_t \subseteq E(H)$ satisfying $B_t \subseteq \bigcup_{e\in\lambda_t} e$; these edges are also called *guards*. The *generalized hypertree width* is

$$\mathrm{ghw}(H) = \min_{(T,(B_t),(\lambda_t))} \max_{t\in V(T)} |\lambda_t|,$$

where the minimum ranges over tree decompositions with such covers. The *fractional hypertree width* fhw instead assigns weights $\gamma_t(e) \geq 0$ to each hyperedge e at each bag B_t, requires

$$\sum_{e\in E(H):\, v\in e} \gamma_t(e) \geq 1 \qquad (t \in V(T),\ v \in B_t), \tag{115}$$

and minimizes $\max_{t\in V(T)} \sum_{e\in E(H)} \gamma_t(e)$. The ordinary *hypertree width* $\mathrm{hw}(H)$ minimizes the same objective as $\mathrm{ghw}(H)$ over rooted tree decompositions that also satisfy the *special condition*

$$\left(\bigcup_{e\in\lambda_t} e\right) \cap \left(\bigcup_{s\in V(T_t)} B_s\right) \subseteq B_t \qquad (t \in V(T)),$$

where T_t is the subtree rooted at t [FGP18, Definitions 2.4–2.7 in the full version].

Marx reduces treewidth to fractional hypertree width in [Mar09, proof of Theorem 5.1]. We use adjacent clone pairs to transfer all three widths on rank-two instances, with the proof given below. Let G be a nonempty graph, and define $G[K_2]$ on $V(G) \times \{1, 2\}$: distinct vertices $(u, i), (v, j)$ are adjacent if and only if $u = v$ or $uv \in E(G)$. Thus each source vertex is replaced by an adjacent pair, and a source edge joins the two pairs completely. We prove that all three widths equal $\mathrm{tw}(G)+1$. The edges of $G[K_2]$ are viewed as size-two hyperedges, with the standard width definitions of [FGP18]. For the upper bound, take a tree decomposition $(T, (B_t))$ of G of width $\mathrm{tw}(G)$ and set, for every $t \in V(T)$,

$$B'_t = B_t \times \{1, 2\}, \qquad \lambda_t = \{\{(v, 1), (v, 2)\} : v \in B_t\}.$$

This gives each of the three widths an upper bound of $\mathrm{tw}(G) + 1$. The guards are contained in the bag, so the special condition for ordinary hypertree width holds as well. In particular,

$$\mathrm{hw}(G[K_2]) \leq \mathrm{tw}(G) + 1. \tag{116}$$

For the lower bound, consider a fractional decomposition of width w with decomposition tree T. By the width definition,

$$\sum_{e\in E(G[K_2])} \gamma_t(e) \leq w \qquad (t \in V(T)). \tag{117}$$

The covering condition and the size-two hyperedges give, at every bag,

$$|B_t| \overset{(115)}{\leq} \sum_{e \in E(G[K_2])} \gamma_t(e)|e \cap B_t| \leq 2 \sum_{e \in E(G[K_2])} \gamma_t(e) \overset{(117)}{\leq} 2w.$$

Define the retained bags by

$$C_t = \{v \in V(G) : (v,1), (v,2) \in B_t\}, \qquad |C_t| \leq |B_t|/2 \leq w \quad (t \in V(T)).$$

Let $T_{(v,i)}$ be the connected subtree of bags containing clone (v,i). The nodes whose retained bags contain v form $T_{(v,1)} \cap T_{(v,2)}$. This intersection is nonempty because the clone pair is an edge, and it is connected because both sets are connected in T. If $uv \in E(G)$, its four clones form a clique, so their sets $T_{(u,1)}, T_{(u,2)}, T_{(v,1)}, T_{(v,2)}$ intersect pairwise. Subtrees of a tree have the Helly property: any finite family that intersects pairwise has a common node. To see this, root the tree, choose in each subtree its node nearest the root, and let z be a deepest chosen node. For every other subtree, an intersection with the subtree that chose z shows that z lies on the path from its chosen node to that intersection, and therefore belongs to it. Hence one bag contains all four clones. The retained bags form a tree decomposition of G with bag size at most w, so

$$\mathrm{tw}(G) + 1 \leq w. \tag{118}$$

Since this holds for every fractional decomposition, the definitions give

$$\begin{aligned} \mathrm{tw}(G) + 1 &\overset{(118)}{\leq} \mathrm{fhw}(G[K_2]) \\ &\leq \mathrm{ghw}(G[K_2]) \leq \mathrm{hw}(G[K_2]) \\ &\overset{(116)}{\leq} \mathrm{tw}(G) + 1, \end{aligned}$$

since fractional covers relax integral covers and hypertree decompositions also impose the special condition. Together with the common upper bound, this proves the three equalities. Thus the NP $r < 12/11$, UGC $r < 4/3$, and established SSE treewidth hardness at every fixed factor transfer even to rank-two hypergraphs. The transfer compares $\mathrm{tw}(G) + 1$, the size of a largest bag. Adding one to both promise bounds is absorbed by Lemma 6.1, since the bounds grow linearly. In the usual query representation, vertices are variables and each hyperedge supplies one relation atom. Their conjunction is a conjunctive query, and size-two hyperedges give binary atoms. Using a distinct relation symbol for each atom therefore gives the same structural width consequences for these queries.

7.5 Binary junction-tree total state space

The width gaps in Sections 3.4 and 5.2 are linear in the output order. Exponentiating bag sizes therefore yields exponential approximation gaps for the total size of dense binary tables.

For an n-vertex graph G with $n \geq 1$, define the total state space (TSS) by

$$\mathrm{TSS}(G) = \min_{(T,(B_t)_{t \in V(T)})} \sum_{t \in V(T)} 2^{|B_t|}.$$

The minimum ranges over tree decompositions of G. Each vertex represents a binary variable. The table for bag B_t has one entry for each of its $2^{|B_t|}$ assignments. Thus this sum is the total number of table entries in the corresponding junction tree. The optimization output is a tree decomposition, with its cost represented as a binary integer; the dense tables need not be materialized. For a decomposition with b bags this integer has $O(n + \log(b + 1))$ bits, so its cost is computable in polynomial time from the decomposition even when the table count is exponential. Bodlaender and Fomin identify the equivalent chordal-completion objective and prove that, unless P = NP, no polynomial-time algorithm approximates binary total state space within any fixed factor [BF02, Lemma 2, Section 8, and Theorem 32]. Bodlaender and Fomin's minimal-decomposition bound gives an optimal-width decomposition with at most n bags [BF02, Lemma 4]. Each bag then contributes at most $2^{\mathrm{tw}(G)+1}$ entries. Conversely, every tree decomposition has a bag of size at least $\mathrm{tw}(G) + 1$, so

$$2^{\mathrm{tw}(G)+1} \le \mathrm{TSS}(G) \overset{\text{[BF02, Lem. 4]}}{\le} n2^{\mathrm{tw}(G)+1}. \tag{119}$$

These bounds give NP hardness within $2^{\chi n}$ for every fixed $0 < \chi < 1/18$, and UGC hardness for every fixed $0 < \chi < 1/6$. For each reduction, subtract the YES treewidth upper bound from the NO treewidth lower bound, denoting these by w_Y, w_N, respectively. Their differences are

$$\begin{aligned} \text{NP:}\quad w_N - w_Y &\overset{\text{Lem. 3.17, (37)}}{=} Rm(1/2 - 2\epsilon) - 3m \\ &= (1/18 - 2\epsilon/9 - 1/(3R))n, \qquad n = 9Rm, \\ \text{UGC:}\quad w_N - w_Y &\overset{\text{Lem. 3.17, Thm. 3.19}}{=} RN(1/6 - 2\zeta) - N/3 \\ &= (1/6 - 2\zeta - 1/(3R))n, \qquad n = RN. \end{aligned} \tag{120}$$

For NP hardness, choose $0 < \epsilon < 1/4$ so that $1/18 - 2\epsilon/9 > \chi$, then take R large enough that $1/(3R) < 1/18 - 2\epsilon/9 - \chi$. For UGC, choose $0 < \zeta < 1/12$ so that $1/6 - 2\zeta > \chi$, then take R large enough that $1/(3R) < 1/6 - 2\zeta - \chi$. In either case, let $\delta > 0$ be the remaining margin, so

$$w_N - w_Y \overset{(120)}{\ge} (\chi + \delta)n. \tag{121}$$

The bag bounds give

$$\begin{aligned} \text{YES:}\quad \mathrm{TSS}(G) &\overset{(119)}{\le} n2^{w_Y+1}, \\ \text{NO:}\quad \mathrm{TSS}(G) &\overset{(119)}{\ge} 2^{w_N+1}. \end{aligned}$$

The ratio of the NO threshold to the YES threshold satisfies

$$\frac{2^{w_N+1}}{n2^{w_Y+1}} = \frac{2^{w_N-w_Y}}{n} \overset{(121)}{\ge} \frac{2^{(\chi+\delta)n}}{n} > 2^{\chi n}$$

for all sufficiently large n, since $2^{\delta n} > n$ then. The source instances may be replaced by any fixed number of disjoint copies without changing their promise fractions. Taking enough copies therefore ensures that every output order lies in this range. Bonnet's earlier linear width gap [Bon25] also gives a fixed positive exponential exponent by the same calculation.

Equivalent formulations. In a *vertex ranking*, every path whose distinct endpoints have the same label has an internal vertex with a larger label. A *centered coloring* gives every nonempty connected induced subgraph a color occurring once. The minimum numbers of labels or colors in these two definitions equal treedepth. These, together with symbolic elimination height, minimum clique number of a trivially perfect completion, maximum hierarchical customizable label size, and maximum CCH vertex search-space size are established treedepth formulations [RRSVS14, BDJ+94, BS22]. These formulations are covered by the same theorem.

8 Open questions

Can treedepth hardness be made superconstant? Our SSE result excludes every fixed factor but leaves an $o(\log n)$ approximation possible, while the upper bound is $O(\log n)$ [Ana26].

Can treedepth's NP-hardness factor exceed 18/17, or can ordinary UGC exclude every fixed factor? For treewidth, our factors are 12/11 and 4/3; SSEH excludes every fixed factor [WAPL15].

For total depth, the 8/7 UGC lower bound remains far from the $O(\sqrt{\log n})$ approximation inherited from average-case graph search [Szy26a], with the confirming-query convention handled by (105). NP-hardness below 63/62 holds even at bounded degree (Corollary 6.25).

Our degree bounds depend on the approximation ratio. Can comparable gaps be proved for graphs of a prescribed small maximum degree?

References

[AKS09] Per Austrin, Subhash Khot, and Muli Safra. Inapproximability of vertex cover and independent set in bounded degree graphs. In *Proceedings of the 24th Annual IEEE Conference on Computational Complexity (CCC 2009)*, pages 74–80. IEEE Computer Society, 2009. Full version: Theory of Computing 7:27–43 (2011). Proof details use the full version at https://theoryofcomputing.org/articles/v007a003/v007a003.pdf.

[Ana26] Aditya Anand. Greedy region growing: Improved approximations for cutwidth, pathwidth, flow-expander decomposition, and more. *arXiv preprint arXiv:2610.17237*, 2026. Submitted 12 October 2026.

[BCK+16] Ivan Bliznets, Marek Cygan, Paweł Komosa, Łukasz Mach, and Michał Pilipczuk. Lower bounds for the parameterized complexity of Minimum Fill-In and other completion problems. In *Proceedings of the Twenty-Seventh Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2016)*, pages 1132–1151. Society for Industrial and Applied Mathematics, 2016. Full version: arXiv:1508.05282.

[BCK+17] Kristóf Bérczi, Karthekeyan Chandrasekaran, Tamás Király, Euiwoong Lee, and Chao Xu. Global and fixed-terminal cuts in digraphs. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2017)*, volume 81 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 2:1–2:20. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2017. Full version: arXiv:1612.00156v2, 6 July 2017. Section references are to this version.

[BDJ+94] Hans L. Bodlaender, Jitender S. Deogun, Klaus Jansen, Ton Kloks, Dieter Kratsch, Haiko Müller, and Zsolt Tuza. Ranking of graphs. In *Graph-Theoretic Concepts in Computer Science, 20th International Workshop, WG '94*, volume 903 of *Lecture Notes in Computer Science*, pages 292–304. Springer, 1994. Full journal version: SIAM J. Discrete Math. 11(1):168–181 (1998).

[BF02] Hans L. Bodlaender and Fedor V. Fomin. Tree decompositions with small cost. In *Algorithm Theory - SWAT 2002, 8th Scandinavian Workshop on Algorithm Theory, Turku, Finland, July 3-5, 2002 Proceedings*, volume 2368 of *Lecture Notes in Computer Science*, pages 378–387. Springer, 2002. Theorem and section references use the expanded version, Discrete Applied Mathematics 145:143–154 (2005). Full text: https://fedorvf.github.io/articles/2005/2005b.pdf.

[BNS25] Édouard Bonnet, Daniel Neuen, and Marek Sokołowski. Treedepth inapproximability and exponential eth lower bound. In *20th International Symposium on Parameterized and Exact Computation (IPEC 2025)*, volume 358 of *Leibniz International Proceedings in Informatics*, pages 17:1–17:10, 2025.

[Bod93] Hans L. Bodlaender. A tourist guide through treewidth. *Acta Cybernetica*, 11(1–2):1–21, 1993.

[Bod06] Hans L. Bodlaender. Treewidth: Characterizations, applications, and computations. In *Graph-Theoretic Concepts in Computer Science, 32nd International Workshop (WG 2006)*, volume 4271 of *Lecture Notes in Computer Science*, pages 1–14. Springer, 2006.

[Bon25] Édouard Bonnet. Treewidth inapproximability and tight ETH lower bound. In *Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC 2025)*, pages 2130–2135. Association for Computing Machinery, 2025. Section and lemma references use the full version, J. ACM (2026), arXiv:2406.11628v2.

[BS22] Johannes Blum and Sabine Storandt. Customizable hub labeling: Properties and algorithms. In *Computing and Combinatorics (COCOON 2022)*, volume 13595 of *Lecture Notes in Computer Science*, pages 345–356. Springer, 2022. Lemma and corollary references are to the full version, https://arxiv.org/abs/2208.08709.

[BSS09] Joshua D. Batson, Daniel A. Spielman, and Nikhil Srivastava. Twice-ramanujan sparsifiers. In *Proceedings of the 41st Annual ACM Symposium on Theory of Computing (STOC 2009)*, pages 255–262. Association for Computing Machinery, 2009. Theorem references follow the full version, arXiv:0808.0163.

[CC17] Moses Charikar and Vaggos Chatziafratis. Approximate hierarchical clustering via sparsest cut and spreading metrics. In *Proceedings of the 28th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2017)*, pages 841–854, 2017. Full version: arXiv:1609.09548.

[Das16] Sanjoy Dasgupta. A cost function for similarity-based hierarchical clustering. In *Proceedings of the 48th Annual ACM Symposium on Theory of Computing (STOC 2016)*, pages 118–127, 2016. Full version: arXiv:1510.05043.

[Die17] Reinhard Diestel. *Graph Theory*, volume 173 of *Graduate Texts in Mathematics*. Springer, 5th edition, 2017.

[DSSV25] Jona Dirks, Nicole Schirrmacher, Sebastian Siebertz, and Alexandre Vigny. Weighted treedepth is NP-complete on graphs of bounded degree. arXiv:2510.18584v2, 2025. Version of 4 February 2026.

[FGP18] Wolfgang Fischl, Georg Gottlob, and Reinhard Pichler. General and fractional hypertree decompositions: Hard and easy cases. In *Proceedings of the 37th ACM SIGMOD-SIGACT-SIGAI Symposium on Principles of Database Systems (PODS 2018)*, pages 17–32. ACM, 2018. Definition numbers refer to the full version, arXiv:2002.05239, a significantly extended version published in JACM 68(5), 2021.

[FHL05] Uriel Feige, Mohammad Taghi Hajiaghayi, and James R. Lee. Improved approximation algorithms for minimum-weight vertex separators. In *Proceedings of the 37th Annual ACM Symposium on Theory of Computing (STOC 2005)*, pages 563–572. Association for Computing Machinery, 2005. Full version: SIAM Journal on Computing 38(2):629–657 (2008).

[GHK+24] Tatsuya Gima, Tesshu Hanaka, Yasuaki Kobayashi, Ryota Murai, Hirotaka Ono, and Yota Otachi. Structural parameterizations of vertex integrity. In *WALCOM: Algorithms and Computation, 18th International Conference and Workshops, WALCOM 2024*, volume 14549 of *Lecture Notes in Computer Science*, pages 406–420. Springer, 2024. Theorem and corollary references use the expanded version, Theor. Comput. Sci. 1024 (2025), article 114954. Full text: https://arxiv.org/abs/2311.05892v3.

[Gru08] Hermann Gruber. Digraph complexity measures and applications in formal language theory. In *Proceedings of the 4th Workshop on Mathematical and Engineering Methods in Computer Science (MEMICS 2008)*, pages 60–67, 2008. Full version: Discrete Math. Theor. Comput. Sci. 14(2):189–204 (2012), DOI 10.46298/dmtcs.583. Theorem references use this version.

[Hås97] Johan Håstad. Some optimal inapproximability results. In *Proceedings of the Twenty-Ninth Annual ACM Symposium on Theory of Computing (STOC 1997)*, pages 1–10. Association for Computing Machinery, 1997.

[HBB+21] Svein Høgemo, Benjamin Bergougnoux, Ulrik Brandes, Christophe Paul, and Jan Arne Telle. On Dasgupta's hierarchical clustering objective and its relation to other graph parameters. In *Fundamentals of Computation Theory (FCT 2021)*, volume 12867 of *Lecture Notes in Computer Science*, pages 287–300, 2021. Theorem references are to the full version, https://arxiv.org/abs/2105.12093.

[HLW06] Shlomo Hoory, Nathan Linial, and Avi Wigderson. Expander graphs and their applications. *Bulletin of the American Mathematical Society*, 43(4):439–561, 2006.

[HPT20] Svein Høgemo, Christophe Paul, and Jan Arne Telle. Hierarchical clusterings of unweighted graphs. In *45th International Symposium on Mathematical Foundations of Computer Science (MFCS 2020)*, volume 170 of *Leibniz International Proceedings in Informatics*, pages 47:1–47:13, 2020.

[HSS03] Elad Hazan, Shmuel Safra, and Oded Schwartz. On the complexity of approximating k-dimensional matching. In *Approximation, Randomization, and Combinatorial Optimization: Algorithms and Techniques, APPROX 2003 and RANDOM 2003*, volume 2764 of *Lecture Notes in Computer Science*, pages 83–97. Springer, 2003. Theorem numbers refer to the full version, On the Hardness of Approximating k-Dimensional Matching, ECCC TR03-020 (2003), https://eccc.weizmann.ac.il/report/2003/020/.

[KR03] Subhash Khot and Oded Regev. Vertex cover might be hard to approximate to within $2-\varepsilon$. In *18th IEEE Annual Conference on Computational Complexity (CCC 2003)*, pages 379–386. IEEE, 2003. Theorem references use the full version, J. Comput. Syst. Sci. 74(3):335–349 (2008).

[LY96] T. W. Lam and F. L. Yue. The NP-completeness of edge ranking. In *International Computer Symposium (ICS 1996)*, 1996. Expanded journal version: Discrete Applied Mathematics 85(1):71–86 (1998). DBLP record: https://dblp.org/rec/journals/dam/LamY98.

[Mar09] Dániel Marx. Approximating fractional hypertree width. In *Proceedings of the Twentieth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2009)*, pages 902–911. Society for Industrial and Applied Mathematics, 2009. Theorem references use the expanded ACM Transactions on Algorithms version, 6(2):29:1–29:17 (2010). Full text: https://www.cs.bme.hu/ dmarx/papers/marx-fractional-talg.pdf.

[NOdM06] Jaroslav Nešetřil and Patrice Ossona de Mendez. Tree-depth, subgraph coloring and homomorphism bounds. *European Journal of Combinatorics*, 27(6):1022–1041, 2006.

[NOdM15] Jaroslav Nešetřil and Patrice Ossona de Mendez. On low tree-depth decompositions. *Graphs and Combinatorics*, 31(6):1941–1963, 2015.

[RRSVS14] Felix Reidl, Peter Rossmanith, Fernando Sánchez Villaamil, and Somnath Sikdar. A faster parameterized algorithm for treedepth. In *Automata, Languages, and Programming (ICALP 2014)*, volume 8572 of *Lecture Notes in Computer Science*, pages 931–942. Springer, 2014. Full version: arXiv:1401.7540.

[RST12] Prasad Raghavendra, David Steurer, and Madhur Tulsiani. Reductions between expansion problems. In *Proceedings of the 27th Conference on Computational Complexity (CCC 2012)*, pages 64–73, 2012. Theorem and section numbers refer to the full version, https://arxiv.org/abs/1011.2586.

[SD26] Michał Szyfelbein and Dariusz Dereniowski. Hierarchical $\mathcal{F}$-clustering: Approximation and hardness of clustering into trees and bounded diameter graphs. arXiv:2607.13217v1, 2026. Full text: https://arxiv.org/abs/2607.13217v1.

[SSV25] Nicole Schirrmacher, Sebastian Siebertz, and Alexandre Vigny. Elimination distance to dominated clusters. In *50th International Symposium on Mathematical Foundations of Computer Science (MFCS 2025)*, volume 345 of *Leibniz International Proceedings in Informatics*, pages 90:1–90:20. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025.

[Szy26a] Michał Szyfelbein. Approximating the average-case graph search problem with non-uniform costs. In *WALCOM: Algorithms and Computation, 20th International Conference and Workshops, WALCOM 2026*, volume 16444 of *Lecture Notes in Computer Science*, pages 96–109. Springer, 2026. Full version: arXiv:2511.06564.

[Szy26b] Michał Szyfelbein. Average case graph searching in non-uniform cost models. arXiv:2603.17916v1, 2026.

[vE17] Martijn van Ee. Some notes on bounded starwidth graphs. *Information Processing Letters*, 125:9–14, 2017.

[WAPL15] Yu (Ledell) Wu, Per Austrin, Toniann Pitassi, and David Liu. Inapproximability of treewidth and related problems (extended abstract). In *Proceedings of the 24th International Joint Conference on Artificial Intelligence (IJCAI 2015)*, pages 4222–4228, 2015. Extended abstract of J. Artif. Intell. Res. 49:569–600 (2014). Theorem and section references use the full version at https://www.cs.toronto.edu/ toni/Papers/treewidth.pdf.

[Yan81] Mihalis Yannakakis. Computing the minimum fill-in is NP-complete. *SIAM Journal on Algebraic Discrete Methods*, 2(1):77–79, 1981.

A Source promises, normalizations, and provenance

The source derivations follow the order in which the paper uses them: the bounded-degree independent-set promise, the multipartite UGC promise, the global SSE promise, and the NP-hardness provenance and parity normalization.

A.1 Degree reduction for the UGC independent-set source

We derive Theorem 3.1 from the weighted density theorem of Austrin, Khot, and Safra. Their bounded-degree reduction uses random sampling [AKS09, Section 4, Steps 2–3]; we obtain bounded degree deterministically through loop removal, vertex unweighting, and a rational implementation of the BSS sparsifier, given in Lemma A.7.

A.1.1 The AKS density theorem

A bipartite *Unique Games instance* $\Lambda = (U, V, E_\Lambda, \Pi, [\ell])$ consists of a bipartite multigraph with left vertices U, right vertices V, and nonempty edge set E_Λ, a label set $[\ell] = \{1, \dots, \ell\}$ for a positive integer ℓ, and a family $\Pi = (\pi_e)_{e \in E_\Lambda}$ of bijections $\pi_e : [\ell] \to [\ell]$. For an edge $e = uv$, we always take $u \in U$ and $v \in V$, so π_e maps labels at u to labels at v. A *labeling* is a map $\lambda : U \cup V \to [\ell]$. It *satisfies* the constraint on $e = uv$ precisely when

$$\lambda(v) = \pi_e(\lambda(u)), \qquad \text{equivalently} \qquad \lambda(u) = \pi_e^{-1}(\lambda(v)).$$

Thus either endpoint's label uniquely determines the label required at the other endpoint. Let $S_\lambda = \{e = uv \in E_\Lambda : \lambda(v) = \pi_e(\lambda(u))\}$. The optimum satisfied fraction is

$$\mathrm{OPT}(\Lambda) = \max_{\lambda : U \cup V \to [\ell]} \frac{|S_\lambda|}{|E_\Lambda|}.$$

All constraints have equal weight; parallel constraints are counted separately and may have different permutations. Write $\mathrm{VOPT}(\Lambda)$ for the largest fraction $|V_0|/|V|$ such that a single labeling satisfies every constraint incident to $V_0 \subseteq V$. For $z \in U \cup V$, put $d_\Lambda(z) = |\{e \in E_\Lambda : z \text{ is an endpoint of } e\}|$, counting parallel constraints separately. The instance is *regular* if there are positive integers d_U, d_V such that

$$d_\Lambda(u) = d_U \quad (u \in U), \qquad d_\Lambda(v) = d_V \quad (v \in V).$$

The two degrees may differ.

Theorem A.1 (Regular Unique Games source [AKS09, Conjecture 2.4] and [KR03])**.** *For every fixed $0 < \gamma < 1/2$, there is a fixed alphabet size ℓ such that it is UGC-hard to distinguish regular bipartite Unique Games instances Λ with alphabet $[\ell]$ satisfying the following cases.*

YES: $\mathrm{VOPT}(\Lambda) \geq 1 - \gamma$. *There is a set $V_0 \subseteq V$ of at least $(1-\gamma)|V|$ right vertices and one labeling satisfying every constraint incident to V_0.*

NO: $\mathrm{OPT}(\Lambda) \leq \gamma$. *Every labeling satisfies at most a γ fraction of the constraints.*

For a vertex- and edge-weighted graph G, let a_v denote a vertex weight and w_e an edge weight, normalized by

$$\sum_{v \in V(G)} a_v = 1, \qquad \sum_{e \in E(G)} w_e = 1.$$

For $S \subseteq V(G)$, put

$$a(S) = \sum_{v \in S} a_v, \qquad e_G(S) = \sum_{e \in E(G[S])} w_e, \qquad \deg_G(v) = \sum_{e \in E(G)} m_v(e) w_e,$$

where $m_v(e)$ is the number of endpoints of e equal to v. Thus a loop is counted once in $e_G(S)$ and twice in $\deg_G(v)$. For $\beta \in [0,1]$ and a real δ, the graph is (β, δ)-*dense* if $a(S) \geq \beta$ implies $e_G(S) \geq \delta$.

Let Ψ be the standard normal cumulative distribution function. For $-1 < \rho < 1$, let X, Y be jointly normal with mean zero, variance one, and covariance ρ, and define

$$\Gamma_\rho(s) = \Pr[X \leq \Psi^{-1}(s),\ Y \leq \Psi^{-1}(s)] \qquad (0 < s < 1).$$

Set $\Gamma_\rho(0) = 0$ and $\Gamma_\rho(1) = 1$. For a bias $q \in (0, 1/2)$, put $\rho = -q/(1-q)$.

Theorem A.2 (AKS density theorem [AKS09, Theorem 3.1])**.** *For every fixed bias $q \in (0, 1/2)$ and error tolerance $\varepsilon_{\mathrm{dens}} > 0$, there is a constant $\gamma_0 = \gamma_0(q, \varepsilon_{\mathrm{dens}}) > 0$ such that, for every $0 < \gamma < \gamma_0$, an algorithm maps a Unique Games instance Λ to a vertex- and edge-weighted graph G with the following properties.*

YES: *If $\mathrm{VOPT}(\Lambda) \geq 1 - \gamma$, then G has an independent set I with $a(I) \geq q - \gamma$.*

NO: *If $\mathrm{OPT}(\Lambda) \leq \gamma$ and Λ is regular, then G is $(\beta, \Gamma_\rho(\beta) - \varepsilon_{\mathrm{dens}})$-dense for every $\beta \in [0,1]$.*

For regular Λ, the weights satisfy $\deg_G(v) = 2a_v$ for every $v \in V(G)$.

The additive NO error $\varepsilon_{\mathrm{dens}}$ and bias q are fixed before the Unique Games promise parameter γ. The running time in Theorem 3.1 of [AKS09] is polynomial in $|U|, |V|, |E_\Lambda|$ and exponential in ℓ. In our application $q, \varepsilon_{\mathrm{dens}}, \gamma$, and ℓ are fixed independently of the input size, so the reduction is polynomial in $|U| + |V| + |E_\Lambda|$.

The factor two agrees with the normalizations, since

$$\sum_{v \in V(G)} \deg_G(v) = 2 \sum_{e \in E(G)} w_e = 2 \sum_{v \in V(G)} a_v = 2.$$

For regular input, each endpoint marginal of the AKS edge distribution is a_v, with a loop contributing twice. Thus the NO conclusion, applied with $\beta = a(S)$, is

$$e_G(S) \overset{\text{Thm. A.2}}{\geq} \Gamma_\rho(a(S)) - \varepsilon_{\mathrm{dens}} \qquad (S \subseteq V(G)). \tag{122}$$

We will use positivity, monotonicity, and convexity of Γ_ρ. For completeness, these follow directly from its definition. The joint normal density is positive, so $\Gamma_\rho(s) > 0$ for $0 < s < 1$. Differentiating the bivariate distribution function along its diagonal, using symmetry of X, Y and the conditional normal law, gives

$$\Gamma_\rho'(s) \overset{\text{conditional normal law}}{=} 2\Psi\left(\sqrt{\frac{1-\rho}{1+\rho}}\,\Psi^{-1}(s)\right). \tag{123}$$

The right-hand side is nonnegative and increasing in s. Consequently Γ_ρ is nondecreasing and convex, including at the endpoints by continuity.

A.1.2 Removing loops and vertex weights

Fix a rational bias q and a loop-loss budget $\varepsilon_{\mathrm{loop}} > 0$. We remove AKS loops before vertex unweighting so that the density argument can select distinct source vertices independently. The next lemma pads the Unique Games alphabet, applies the unchanged AKS construction, and deletes its loops. Padding bounds the lost edge weight.

Lemma A.3 (Removing loops by alphabet padding)**.** *Fix $\varepsilon_{\mathrm{dens}}, \varepsilon_{\mathrm{loop}} > 0$ and $0 < \gamma < \min\{1/2, \gamma_0(q, \varepsilon_{\mathrm{dens}})\}$. Let Λ be a regular instance satisfying the promise of Theorem A.1. A fixed alphabet padding, the AKS construction, and deletion of all loops give a loopless graph G^- with vertex weights a_v and edge weights w_e satisfying*

$$\sum_{v \in V(G^-)} a_v = 1, \qquad \sum_{e \in E(G^-)} w_e \le 1, \qquad \deg_{G^-}(v) \le 2a_v \quad (v \in V(G^-)).$$

In the YES case there is an independent set I with $a(I) \ge q - \gamma$. In the NO case,

$$e_{G^-}(S) \ge \Gamma_\rho(a(S)) - \varepsilon_{\mathrm{dens}} - \varepsilon_{\mathrm{loop}} \qquad (S \subseteq V(G^-)). \tag{124}$$

The reduction is deterministic and polynomial for fixed parameters.

Proof. Put $c = q^2 + (1-q)^2 \in (0,1)$. Choose a positive integer t such that

$$c^t < \varepsilon_{\mathrm{loop}}. \tag{125}$$

Such a fixed t exists since $c < 1$. Define $\Lambda^{[t]}$ on the same sets U, V, E_Λ, with alphabet $[\ell] \times [t]$ and constraint $\pi_e^{[t]}(i,j) = (\pi_e(i), j)$ on each edge e. A padded constraint is satisfied exactly when the first-coordinate labels satisfy the original constraint and the second-coordinate labels agree. Projecting a padded labeling to its first coordinate, and extending an original labeling with one common second coordinate, give

$$\mathrm{OPT}(\Lambda^{[t]}) = \mathrm{OPT}(\Lambda), \qquad \mathrm{VOPT}(\Lambda^{[t]}) = \mathrm{VOPT}(\Lambda).$$

The source graph and its regularity are unchanged. Apply Theorem A.2 to $\Lambda^{[t]}$ with the same parameters $q, \rho, \varepsilon_{\mathrm{dens}}, \gamma$: padding changes the input alphabet and constraints, not the AKS construction. Call its weighted output G. Delete its loops to obtain G^-, retaining every vertex, every vertex weight, and every nonloop edge weight. The remaining edge weights are not renormalized.

Following the sampling idea in [AKS09, proof of Theorem 3.1], we specify the padded output's weights explicitly. Its vertex set is $W = V \times \{0,1\}^{[\ell]\times[t]}$. For $u \in U$, put $E_\Lambda(u) = \{e = uv \in E_\Lambda : v \in V\}$. Choose a random triple $(\mathbf{u}, \mathbf{e}_1, \mathbf{e}_2)$. For $u \in U$ and $e_1, e_2 \in E_\Lambda(u)$, define its outcome event and prescribe its probability by

$$\mathcal{C}_{u,e_1,e_2} = \{(\mathbf{u}, \mathbf{e}_1, \mathbf{e}_2) = (u, e_1, e_2)\},$$
$$\Pr[\mathcal{C}_{u,e_1,e_2}] = \frac{1}{|U| d_\Lambda(u)^2}.$$

Thus parallel constraints are separate choices, and the two constraints may coincide. Independently of this triple, sample strings $X, Y \in \{0,1\}^{[\ell]\times[t]}$ whose coordinate pairs are independent and, for every $k \in [\ell] \times [t]$, have distribution

$$\Pr[X_k = 0, Y_k = 0] = 1 - 2q, \quad \Pr[X_k = 1, Y_k = 1] = 0,$$
$$\Pr[X_k = 0, Y_k = 1] = q, \qquad \Pr[X_k = 1, Y_k = 0] = q.$$

For a string $x = (x_k)_{k\in[\ell]\times[t]}$ and a bijection $\sigma : [\ell]\times[t] \to [\ell]\times[t]$, define the *reindexed string*

$$x \circ \sigma := (x_{\sigma(k)})_{k\in[\ell]\times[t]}.$$

Here $\circ$ denotes function composition after viewing the string x as the map $k \mapsto x_k$: coordinate k of the new string receives the bit at coordinate $\sigma(k)$ of the original string. Writing $\mathbf{e}_i = (\mathbf{u}, \mathbf{v}_i)$ for $i \in \{1,2\}$, define the random endpoints by

$$Z_1 = (\mathbf{v}_1, X \circ (\pi^{[t]}_{\mathbf{e}_1})^{-1}), \qquad Z_2 = (\mathbf{v}_2, Y \circ (\pi^{[t]}_{\mathbf{e}_2})^{-1}).$$

Their ordered-pair probabilities are

$$P(z,z') = \Pr[Z_1 = z,\ Z_2 = z'] \qquad (z, z' \in W).$$

Assign vertex weights by

$$a_{(v,x)} = \frac{1}{|V|} \prod_{k\in[\ell]\times[t]} q^{x_k}(1-q)^{1-x_k} \qquad ((v,x) \in W).$$

For $v \in V$ and $i \in \{1,2\}$, regularity gives

$$\Pr[\mathbf{v}_i = v] \overset{\text{triple distribution}}{=} \sum_{e\in E_\Lambda:\, e=uv} \frac{1}{|U|d_U} = \frac{d_V}{|U|d_U} \overset{|U|d_U=|V|d_V}{=} \frac{1}{|V|}.$$

Conditional on each sampled constraint, reindexing preserves the same product bit distribution. Hence $\Pr[Z_1 = z] = \Pr[Z_2 = z] = a_z$ for $z \in W$. For an unordered pair zz' (allowing $z = z'$), assign edge weight

$$w_{zz'} = \begin{cases} P(z,z') + P(z',z), & z \neq z', \\ P(z,z), & z = z', \end{cases} \qquad E(G) = \{zz' : z, z' \in W,\ w_{zz'} > 0\}.$$

These are the AKS weights for $\Lambda^{[t]}$; the only subsequent change is deletion of the loops, without renormalization. In particular, for every $z \in W$,

$$\deg_G(z) = \sum_{z'\in W\setminus\{z\}} \big(P(z,z') + P(z',z)\big) + 2P(z,z) \overset{\text{endpoint marginals}}{=} 2a_z.$$

The sampler specifies exact weights: the reduction enumerates its outcomes rather than drawing an edge. Indeed, if $q = a_0/a_1$ and $R = \ell t$, there are $|U|d_U^2 3^R$ nonzero outcomes, whose probabilities have common denominator $|U|d_U^2 a_1^R$. Summing these rational probabilities for each unordered endpoint pair computes all weights deterministically, in polynomial time and with polynomial bit length for fixed q, R.

Fix $u \in U$ and $e_1, e_2 \in E_\Lambda(u)$, and abbreviate $\mathcal{C} = \mathcal{C}_{u,e_1,e_2}$. Put $i_s = \pi_{e_s}^{-1}(1)$ for $s \in \{1,2\}$. Padding gives

$$(\pi^{[t]}_{e_s})^{-1}(1,j) = (i_s, j) \qquad (s \in \{1,2\},\ j \in [t]).$$

Thus, conditional on $\mathcal{C}$, the bits at coordinate $(1,j)$ of the two endpoint strings are $X_{(i_1,j)}$ and $Y_{(i_2,j)}$. Define their agreement event by

$$A_j = \{X_{(i_1,j)} = Y_{(i_2,j)}\} \qquad (j \in [t]).$$

When $i_1 = i_2$, the displayed bit-pair distribution gives agreement probability $(1-2q)+0$. When $i_1 \neq i_2$, the two bits belong to distinct independent coordinate pairs and each equals one with probability q. Hence

$$\Pr[A_j \mid \mathcal{C}] = \begin{cases} 1-2q, & i_1 = i_2, \\ (1-q)^2 + q^2 = c, & i_1 \neq i_2 \end{cases} \leq c, \tag{126}$$

where $c-(1-2q) = 2q^2 > 0$. A loop requires agreement at every coordinate $(1, j)$, so it implies all events A_j. Conditional on $\mathcal{C}$, these events depend on disjoint sets of the independent coordinate pairs (X_k, Y_k): event A_j uses only indices (i_1, j) and (i_2, j). Consequently

$$\begin{aligned} \Pr[Z_1 = Z_2 \mid \mathcal{C}] &\leq \Pr\left[\bigcap_{j=1}^{t} A_j \,\middle|\, \mathcal{C}\right] \\ &\overset{\text{independence}}{=} \prod_{j=1}^{t} \Pr[A_j \mid \mathcal{C}] \overset{(126)}{\leq} c^t. \end{aligned} \tag{127}$$

This bound holds for every sampled triple. Averaging over the triple, and using $w_{zz} = P(z, z)$, gives, for the set L of loops,

$$\sum_{e \in L} w_e = \Pr[Z_1 = Z_2] \overset{(127)}{\leq} c^t. \tag{128}$$

Thus, in the NO case, every $S \subseteq V(G)$ satisfies

$$\begin{aligned} e_{G^-}(S) &= e_G(S) - \sum_{e \in L \cap E(G[S])} w_e \\ &\overset{(122),\,(128)}{\geq} \Gamma_\rho(a(S)) - \varepsilon_{\text{dens}} - c^t \\ &\overset{(125)}{\geq} \Gamma_\rho(a(S)) - \varepsilon_{\text{dens}} - \varepsilon_{\text{loop}}. \end{aligned}$$

Deleting loops preserves the YES independent set and can only decrease weighted degrees. The normalization and degree identity in Theorem A.2 therefore give the displayed bounds. The padded alphabet size $R = \ell t$ is fixed, so the AKS runtime remains polynomial. Alphabet padding and loop deletion are deterministic. □

A.1.3 Vertex unweighting and uniform degree bounds

Write the padded alphabet size as $R = \ell t$ and the fixed bias as $q = a_0/a_1$, with positive integers $a_0 < a_1$. For a binary string $x \in \{0,1\}^R$, put $|x| = \sum_{k=1}^{R} x_k$. The vertex weights in the AKS construction [AKS09, proof of Theorem 3.1], indexed by right vertices $v \in V$ and strings x, have the form

$$a_{(v,x)} \overset{\text{[AKS09, proof of Thm. 3.1]}}{=} \frac{a_0^{|x|}(a_1 - a_0)^{R-|x|}}{|V| a_1^R}, \qquad v \in V, \quad x \in \{0,1\}^R.$$

Thus $N_{\text{cl}} = |V| a_1^R$ is polynomial in the Unique Games instance, since a_1 and R are fixed. The integers $m_u = N_{\text{cl}} a_u$ are positive and satisfy $\sum_{u \in V(G^-)} m_u = N_{\text{cl}}$. We use the clone-replication operation of [AKS09, Section 4, Step 1] with these exact multiplicities: its preliminary rounding is omitted, and edge weights are retained instead of being converted to parallel unit edges. Replace each original vertex u by an independent *clone block*

$$C_u = \{(u, i) : i \in [m_u]\}.$$

For each pair uv joined in G^-, combine parallel edge weights as $w_{uv} = \sum_{e \in E(G^-):\, e=uv} w_e$. Define the clone graph G_1 by

$$\begin{aligned} V(G_1) &= \bigcup_{u \in V(G^-)} C_u, \\ E(G_1) &= \{(u,i)(v,j) : uv \in E(G^-),\ i \in [m_u],\ j \in [m_v]\}, \\ a_{(u,i)} &= \frac{1}{N_{\mathrm{cl}}}, \qquad w_{(u,i)(v,j)} = \frac{w_{uv}}{m_u m_v}. \end{aligned}$$

Thus G_1 is edge-weighted and vertex-unweighted. Its weights satisfy

$$|V(G_1)| = N_{\mathrm{cl}}, \qquad \sum_{e \in E(G_1)} w_e \le 1, \qquad \deg_{G_1}(v) \le \frac{2}{N_{\mathrm{cl}}} \quad (v \in V(G_1)).$$

For every clone (u,i),

$$\deg_{G_1}((u,i)) = \frac{\deg_{G^-}(u)}{m_u} \overset{\text{Lem. A.3}}{\le} \frac{2a_u}{N_{\mathrm{cl}} a_u} = \frac{2}{N_{\mathrm{cl}}}. \tag{129}$$

Each source edge retains its total weight. For a YES independent set $I \subseteq V(G^-)$, the independent set $I_1 = \bigcup_{u \in I} C_u$ satisfies

$$\frac{|I_1|}{N_{\mathrm{cl}}} = \sum_{u \in I} \frac{m_u}{N_{\mathrm{cl}}} = a(I).$$

Lemma A.4 (Density after vertex unweighting [AKS09, Section 4, Step 1])**.** *If G^- satisfies* (124)*, then its clone graph G_1 satisfies, for every $S' \subseteq V(G_1)$,*

$$e_{G_1}(S') \ge \Gamma_\rho(|S'|/N_{\mathrm{cl}}) - \varepsilon_{\mathrm{dens}} - \varepsilon_{\mathrm{loop}}. \tag{130}$$

Proof. For $S' \subseteq V(G_1)$, put

$$f_u = \frac{|S' \cap C_u|}{m_u} \qquad (u \in V(G^-)).$$

Form $S \subseteq V(G^-)$ by independent choices with $\Pr[u \in S] = f_u$ for every $u \in V(G^-)$. Loop removal in Lemma A.3 makes the endpoints of each edge distinct, so their selection events are independent. For an edge uv of weight w_{uv}, the clone pairs in S' contribute $(m_u f_u)(m_v f_v) w_{uv}/(m_u m_v) = w_{uv} f_u f_v$. Since $a_u = m_u/N_{\mathrm{cl}}$, the expected source weight and internal edge weight are

$$\begin{aligned} \mathbb{E}[a(S)] &\overset{\text{linearity}}{=} \sum_{u \in V(G^-)} a_u f_u = \frac{|S'|}{N_{\mathrm{cl}}}, \\ \mathbb{E}[e_{G^-}(S)] &\overset{\text{independence}}{=} \sum_{uv \in E(G^-)} w_{uv} f_u f_v = e_{G_1}(S'). \end{aligned} \tag{131}$$

Applying the source density bound and then Jensen's inequality for the convex function Γ_ρ gives

$$\begin{aligned} e_{G_1}(S') &\overset{(131)}{=} \mathbb{E}[e_{G^-}(S)] \\ &\overset{(124)}{\ge} \mathbb{E}[\Gamma_\rho(a(S))] - \varepsilon_{\mathrm{dens}} - \varepsilon_{\mathrm{loop}} \\ &\overset{\text{Jensen}}{\ge} \Gamma_\rho(\mathbb{E}[a(S)]) - \varepsilon_{\mathrm{dens}} - \varepsilon_{\mathrm{loop}} \\ &\overset{(131)}{=} \Gamma_\rho(|S'|/N_{\mathrm{cl}}) - \varepsilon_{\mathrm{dens}} - \varepsilon_{\mathrm{loop}}. \end{aligned}$$

□

For the normalized promise, fix a rational $0 < \eta < 1/10$. Let N denote the number of output vertices. The positive rational constant ζ_η below may depend on η.

Lemma A.5 (Normalized density promise)**.** *There is a choice of ζ_η for which it is UGC-hard to distinguish simple edge-weighted graphs G on N vertices, with positive rational edge weights satisfying*

$$\sum_{e \in E(G)} w_e \le 1, \qquad \deg_G(v) \le \frac{2}{N} \quad (v \in V(G)),$$

and the following promise.

YES: *There is an independent set of size at least $(1/2 - \eta)N$.*

NO: *Every $S \subseteq V(G)$ with $|S| \ge \eta N$ satisfies $e_G(S) \ge \zeta_\eta$.*

The reduction is deterministic and polynomial time.

Proof. Choose the fixed parameters in this order:

1. Set $q = 1/2 - \eta/4$ and $\rho = -q/(1-q)$.
2. Choose a positive rational g with $4g < \Gamma_\rho(\eta)$, and set $\varepsilon_{\mathrm{dens}} = \varepsilon_{\mathrm{loop}} = g$.
3. Choose $0 < \gamma < \min\{\gamma_0(q, g), \eta/4\}$ and take the source alphabet size ℓ from Theorem A.1. With $c = q^2 + (1-q)^2$, choose an integer $t \ge 1$ with $c^t < g$ as in (125), and put $R = \ell t$.

These parameters are fixed before the input size varies. Apply the regular source of Theorem A.1, then the loop removal and clone construction above. The YES independent-set density from Lemma A.3 is at least $q - \gamma \ge 1/2 - \eta/2 \ge 1/2 - \eta$. For $|S| \ge \eta N$, equation (130) and monotonicity give

$$\begin{aligned} e_{G_1}(S) &\overset{(130)}{\ge} \Gamma_\rho(|S|/N) - 2g \\ &\overset{(123)}{\ge} \Gamma_\rho(\eta) - 2g > 2g > g. \end{aligned}$$

Thus we may take $\zeta_\eta = g$.

The clone graph is simple after parallel edge weights are combined. Each source edge retains its total weight, and equation (129) supplies the degree bound. The exact sampler enumeration above gives rational edge weights of polynomial bit length. Each clone multiplicity is at most a_1^R, so dividing an edge weight by $m_u m_v$ adds only a fixed number of bits. For fixed η, the alphabet sizes, bias denominator, and padding factor are constants, so the displayed clone count and all edge descriptions are polynomial. The reduction is deterministic. □

A.1.4 Deterministic sparsification of a dense-set promise

Spectral sparsification bounds the support size. Discarding edges below a weight threshold then bounds ordinary degree: the support size controls the discarded weight, and the threshold converts the weighted degree bound to an ordinary one. We show that every large vertex set retains positive internal weight.

Use the notation of Section 4: $w(F) = \sum_{e \in F} w_e$ and $\mathrm{vol}_G(S) = \sum_{v \in S} \deg_G(v)$. Write $\mathbf{1}_S \in \mathbb{R}^{V(G)}$ for the indicator vector of S. For a loopless weighted graph G, let w_{uv} be the total weight of edges

between distinct vertices u, v. Its *Laplacian* is the symmetric matrix with $L_G(u,u) = \deg_G(u)$ and $L_G(u,v) = -w_{uv}$ for $u \neq v$. For $x \in \mathbb{R}^{V(G)}$, expanding the quadratic form gives

$$x^{\mathsf{T}} L_G x = \sum_{e=uv \in E(G)} w_e (x_u - x_v)^2. \tag{132}$$

For $S \subseteq V(G)$, the quadratic form counts boundary edge weight, so

$$\begin{aligned} \mathbf{1}_S^{\mathsf{T}} L_G \mathbf{1}_S &\overset{(132)}{=} w(\delta_G(S)), \\ \mathbf{1}_{\{v\}}^{\mathsf{T}} L_G \mathbf{1}_{\{v\}} &\overset{(132)}{=} \deg_G(v). \end{aligned} \tag{133}$$

The notation $L_G \preceq L_H$ means that $x^{\mathsf{T}} L_G x \leq x^{\mathsf{T}} L_H x$ for every vector x. A *spectral sparsifier* is a weighted subgraph with fewer edges whose Laplacian preserves these quadratic forms within multiplicative bounds. An internal edge contributes twice to the volume, while a crossing edge contributes once to both the volume and the cut. Hence

$$2e_G(S) = \mathrm{vol}_G(S) - w(\delta_G(S)). \tag{134}$$

Theorem A.6 (BSS sparsification [BSS09, Theorem 1.1])**.** *For every $d > 1$, a loopless undirected weighted graph G on $N \geq 1$ vertices, with positive edge weights, has a reweighted spanning subgraph G' with at most $\lceil d(N-1) \rceil$ edges such that*

$$L_G \preceq L_{G'} \preceq \frac{d + 1 + 2\sqrt{d}}{d + 1 - 2\sqrt{d}} L_G.$$

For fixed d, the construction is deterministic and polynomial time.

The BSS barrier argument also gives the following version with rational output and an explicit bound on bit complexity. We use the barrier lemmas of [BSS09, Lemmas 3.3–3.5], leaving a strict interval for each update and rounding within that interval. This small loss in the spectral factor makes the subsequent edge threshold an exact rational comparison.

Lemma A.7 (Rational BSS implementation)**.** *Fix a rational $0 < \theta \leq 1/12$, and put $d = \lceil \theta^{-2} \rceil$. Given a loopless graph G on $N \geq 1$ vertices with nonnegative rational edge weights, a deterministic algorithm produces a rationally weighted spanning subgraph H with at most dN edges and*

$$L_G \preceq L_H \preceq (1 + 7\theta) L_G.$$

The running time and output bit length are polynomial in the binary input size, for fixed θ.

Proof. Discard zero-weight edges, combine parallel edge weights, and retain isolated vertices. In each nontrivial connected component, delete one row and the corresponding column of its Laplacian. Let B be the block diagonal matrix of these reduced Laplacians, and let r be its dimension. If $r = 0$, output the edgeless graph on the same vertices. Otherwise B is rational and positive definite. Number the edges by $[m]$, choose an orientation of each edge, and let b_i be its incidence vector with the deleted coordinates omitted. Then

$$B \overset{(132)}{=} \sum_{i=1}^{m} w_i b_i b_i^{\mathsf{T}}. \tag{135}$$

Inequalities for reduced Laplacians extend to full Laplacians: subtracting the coordinate of the deleted vertex from every coordinate in its component leaves every edge difference unchanged.

For analysis only, define $v_i = \sqrt{w_i} B^{-1/2} b_i$ for $i \in [m]$. Equation (135) gives $\sum_{i=1}^m v_i v_i^{\mathsf{T}} = I_r$, so $\|v_i\| \le 1$. These vectors and the matrix square root will not be computed. The algorithm adds positive multiples of $v_i v_i^{\mathsf{T}}$, starting with $A_0 = 0$, while keeping its eigenvalues between two moving barriers. For a symmetric matrix A whose eigenvalues lie strictly between l and u, define the potentials

$$\Phi^u(A) = \operatorname{tr}((uI_r - A)^{-1}), \qquad \Phi_l(A) = \operatorname{tr}((A - lI_r)^{-1}).$$

Set

$$\delta_L = 1, \qquad \delta_U = \frac{1}{1-3\theta}, \qquad l_j = -\frac{r}{\theta} + j, \qquad u_j = \frac{r}{\theta} + j\delta_U \quad (0 \le j \le dr). \tag{136}$$

The initial potentials both equal θ. Our invariant at step j is

$$l_j I_r \prec A_j \prec u_j I_r, \qquad \Phi^{u_j}(A_j) \le \theta, \qquad \Phi_{l_j}(A_j) \le \theta. \tag{137}$$

Here $A \prec B$ means that $B - A$ is positive definite.

We specify the permissible update interval from the BSS barrier lemmas. For $0 \le j < dr$, put $u' = u_j + \delta_U$, $l' = l_j + 1$, and

$$D_U = \Phi^{u_j}(A_j) - \Phi^{u'}(A_j), \qquad D_L = \Phi_{l'}(A_j) - \Phi_{l_j}(A_j).$$

The lower potential bound implies $\lambda_{\min}(A_j) - l_j \ge 1/\theta > 1$, so l' remains below the spectrum and both differences are positive. For $i \in [m]$, define

$$\begin{aligned} U_j(v_i) &= \frac{v_i^{\mathsf{T}}(u'I_r - A_j)^{-2} v_i}{D_U} + v_i^{\mathsf{T}}(u'I_r - A_j)^{-1} v_i, \\ L_j(v_i) &= \frac{v_i^{\mathsf{T}}(A_j - l'I_r)^{-2} v_i}{D_L} - v_i^{\mathsf{T}}(A_j - l'I_r)^{-1} v_i. \end{aligned} \tag{138}$$

By [BSS09, Lemmas 3.3–3.4], an update $A_{j+1} = A_j + t v_i v_i^{\mathsf{T}}$ preserves (137) with the shifted barriers whenever $t > 0$ and $U_j(v_i) \le 1/t \le L_j(v_i)$. The sum estimates in [BSS09, proof of Lemma 3.5], applied with both potential bounds equal to θ, apply because $\delta_L \theta < 1$ and $1/\delta_U + \theta = 1 - 2\theta < 1 - \theta = 1/\delta_L - \theta$. They give a strict margin:

$$\begin{aligned} \sum_{i=1}^m L_j(v_i) - \sum_{i=1}^m U_j(v_i) &\overset{\text{[BSS09, Lem. 3.5]}}{\ge} (1/\delta_L - \theta) - (1/\delta_U + \theta) \\ &\overset{(136)}{=} \theta. \end{aligned} \tag{139}$$

Thus at least one edge has $L_j(v_i) - U_j(v_i) \ge \theta/m$. Choose the least such index i. We next show that rounding a midpoint update on a fixed dyadic mesh keeps its reciprocal within this interval.

Write $\lambda_1, \dots, \lambda_r$ for the eigenvalues of A_j. Each summand in either potential is positive, so

$$u_j - \lambda_a \overset{(137)}{\ge} 1/\theta, \qquad \lambda_a - l_j \overset{(137)}{\ge} 1/\theta \quad (a \in [r]). \tag{140}$$

Put $K = 2/\theta + 2d + 2$ and $C = 4K^2 + 1$, constants for fixed θ. Since $j < dr$, $r \ge 1$, and $1 \le \delta_U \le 4/3$, all positive denominator factors in the following sums are at most Kr:

$$\begin{aligned} D_U &\overset{\text{spectral decomposition}}{=} \delta_U \sum_{a=1}^r \frac{1}{(u_j - \lambda_a)(u' - \lambda_a)} \ge \frac{1}{K^2 r}, \\ D_L &\overset{\text{spectral decomposition}}{=} \sum_{a=1}^r \frac{1}{(\lambda_a - l')(\lambda_a - l_j)} \ge \frac{1}{K^2 r}. \end{aligned} \tag{141}$$

For example, $u' - \lambda_a < u' - l_j = 2r/\theta + j(\delta_U - 1) + \delta_U \le Kr$; the remaining factors are smaller. Equation (140) also bounds the operator norms of the upper and lower shifted inverses by θ and $\theta/(1-\theta) \le 2\theta$, respectively. Using $\|v_i\| \le 1$ in (138) therefore gives

$$\begin{aligned} 0 < U_j(v_i) &\overset{(141)}{\le} \theta^2 K^2 r + \theta \le Cr, \\ L_j(v_i) &\overset{(141)}{\le} 4\theta^2 K^2 r \le Cr. \end{aligned} \tag{142}$$

For the selected edge define $s = (L_j(v_i) + U_j(v_i))/2$ and $t = 1/s$. The margin and these upper bounds imply

$$\frac{\theta}{2m} \le s \le Cr, \qquad \frac{1}{Cr} \le t \le \frac{2m}{\theta}. \tag{143}$$

Moreover, s has distance at least $\theta/(2m)$ from both endpoints of the permissible reciprocal interval.

Choose the least nonnegative integer b for which the mesh $h = 2^{-b}$ satisfies

$$h \le \min\left\{ \frac{1}{2Cr}, \frac{\theta}{16mC^2r^2} \right\}. \tag{144}$$

This mesh is fixed for the entire run. Round t to the nearest multiple t' of h, breaking ties toward the smaller multiple. Then $|t' - t| \le h/2$ and $t' \ge 1/(2Cr) > 0$. Consequently

$$\left| \frac{1}{t'} - \frac{1}{t} \right| = \frac{|t' - t|}{tt'} \overset{(143)}{\le} C^2 r^2 h \overset{(144)}{\le} \frac{\theta}{16m}. \tag{145}$$

The reciprocal $1/t'$ stays strictly inside the permissible interval. The BSS barrier lemmas therefore preserve the invariant after the rounded update $A_{j+1} = A_j + t' v_i v_i^{\mathsf{T}}$.

After $k = dr$ updates, $l_k = r(d - 1/\theta) > 0$, and the spectral ratio satisfies

$$\begin{aligned} \frac{u_k}{l_k} &\overset{(136)}{=} \frac{d/(1-3\theta) + 1/\theta}{d - 1/\theta} \\ &\overset{d \ge \theta^{-2}}{\le} \frac{1/(1-3\theta) + \theta}{1-\theta} \overset{0 < \theta \le 1/12}{\le} 1 + 7\theta. \end{aligned} \tag{146}$$

For the last inequality, clearing the positive denominator leaves $\theta(2 - 22\theta + 21\theta^2) \ge 0$, since $2 - 22\theta \ge 1/6$. Let a_i be the sum of the rounded steps assigned to edge $i \in [m]$. Define H on the original vertices by

$$E(H) = \{ i \in [m] : a_i > 0 \}, \qquad w_H(i) = \frac{a_i w_i}{l_k} \quad (i \in E(H)).$$

The invariant gives $I_r \preceq A_k/l_k$, and (146) gives $A_k/l_k \preceq (1 + 7\theta) I_r$. Undoing the change of coordinates and restoring the deleted coordinates proves the required Laplacian inequalities. At most $dr \le dN$ different edges have been selected.

It remains to implement these updates using rational arithmetic with polynomial bit length. Store

$$P_j = \sum_{i=1}^{m} a_i w_i b_i b_i^{\mathsf{T}},$$

where a_i is the sum of the rounded steps already assigned to edge i. Then $A_j = B^{-1/2} P_j B^{-1/2}$, and direct matrix multiplication gives the rational identities

$$\begin{aligned} \Phi^u(A_j) &= \operatorname{tr}\big(B(uB - P_j)^{-1}\big), \\ v_i^{\mathsf{T}}(uI_r - A_j)^{-1} v_i &= w_i b_i^{\mathsf{T}} (uB - P_j)^{-1} b_i, \\ v_i^{\mathsf{T}}(uI_r - A_j)^{-2} v_i &= w_i b_i^{\mathsf{T}} (uB - P_j)^{-1} B (uB - P_j)^{-1} b_i. \end{aligned} \tag{147}$$

The lower-barrier formulas replace $uB - P_j$ by $P_j - lB$. Every inverse used here is nonsingular by the invariant and (140). Thus the potentials, candidate intervals, midpoint reciprocals, and dyadic rounding are all computable by exact rational operations, without computing $B^{-1/2}$ or any v_i.

For fixed θ, the least mesh exponent in (144) is $b = O(\log(2mr))$. Each stored a_i is a multiple of the same mesh and is at most $dr(2m/\theta + h)$ by (143). Its numerator and denominator therefore have polynomial bit length. A common denominator for the original weights is their denominator product, whose bit length is at most the sum of their input bit lengths. It follows that B and every stored P_j have polynomial bit length. Exact matrix inversion and rational comparisons take polynomial time, for example by fraction-free elimination; determinant bounds give polynomial bit length for the resulting inverse entries. There are $dr = O(N)$ iterations, each scanning m candidates. The final scaling by the positive rational l_k also has polynomial bit length. This proves the claimed Turing running time and output bound. □

Lemma A.8. *Fix positive rational constants C_0, ζ, η. Let G be a loopless graph on $N \geq 1$ vertices with positive rational edge weights and parallel edge weights already combined, satisfying*

$$\sum_{e \in E(G)} w_e \leq 1, \qquad \deg_G(v) \leq \frac{C_0}{N} \quad (v \in V(G)). \tag{148}$$

There is a deterministic polynomial-time construction of a simple unweighted graph J with $V(J) = V(G)$ and $E(J) \subseteq E(G)$, whose maximum degree is bounded by a constant depending only on C_0, ζ. If, in addition,

$$e_G(S) \geq \zeta \quad (S \subseteq V(G),\ |S| \geq \eta N), \tag{149}$$

then every independent set of J has size less than ηN.

Proof. Set the fixed rational $\theta = \min\{1/12, \zeta/14\}$, and put $d = \lceil \theta^{-2} \rceil$. Apply Lemma A.7 to obtain a rationally reweighted subgraph G' with

$$\begin{gathered} |E(G')| \overset{\text{Lem. A.7}}{\leq} dN, \\ L_G \overset{\text{Lem. A.7}}{\preceq} L_{G'} \overset{\text{Lem. A.7}}{\preceq} (1+7\theta) L_G. \end{gathered} \tag{150}$$

Apply the sandwich to the indicator vectors in (133). For each $v \in V(G)$,

$$\begin{aligned} \deg_G(v) &\overset{(150)}{\leq} \deg_{G'}(v) \\ &\overset{(150)}{\leq} (1+7\theta)\deg_G(v) \overset{(148)}{\leq} (1+7\theta)C_0/N. \end{aligned} \tag{151}$$

Write w'_e for the rational weight of $e \in E(G')$. Define the simple unweighted graph J by $V(J) = V(G)$ and

$$E(J) = \{e \in E(G') : w'_e \geq \zeta/(4dN)\}. \tag{152}$$

Every retained edge has weight at least $\zeta/(4dN)$, so for every $v \in V(J)$,

$$\deg_J(v) \overset{(152)}{\leq} \frac{4dN}{\zeta} \deg_{G'}(v) \overset{(151)}{\leq} \frac{4(1+7\theta)C_0 d}{\zeta}. \tag{153}$$

This constant bound and the construction do not require the density assumption. Lemma A.7 gives rational weights of polynomial bit length, so all threshold comparisons are exact rational operations taking polynomial time.

Now assume (149). Summing the lower degree bounds and testing the sandwich on $\mathbf{1}_S$ give, respectively,

$$\begin{aligned} \mathrm{vol}_{G'}(S) &\overset{(151)}{\geq} \mathrm{vol}_G(S), \\ w(\delta_{G'}(S)) &\overset{(150),\,(133)}{\leq} (1+7\theta)w(\delta_G(S)). \end{aligned} \tag{154}$$

Every cut has weight at most the total edge weight, so

$$w(\delta_G(S)) \leq \sum_{e\in E(G)} w_e \overset{(148)}{\leq} 1. \tag{155}$$

Also $\theta \leq \zeta/14$ gives $7\theta/2 \leq \zeta/4$. Therefore, whenever $|S| \geq \eta N$,

$$\begin{aligned} 2e_{G'}(S) &\overset{(134)}{=} \mathrm{vol}_{G'}(S) - w(\delta_{G'}(S)) \\ &\overset{(154)}{\geq} \mathrm{vol}_G(S) - (1+7\theta)w(\delta_G(S)) \\ &\overset{(134)}{=} 2e_G(S) - 7\theta w(\delta_G(S)), \\ e_{G'}(S) &\geq e_G(S) - \tfrac{7\theta}{2} w(\delta_G(S)) \\ &\overset{(149),\,(155)}{\geq} \zeta - \tfrac{7\theta}{2} \geq 3\zeta/4. \end{aligned} \tag{156}$$

The discarded edges satisfy

$$\sum_{e\in E(G')\setminus E(J)} w'_e \overset{(152)}{\leq} |E(G')|\frac{\zeta}{4dN} \overset{(150)}{\leq} \frac{\zeta}{4}. \tag{157}$$

Thus every $S \subseteq V(G)$ with $|S| \geq \eta N$ satisfies

$$\begin{aligned} \sum_{e\in E(J[S])} w'_e &\geq e_{G'}(S) - \sum_{e\in E(G')\setminus E(J)} w'_e \\ &\overset{(156),\,(157)}{\geq} \frac{3\zeta}{4} - \frac{\zeta}{4} = \frac{\zeta}{2} > 0. \end{aligned}$$

Thus $J[S]$ contains an edge, so every independent set of J has size less than ηN.

□

Theorem 3.1. *For every fixed rational $0 < \eta < 1/10$, there is a constant d_η such that the following promise problem is Unique Games-hard. The input is a simple unweighted graph G on $n \geq 1$ vertices of maximum degree at most d_η. Distinguish the following cases:*

YES: $\alpha(G) \geq (1/2 - \eta)n$.

NO: $\alpha(G) < \eta n$.

The reduction is deterministic and polynomial time.

Proof of Theorem 3.1. Use the parameters from Lemma A.5. Let Λ be the regular input of Theorem A.1, $t \geq 1$ the padding integer of Lemma A.3, and $\Lambda^{[t]}$ the padded instance. Write G for its AKS output, G^- for its loop deletion, and G_1 for the clone graph. Apply Lemma A.8 to G_1 with $C_0 = 2$ and $\zeta = \zeta_\eta$, obtaining a rationally reweighted subgraph G' and a thresholded simple unweighted graph J. The construction and its degree guarantee apply in both cases; only the independent-set bound uses the additional NO density premise. The two parts of the reduction are (158) and (159):

$$\Lambda \xrightarrow{\text{padding, Lemma A.3}} \Lambda^{[t]} \xrightarrow{\text{AKS, Theorem A.2}} G \xrightarrow{\text{delete loops, Lemma A.3}} G^-. \tag{158}$$

$$G^- \xrightarrow{\text{clones, Lemma A.4}} G_1 \xrightarrow{\text{rational BSS, Lemma A.7}} G' \xrightarrow{\text{threshold, Lemma A.8}} J. \tag{159}$$

Here G_1, G', J have the same $N = N_{\mathrm{cl}}$ vertices. Lemma A.5 supplies the required weighted promise for G_1. Deleting edges preserves the YES independent set. In the NO case, every set of at least ηN vertices still contains an edge, so every independent set has size less than ηN. The output degree depends only on η. Both constructions are deterministic and polynomial time. □

A.2 The balanced multipartite source

Theorem 3.19. *For every fixed integer $k \geq 3$ and fixed rational $0 < \zeta < 1/12$, there is a constant $D_{k,\zeta}$ such that the following promise problem is Unique Games-hard. The input is a simple unweighted k-partite graph G on $N \geq k$ vertices, of maximum degree at most $D_{k,\zeta}$, with a specified partition into k independent parts each of size N/k.*

YES: $\mathrm{vc}(G) \leq (1/2 + \zeta)N$.

NO: $\mathrm{vc}(G) \geq (1 - 1/k - \zeta)N$.

The reduction is deterministic and polynomial time.

Proof of Theorem 3.19. Apply the bounded-degree source of Theorem 3.1 with $\eta = \zeta/2$. This is an admissible fixed rational parameter, since $0 < \zeta < 1/12$ gives $0 < \eta < 1/24 < 1/10$. For a source graph G on n vertices, form $P_k(G)$ as in Lemma 3.18. It has $N = kn$ vertices, with $n = N/k$ vertices in each of its k independent layers. The degree identity in the same lemma gives

$$\Delta(P_k(G)) \overset{\text{Lem. 3.18}}{=} (k-1)\Delta(G) \overset{\text{Thm. 3.1}}{\leq} (k-1)d_{\zeta/2} =: D_{k,\zeta},$$

which is constant for fixed k, ζ.

The independent-set bounds in Lemma 3.18 are

$$k\alpha(G) \overset{\text{Lem. 3.18}}{\leq} \alpha(P_k(G)) \overset{\text{Lem. 3.18}}{\leq} n + (k-1)\alpha(G). \tag{160}$$

The source bounds are

$$\text{YES:}\quad \alpha(G) \overset{\text{Thm. 3.1}}{\geq} (1/2-\eta)n, \qquad \text{NO:}\quad \alpha(G) \overset{\text{Thm. 3.1}}{<} \eta n. \tag{161}$$

Taking complements and using $N = kn$ therefore gives both cover thresholds:

$$\begin{aligned}\text{YES:}\quad \operatorname{vc}(P_k(G)) &\overset{(160),(161)}{\leq} N - k(1/2-\eta)n = (1/2+\eta)N \leq (1/2+\zeta)N,\\ \text{NO:}\quad \operatorname{vc}(P_k(G)) &\overset{(160),(161)}{>} N - n - (k-1)\eta n\\ &= \left(1 - \frac{1}{k} - \frac{k-1}{k}\eta\right)N \geq \left(1 - \frac{1}{k} - \zeta\right)N.\end{aligned}$$

Here $\operatorname{vc}(P_k(G)) \overset{(1)}{=} N - \alpha(P_k(G))$ and $(k-1)\eta/k \leq \eta = \zeta/2 < \zeta$, so the strict NO estimate implies the stated non-strict threshold. The product has no loops because edges require distinct layer indices, and no parallel edges because G is simple. Its k is fixed, so its size is polynomial in that of G. Both the independent-set source reduction and the product construction are deterministic. Writing Λ for the Unique Games input, this normalization is

$$\Lambda \xrightarrow{\text{Theorem 3.1, } \eta = \zeta/2} G \xrightarrow{\text{Lemma 3.18}} P_k(G).$$

□

A.3 The global SSE partition promise

A.3.1 The source promises and Gaussian normalization

Section 4 defines $\mathrm{SSE}(\eta, \delta)$, SSEH, and SSE-hardness. For fixed $0 < \eta < 1/2$, $M_{\mathrm{SSE}} > 1$, and $0 < \delta < 1/M_{\mathrm{SSE}}$, the input to $\mathrm{SSE}(\eta, \delta, M_{\mathrm{SSE}})$ is an unweighted regular graph G of positive degree. Its promises are

$$\begin{aligned}\text{YES:}\quad & \exists A \subseteq V(G): \quad \mu_G(A) = \delta, \quad \Phi_G(A) \leq \eta,\\ \text{NO:}\quad & \Phi_G(A) > 1-\eta \quad \text{for every } A \subseteq V(G) \text{ with}\\ & \frac{\delta}{M_{\mathrm{SSE}}} < \mu_G(A) < M_{\mathrm{SSE}}\delta.\end{aligned}$$

RST relates these promises in Proposition 5.8 [RST12, Problem 5.7, Proposition 5.8, and Appendix A.2]. The finite variant below accounts for sampling subsets of an exact cardinality by including a factor of two.

Lemma A.9 (Finite strengthening of SSE)**.** *For fixed rational $0 < \eta < 1/2$, integer $M_{\mathrm{SSE}} > 1$, and $0 < \delta < \eta/(2M_{\mathrm{SSE}})$, there is a deterministic polynomial-time reduction from $\mathrm{SSE}(\eta/(2M_{\mathrm{SSE}}), \delta)$ to $\mathrm{SSE}(\eta, \delta, M_{\mathrm{SSE}})$.*

Proof. Write $M = M_{\mathrm{SSE}}$, let G_0 be the input graph and $n = |V(G_0)|$, and abbreviate $\mu = \mu_{G_0}$ and $\Phi = \Phi_{G_0}$. If $s = \delta n$ is an integer with $s \geq 2$, keep the graph. The YES promise is preserved because $\eta/(2M) \leq \eta$. For the NO promise, suppose instead that a set S of size t has $s/M < t < Ms$ and $\Phi(S) \leq 1-\eta$. Let (X, Y) be a uniformly chosen ordered edge, counting both orientations of every edge occurrence, and define $I(S) = \Pr[X \in S,\ Y \in S]$. Regularity gives

$$I(S) = \mu(S)(1 - \Phi(S)) \geq \eta t/n. \tag{162}$$

If $t \le s$, enlarge S to a set T of size s. Then $I(T)/\mu(T) \overset{(162)}{\ge} \eta t/s > \eta/M$. If $t > s$, choose T uniformly among the s-subsets of S. For an edge occurrence e inside S, its retention probability is

$$\Pr[e \in E(G_0[T])] = \begin{cases} s(s-1)/(t(t-1)), & e \text{ is not a loop,} \\ s/t, & e \text{ is a loop.} \end{cases}$$

Since $s/t \ge s(s-1)/(t(t-1))$,

$$\mathbb{E}[I(T)] \ge \frac{s(s-1)}{t(t-1)} I(S). \tag{163}$$

$$\begin{aligned} \mathbb{E}\left[\frac{I(T)}{\mu(T)}\right] &\overset{(163)}{\ge} \frac{n(s-1)}{t(t-1)} I(S) \\ &\overset{(162)}{\ge} \eta \frac{s-1}{t-1} > \frac{\eta}{2M}, \end{aligned}$$

where $s - 1 \ge s/2$ and $t - 1 < Ms$. Thus in either case some set T of measure δ has expansion less than $1 - \eta/(2M)$, contradicting the basic NO promise.

If δn is not an integer, the basic YES case is impossible and the basic NO condition is vacuous. Output a fixed strengthened NO instance in this case. If $\delta n < 2$ is an integer, then $n < 2/\delta$ is bounded for fixed δ; enumerate all subsets of size δn to decide whether the basic YES condition holds, and output a fixed strengthened YES or NO instance accordingly. For the fixed NO output, use K_2. Since $M\delta < \eta/2 < 1/4$, no vertex set has measure in the required open interval. If $\delta = p/h$ for integers $1 \le p < h$, a disjoint union of h copies of K_2 is a strengthened YES instance: take the vertices in p whole components. If δ is irrational, the basic YES case is always impossible, so only the fixed NO output is needed. All these exceptions are handled in polynomial time for the fixed parameters. □

Let Ψ be the standard normal cumulative distribution function. For $-1 < \xi < 1$ and standard jointly normal X, Y of correlation ξ,

$$\Gamma_\xi(\mu) = \Pr[X \le \Psi^{-1}(\mu),\ Y \le \Psi^{-1}(\mu)] \qquad (0 < \mu < 1).$$

Set $\Gamma_\xi(0) = 0$ and $\Gamma_\xi(1) = 1$, the continuous endpoint values. The Gaussian graph $\mathcal{G}(\xi)$ has standard normal vertex measure and ordered-edge distribution (X, Y). RST defines the same function using upper tails. Simultaneously replacing X, Y by $-X, -Y$ shows that the conventions agree. Its Gaussian expansion profile is [RST12, Section 2]

$$\Phi_{\mathcal{G}(\xi)}(\mu) = 1 - \frac{\Gamma_\xi(\mu)}{\mu}.$$

RST Theorem 3.5 gives a NO expansion bound $1 - (\Gamma_{1-\lambda/2}(\mu) + \gamma)/\mu$ for its error parameters $\lambda, \gamma > 0$. The next lemma supplies the required Gaussian estimate.

Lemma A.10 (Gaussian separation at medium measure)**.** *There are absolute constants $a, \varepsilon_{\mathrm{G}} > 0$ such that, for $0 < \varepsilon < \varepsilon_{\mathrm{G}}$, $\varepsilon' \in [\varepsilon/2, \varepsilon]$, and $1/10 \le \mu \le 1/2$, $\mu - \Gamma_{1-\varepsilon'/2}(\mu) \ge a\sqrt{\varepsilon}$.*

Proof. Let X, Z be independent standard normal random variables, put $\xi = 1 - \varepsilon'/2$, and write $Y = \xi X + \sqrt{1-\xi^2}\, Z$. Then X, Y are standard normal with correlation ξ. If $\Pr[X \le t] = \mu \in [1/10, 1/2]$, the threshold t lies in a fixed bounded interval with $t \le 0$. Moreover, $\mu - \Gamma_\xi(\mu) = \Pr[X \le t < Y]$. We may decrease ε_{G} so that $\varepsilon_{\mathrm{G}} \le 1$. Then $0 \le \xi < 1$ and $\sigma := \sqrt{1-\xi^2} \le 1$. Put $t_0 = \Psi^{-1}(1/10)$. The interval $[t-\sigma, t]$ is contained in the fixed interval $[t_0 - 1, 0]$. Let $c_* := \Pr[Z > 1] \min_{x \in [t_0-1, 0]} \varphi(x) > 0$,

where φ is the standard normal density. On the event $t-\sigma \le X \le t$ and $Z>1$, we have $X \le t \le 0$ and

$$Y = \xi X + \sigma Z > X + \sigma \ge t,$$

since $\xi \le 1$. The interval event for X has probability at least $\sigma \min_{x\in[t_0-1,0]} \varphi(x)$, and it is independent of $Z>1$. Hence

$$\mu - \Gamma_\xi(\mu) \ge c_*\sigma \ge \frac{c_*}{2}\sqrt{\varepsilon},$$

where $1-\xi^2 = (1-\xi)(1+\xi) \ge 1-\xi = \varepsilon'/2 \ge \varepsilon/4$. Taking $a = c_*/2$ proves the lemma. □

A.3.2 The finite RST source interface

We use the following two properties of the finite construction [RST12, Lemmas 6.2 and 6.12 and Section 6.3]. Let G_0 be an unweighted regular graph with n vertices and positive degree. Fix integers $q \ge 2$ and $R, k \ge 1$ with $k \mid R$ and $R/k > 4$, and parameters $\rho, \beta, \varepsilon_V, \eta \in (0,1)$. The output graph H has vertex set

$$V_H = V(G_0)^R \times [q]^R \times \{\bot, \top\}^R.$$

Define $\nu_\beta(\top) = \beta$ and $\nu_\beta(\bot) = 1-\beta$. The endpoint marginal is

$$\mu_H(\{(a,x,z)\}) = \frac{1}{(nq)^R} \prod_{i=1}^{R} \nu_\beta(z_i) \qquad ((a,x,z) \in V_H).$$

The sampler below defines an ordered-edge distribution P on $V_H \times V_H$. For $A \subseteq V_H$, put $P(A,A) = \sum_{(u,v)\in A\times A} P(u,v)$. The distribution is symmetric, with endpoint marginal $\mu_H(A) = \sum_{(u,v)\in A\times V_H} P(u,v)$.

Completeness, RST Lemma 6.2: If some $S \subseteq V(G_0)$ satisfies

$$\frac{k}{10\beta R} \le \mu_{G_0}(S) \le \frac{k}{\beta R}, \qquad \Phi_{G_0}(S) \le \eta,$$

then $V_H = V_1 \dot\cup \cdots \dot\cup V_q$ with $\mu_H(V_i) = 1/q$ and

$$\begin{aligned} \Phi_H(V_i) &\le 2u + u^2 + \frac{(1-\rho^2)\beta}{\rho^2} + e_k \qquad (i \in [q]), \\ u &= 1 - \rho^2 + \eta + 2\varepsilon_V. \end{aligned} \tag{164}$$

Here $0 \le e_k \le K_0 2^{-b_0 k}$ for absolute constants $K_0 \ge 1$ and $b_0 > 0$, independent of q; the exceptional measure decays exponentially in the number k of blocks. The partition is equivariant under adding one modulo q to every color coordinate: this shift maps class i to class $i+1$ modulo q. It preserves the endpoint measure, so all classes have the same measure and hence measure $1/q$.

Soundness, RST Lemma 6.12: For every fixed β, q, ρ there is a constant $\zeta > 0$ such that either

$$P(A,A) \le \Gamma_{\rho^2}(\mu_H(A)) + 5\beta^{1/3} \qquad \text{for every } A \subseteq V_H, \tag{165}$$

or some $S \subseteq V(G_0)$ satisfies

$$\frac{\zeta}{R} \le \mu_{G_0}(S) \le \frac{3k}{\varepsilon_V R}, \qquad \Phi_{G_0}(S) \le 1 - \frac{\zeta}{k}.$$

We may take ζ to be a positive rational: replacing it by a smaller positive number only widens this interval and weakens the asserted expansion bound.

The weak inequalities above suffice for us; the cited soundness lemma gives a strict inequality in its first alternative.

Here is why the exceptional term in completeness is uniform in q. In the construction of RST Lemma 6.2, a block is usable when exactly one coordinate belongs to the source set S and has marker $\top$. Put $L = R/k > 4$ and $p = \beta\mu_{G_0}(S)$. The completeness window gives $Lp \in [1/10, 1]$. Independence within and between blocks yields

$$\Pr[\text{a block is usable}] = Lp(1-p)^{L-1} \geq \frac{1}{10}(1-1/L)^{L-1} \geq \frac{1}{40}.$$

For the last inequality, $\log(1-x) \geq -x/(1-x)$ for $0 < x < 1$ gives $(1-1/L)^{L-1} \geq e^{-1} > 1/4$. The set B of vertices with no usable block therefore has endpoint measure $r := \mu_H(B) \leq (39/40)^k$. On its complement, RST assigns the class using the color of the unique active coordinate in the first usable block. Extend this assignment to B equivariantly: choose one representative of each orbit of the simultaneous cyclic color shift, assign it a class, and assign its shifts the corresponding shifted classes. Every orbit has size q, so this defines a partition and gives each class measure $1/q$.

The ordered-edge distribution is invariant under applying the same color shift to both endpoints, and membership in B is color invariant. For a specified endpoint, the event that it lies in B and the first endpoint lies in class i consequently has probability r/q. A union bound over the two endpoints gives at most $2r/q$. After division by the class measure $1/q$, the additional expansion is at most $2r$. Thus the exceptional loss can be bounded by

$$e_k \leq 2(39/40)^k = 2\,2^{-k\log_2(40/39)},$$

independently of q. These choices give explicit admissible constants $K_0 = 2$ and $b_0 = \log_2(40/39)$ in (164).

Following the sampler of RST [RST12, Figure 1 and the preceding definitions in Sections 5 and 6], we define P as follows. On a finite set Ω with probability measure ν, the noise kernel of parameter $t \in [0,1]$ is

$$N_t^\nu(s,s') = t\mathbf{1}_{\{s=s'\}} + (1-t)\nu(\{s'\}) \qquad (s,s' \in \Omega).$$

Thus the input is retained with probability t and is otherwise resampled from ν. Tensor noise uses this kernel independently in each coordinate. Divide $[R]$ into k consecutive blocks of size R/k, and define

$$I_j = \{(j-1)R/k + 1, \ldots, jR/k\} \quad (j \in [k]),$$
$$\Pi_k = \{\pi \in S_R : \pi(I_j) = I_j \text{ for all } j \in [k]\},$$

where S_R is the set of permutations of $[R]$. We use the action $(\pi a)_i = a_{\pi^{-1}(i)}$, applied to every vector in a triple. Write $d_0 > 0$ for the common degree of G_0, and a_{ab} for the number of edge occurrences joining a and b, counting a loop twice when $a = b$. Its random-walk kernel is $T_0(a,b) = a_{ab}/d_0$. For the noise steps, define the base measures

$$\nu_V(\{v\}) = 1/n \quad (v \in V(G_0)), \qquad \nu_{\mathrm{col}}(\{(x,z)\}) = \nu_\beta(z)/q \quad ((x,z) \in [q] \times \{\bot, \top\}).$$

The sampler performs the following operations. Conditional on their stated inputs, all noise and resampling draws are independent across coordinates and branches; all other random choices are independent.

1. Draw $A_0, B_0, C_0 \in V(G_0)^R$ according to

$$\Pr[A_0 = a, B_0 = b, C_0 = c] = \frac{1}{n^R}\prod_{i=1}^{R} T_0(a_i,b_i)T_0(a_i,c_i).$$

2. Apply tensor noise $(N_{1-\varepsilon_V}^{\nu_V})^{\otimes R}$ to B_0, C_0, obtaining $\widetilde{B}, \widetilde{C}$. Independently choose π_B, π_C uniformly from Π_k.

3. Draw $(x_0, z_0) \sim \nu_{\mathrm{col}}^{\otimes R}$. Independently apply tensor noise $(N_\rho^{\nu_{\mathrm{col}}})^{\otimes R}$ twice to obtain (x_B, z_B) and (x_C, z_C).

4. For every coordinate with $(z_B)_i = \perp$, redraw the pair $(\widetilde{B}_i, (x_B)_i)$ from the uniform measure on $V(G_0) \times [q]$; leave other coordinates unchanged. Do the same for the C branch using z_C. Apply π_B and π_C to all coordinates of the respective resulting triples, and call the outputs Y_B, Y_C. Define
$$P(u, v) = \Pr[Y_B = u,\ Y_C = v] \qquad (u, v \in V_H).$$

The operations preserve the product marginal, and exchanging branches proves $P(u, v) = P(v, u)$.

To pass from this ordered distribution to the weighted graph notation of Section 4, give an unordered pair of distinct vertices u, v weight $w_{uv} = P(u, v) + P(v, u)$, and give a loop at v weight $w_{vv} = P(v, v)$. Omit pairs of weight zero. Then

$$W = 1, \qquad \deg_H(v) = 2 \sum_{u \in V_H} P(v, u), \qquad \mu_H(A) = \frac{\mathrm{vol}_H(A)}{2W},$$

$$\frac{w(\delta_H(A))}{W} = 2 \sum_{(u,v) \in A \times (V_H \setminus A)} P(u, v).$$

Thus the expansion in the source interface is precisely

$$\Phi_H(A) = 1 - \frac{P(A, A)}{\mu_H(A)} \qquad (\mu_H(A) > 0). \tag{166}$$

A loop contributes twice its weight to the degree and nothing to the cut.

Theorem A.11 (Global SSE partition gap, weighted normalized form [RST12, Theorem 3.5])**.** *There are absolute constants $c > 0$ and $\varepsilon_0 > 0$ such that, for every fixed integer $q \geq 2$ and every fixed $0 < \varepsilon < \varepsilon_0$, the following promise on finite undirected graphs with positive rational edge weights is SSE-hard. Parallel edges and loops are allowed, with $W > 0$.*

YES: *There is a partition $V(G) = V_1 \dot\cup \cdots \dot\cup V_q$ with $\mu_G(V_i) = 1/q$ and $\Phi_G(V_i) \leq 2\varepsilon$ for every i.*

NO: *Every $A \subseteq V(G)$ with $\mu_G(A) \in [1/10, 9/10]$ satisfies $w(\delta_G(A)) \geq c\sqrt{\varepsilon}\, W$.*

Proof of Theorem A.11. Let a be the absolute constant from Lemma A.10, and take $\varepsilon_0 \leq \min\{\varepsilon_{\mathrm{G}}, 1/9\}$. Fix q and ε as in the statement, and choose the parameters in this order.

1. Choose a rational ρ with $\sqrt{1 - \varepsilon/2} < \rho < \sqrt{1 - \varepsilon/4}$, and put $\varepsilon' = 2(1 - \rho^2)$. Thus $\varepsilon/2 < \varepsilon' < \varepsilon$ and $\rho^2 = 1 - \varepsilon'/2$.

2. Choose a rational $0 < \gamma \leq \min\{1/2, a\sqrt{\varepsilon}/20\}$, and set
$$\beta = \min\{\varepsilon', (\gamma/10)^3\}, \qquad \varepsilon_V = \varepsilon'^2.$$
Then $\beta \leq \varepsilon'$ and $5\beta^{1/3} \leq \gamma/2$.

3. Choose an integer $k \geq 1$ such that $K_0 2^{-b_0 k} \leq \varepsilon'^2$, using the absolute constants in (164). Take a positive rational ζ no larger than the soundness constant for these (β, q, ρ).

4. Choose a rational $0 < \eta \le \min\{\varepsilon'^2, \zeta/(2k)\}$, and an integer

$$M_{\mathrm{SSE}} > \max\{1, k/(\beta\zeta), 6\beta/\varepsilon_V\}.$$

   Put $\eta_* = \eta/(2M_{\mathrm{SSE}})$.

5. For each fixed $\delta \in (0, \eta_*)$, begin with $\mathrm{SSE}(\eta_*, \delta)$ and apply Lemma A.9. Its output is an instance of $\mathrm{SSE}(\eta, \delta, M_{\mathrm{SSE}})$. Use the integer

$$R = k \left\lfloor \frac{1}{\beta\delta} \right\rfloor$$

   in the finite sampler.

$\eta_* > 0$ is fixed before δ, as SSE-hardness requires, and $k \mid R$. Since $\beta \le \varepsilon' < 1/9$ and $\delta < \eta_* \le \varepsilon'^2$, we have $\beta\delta < 1/10$ and hence $R/k > 4$. Also $\beta\delta < 1/2$, so flooring changes the unrounded value by at most a factor of two:

$$\frac{k}{2\beta\delta} \le R \le \frac{k}{\beta\delta}. \tag{167}$$

Consequently a source YES set of measure δ lies in the completeness window:

$$\frac{k}{10\beta R} \overset{(167)}{\le} \frac{\delta}{5} < \delta \overset{(167)}{\le} \frac{k}{\beta R}.$$

For soundness, the possible decoded interval lies strictly inside the strengthened NO window, because

$$\frac{\zeta}{R} \overset{(167)}{\ge} \frac{\beta\zeta\delta}{k} > \frac{\delta}{M_{\mathrm{SSE}}}, \qquad \frac{3k}{\varepsilon_V R} \overset{(167)}{\le} \frac{6\beta\delta}{\varepsilon_V} < M_{\mathrm{SSE}}\delta.$$

The decoded expansion bound is $1 - \zeta/k < 1 - \eta$, so the second alternative in RST Lemma 6.12 contradicts the strengthened NO promise.

YES case. RST Lemma 6.2 gives a partition of the whole output vertex set into q classes of measure $1/q$. For u in (164), $u \le \varepsilon'/2 + 3\varepsilon'^2 \le \varepsilon'$, since $\varepsilon' \le 1/6$. Also $\rho^2 \ge 1/2$ and $\beta \le \varepsilon'$ give $(1-\rho^2)\beta/\rho^2 \le \varepsilon'^2$, and our choice of k gives $e_k \le \varepsilon'^2$. Thus the full expansion bound is

$$\begin{aligned}\Phi_G(V_i) &\overset{(164)}{\le} 2u + u^2 + \frac{(1-\rho^2)\beta}{\rho^2} + e_k \\ &\le \varepsilon' + 9\varepsilon'^2 \le 2\varepsilon' \le 2\varepsilon.\end{aligned}$$

The upper threshold on ε is independent of q.

NO case. The decoded alternative has been excluded above, so (165) and $5\beta^{1/3} \le \gamma/2 \le \gamma$ give, for every set of positive volume,

$$\Phi_G(A) \overset{(166),\,(165)}{\ge} 1 - \frac{\Gamma_{1-\varepsilon'/2}(\mu_G(A)) + \gamma}{\mu_G(A)}. \tag{168}$$

For $\mu = \mu_G(A) \in [1/10, 1/2]$, multiply by 2μ and use $\mathrm{vol}_G(A) = 2W\mu$ and Lemma A.10:

$$\begin{aligned}\frac{w(\delta_G(A))}{W} = 2\mu\Phi_G(A) &\overset{(168)}{\ge} 2\big(\mu - \Gamma_{1-\varepsilon'/2}(\mu)\big) - 2\gamma \\ &\overset{\text{Lem. A.10}}{\ge} 2a\sqrt{\varepsilon} - 2\gamma \ge \frac{a}{10}\sqrt{\varepsilon}.\end{aligned}$$

Here $2\gamma \le (a/10)\sqrt{\varepsilon}$. For $\mu_G(A) \in [1/2, 9/10]$, apply the same bound to its complement, whose measure is $1 - \mu_G(A) \in [1/10, 1/2]$ and whose cut is $\delta_G(V(G) \setminus A) = \delta_G(A)$ as multisets. This proves the NO bound throughout $[1/10, 9/10]$ with $c = a/10$.

The finite sampler has rational probabilities and hence rational edge weights. Its graph satisfies

$$|V(G)| = (2qn)^R, \qquad |E(G)| \le \frac{(2qn)^{2R} + (2qn)^R}{2},$$

where the edge count includes loops. These bounds are polynomial in n for each fixed (q, ε, δ), since R is then fixed. Omitting zero-weight pairs does not change the distribution or either promise.

The reduction computes these weights exactly, rather than sampling edges. Enumerate the finitely many choices in the sampler and add their rational probabilities to the corresponding unordered edge records. For fixed (q, ε, δ), all noise denominators, R, and the permutation counts are constants. The host choices range over at most $(n+|E(G_0)|)^{O(R)}$ possibilities. Their probabilities have a common denominator dividing $C_0 n^{C_1 R} d_0^{2R}$ for constants C_0, C_1 depending only on the fixed parameters, where d_0 is the input degree. Its bit length is $O(R \log(nd_0))$ plus a parameter-dependent constant. Exact addition therefore computes all weights in deterministic polynomial time and with polynomial encoding length. □

A.3.3 Rounding to an unweighted host

The following conversion proves Lemma 4.1 by replacing weighted edges with parallel unweighted copies. The source promise uses the wider interval $[1/10, 9/10]$ rather than the interval $[1/4, 3/4]$ required by Lemma 4.5 to absorb additive errors in volumes and cuts. Section 6.5 later reduces the line graph's degree through incidence clique factors.

Lemma 4.1. *There are absolute constants $c > 0$ and $\varepsilon_0 > 0$ such that, for every fixed integer $q \ge 2$ and every fixed rational $0 < \varepsilon < \varepsilon_0$, the following promise on unweighted multigraphs G with $m > 0$ edges, allowing loops, is SSE-hard:*

YES: *There is a partition $V(G) = V_1 \dot\cup \cdots \dot\cup V_q$ such that $(1-\varepsilon)/q \le \mu_G(V_i) \le (1+\varepsilon)/q$ and $\Phi_G(V_i) \le 3\varepsilon$ for every i.*

NO: *Every $A \subseteq V(G)$ with $\mu_G(A) \in [1/4, 3/4]$ satisfies $|\delta_G(A)| \ge (c/2)\sqrt{\varepsilon}\, m$.*

Proof of Lemma 4.1. Let c and ε_0 be the constants from Theorem A.11. Decrease ε_0 so that every $0 < \varepsilon < \varepsilon_0$ satisfies $\varepsilon \le 1/10$ and $\varepsilon^2 \le (c/2)\sqrt{\varepsilon}$. Fix $q \ge 2$ and a rational ε in this range, and set $\theta = \min\{\varepsilon^2, \varepsilon/(2q)\}$.

Construction. Let G be a weighted source instance with $k = |E(G)| \ge 1$ and $W = w(E(G))$, counting parallel edges separately. Set $M = \lceil k/\theta \rceil$ and $r_e = \lfloor M w_e / W \rfloor$. Form the unweighted multigraph G^* on $V(G)$ by replacing each edge e by r_e copies with the same endpoints, with no other edges. A loop is replaced by loops, and an edge with $r_e = 0$ is removed. Write $m = \sum_{e \in E(G)} r_e$ and output G^*.

Rounding estimates. For each input edge e, write $\omega_e = M w_e / W - r_e \in [0, 1)$. Since $k \le \theta M$ and $\omega_e < 1$ for each of the k input edges,

$$(1-\theta)M \le M - k < M - \sum_{e \in E(G)} \omega_e = m \le M. \tag{169}$$

Thus $m > 0$, and G^* has at most $M \le k/\theta + 1 \le (1 + 1/\theta)k$ edges. The vertex set is unchanged, including vertices whose rounded degree becomes zero. Since θ is fixed for each (q, ε), the output

has polynomial size. The rationally encoded weights allow W, M, and every floor r_e to be computed exactly in polynomial time, without expanding a common denominator into edge copies.

For a fixed vertex set A, an edge contributes zero, one, or two endpoint incidences to its volume. A loop in A contributes two. Since $w_e/W - r_e/M = \omega_e/M$, edge e contributes between zero and ω_e/M to the difference of normalized volumes. Summing gives

$$0 \le \frac{\mathrm{vol}_G(A)}{2W} - \frac{\mathrm{vol}_{G^*}(A)}{2M} \le \frac{1}{M}\sum_{e\in E(G)} \omega_e < \frac{k}{M} \le \theta. \tag{170}$$

For the cut, only crossing edges contribute. Thus

$$|\delta_{G^*}(A)| = M\frac{w(\delta_G(A))}{W} - \sum_{e\in\delta_G(A)} \omega_e \ge M\left(\frac{w(\delta_G(A))}{W} - \theta\right), \tag{171}$$

because $\sum_{e\in\delta_G(A)} \omega_e < k \le \theta M$.

YES case. Use the source partition $(V_1, \dots, V_q)$. Since $\mu_G(V_i) \overset{\text{Thm. A.11}}{=} 1/q$, equation (170) and $(1-\theta)M \le m \le M$ give

$$\begin{aligned}
\mu_{G^*}(V_i) &\overset{(170),(169)}{\ge} 1/q - \theta \ge (1-\varepsilon/2)/q \ge (1-\varepsilon)/q,\\
\mu_{G^*}(V_i) &\overset{(170),(169)}{\le} \frac{1/q}{1-\theta} \le \frac{1+\varepsilon}{q}.
\end{aligned}$$

The lower bound uses $\theta \le \varepsilon/(2q)$; the upper bound uses $\theta \le \varepsilon/(2q) \le \varepsilon/(1+\varepsilon)$. The source bounds give

$$w(\delta_G(V_i)) = \Phi_G(V_i)\,\mathrm{vol}_G(V_i) \overset{\text{Thm. A.11}}{\le} 2\varepsilon \cdot 2W/q = 4\varepsilon W/q. \tag{172}$$

Since $r_e \le Mw_e/W$, summing the rounded edge multiplicities over the cut gives

$$|\delta_{G^*}(V_i)| \le Mw(\delta_G(V_i))/W \overset{(172)}{\le} 4\varepsilon M/q. \tag{173}$$

Consequently

$$\begin{aligned}
\Phi_{G^*}(V_i) &\overset{(173),(170)}{\le} \frac{4\varepsilon M/q}{2M(1/q-\theta)} = \frac{2\varepsilon}{1-q\theta}\\
&\le \frac{2\varepsilon}{1-\varepsilon/2} < 3\varepsilon.
\end{aligned}$$

The last inequality uses $\varepsilon \le 1/10$.

NO case. Let A be any set with $\mu_{G^*}(A) \in [1/4, 3/4]$. Since $1-\theta \le m/M \le 1$, equation (170) gives

$$\frac{1-\theta}{4} \overset{(169)}{\le} \frac{m}{M}\mu_{G^*}(A) \overset{(170)}{\le} \mu_G(A) \overset{(170)}{\le} \frac{m}{M}\mu_{G^*}(A) + \theta \overset{(169)}{\le} \frac{3}{4} + \theta.$$

For $\theta \le 1/10$, this interval lies in $[9/40, 17/20] \subset [1/10, 9/10]$. The source NO promise gives

$$\frac{w(\delta_G(A))}{W} \overset{\text{Thm. A.11}}{\ge} c\sqrt{\varepsilon}. \tag{174}$$

Applying (171), $\theta \le (c/2)\sqrt{\varepsilon}$, and $M \ge m$ gives

$$|\delta_{G^*}(A)| \overset{(171),(174)}{\ge} M(c\sqrt{\varepsilon} - \theta) \ge (c/2)\sqrt{\varepsilon}\,M \overset{(169)}{\ge} (c/2)\sqrt{\varepsilon}\,m.$$

Section 4 constructs the simple line graph $\mathcal{L}(G^*)$ in $O(M^2)$ time: it has $m \le M$ vertices, one per edge occurrence, including parallel edges and loops. □

For reference, let G_0 be a basic SSE instance, G_1 the strengthened instance supplied by Lemma A.9, H the weighted sampler output in Theorem A.11, and G^* its rounded unweighted host. The source normalization is

$$G_0 \xrightarrow{\text{Lemma A.9}} G_1 \xrightarrow{\text{[RST12, Lemmas 6.2 and 6.12]}} H \xrightarrow{\text{Lemma 4.1}} G^*.$$

Equations (170) and (171) account for the last arrow's losses.

A.4 Provenance of the NP-hardness construction

The reduction in Section 5.1 passes from a constraint system to tripartite Vertex Cover and then to Treedepth, following Bonnet, Neuen, and Sokołowski [BNS25, Sections 3 and 4]. Their truth-block and assignment-witness construction adapts Bonnet's earlier treewidth reduction [Bon25, full version, Section 3.1]. We inherit two opposite truth blocks per variable, complete bipartite edges between them, and witnesses for satisfying local assignments. For each variable in its equation, a witness is adjacent to the truth block assigning the value opposite to the witness's represented value. Variable-dependent block sizes also occur in Bonnet's full version [Bon25, Section 3.5].

Here the source predicates are three-variable parity equations. Each has four satisfying assignments. We add a star on their four witnesses, place its monochromatic witness 000 or 111 in a truth part, and place the other three in the third part. We choose the block size $5d_x/6$ by analyzing arbitrary partial assignments: uniformly completing the unassigned variables bounds the excess witnesses of an equation by $5r_j/6$, where r_j is its number of unassigned variables. This is the soundness argument proved in Lemma 5.3. The corresponding argument in [BNS25, proof of Lemma 10] uses variable-by-variable exchanges in a vertex cover. Pairing the source with a complemented copy then makes all three parts have size $3m$. These are the modifications to the earlier encoding used in this proof.

The anchor graph is the one in [BNS25, Lemma 6]. We use their identity as a black box at its stated anchor length, together with the cover-first upper bound and Claims 7 and 9 of their proof. The main-text corollary extends the identity to anchors as short as the largest part. Its derivation is given in Appendix B.3. For our balanced source graph this allows $\ell = 3m$ even though $\mathrm{vc}(G_F) \overset{\text{Lem. 5.3}}{\geq} 11m/2$. Combined with the parity-witness bound, the anchor identity yields the factor below 18/17.

A.5 The parity source normalization

Lemma A.12 (Balancing parity right-hand sides)**.** *A system F_0 of $m_0 \geq 1$ parity equations, each on three distinct Boolean variables, can be paired with a system F_0' on fresh variables so that their union has $2m_0$ equations, optimum $2\,\mathrm{OPT}(F_0)$, and m_0 equations with each right-hand side. Each variable and its copy retain the original variable's occurrence count.*

Proof. Make a copy F_0' on fresh variables, replacing every equation $x + y + z = a \pmod 2$, with $a \in \{0, 1\}$, by $x' + y' + z' = 1 - a$. For an assignment σ to F_0, assign $x' = 1 - \sigma(x)$ in F_0'. Since each equation contains three variables,

$$(1 - \sigma(x)) + (1 - \sigma(y)) + (1 - \sigma(z)) \equiv 1 - (\sigma(x) + \sigma(y) + \sigma(z)) \pmod 2.$$

Thus the copied equation is satisfied if and only if the original one is satisfied. Complementing all variable values is a bijection between assignments of the two systems, so $\mathrm{OPT}(F_0') = \mathrm{OPT}(F_0)$.

Because the variable sets are disjoint, any assignment to the union satisfies at most $\mathrm{OPT}(F_0) + \mathrm{OPT}(F_0') = 2\,\mathrm{OPT}(F_0)$ equations; independent optimal assignments attain this bound. The union

has $2m_0$ equations. Every original equation is paired with one having the opposite right-hand side, so each right-hand side occurs in m_0 equations. Copying to fresh variables preserves each occurrence count and the distinctness of the three variables in every equation. □

Lemma A.13 (Repeating parity equations)**.** *Let F_1 be an unweighted system of $m_1 \geq 1$ equations, and let F repeat every equation of F_1 six times. Then F has $m = 6m_1$ equations and $\mathrm{OPT}(F) = 6\,\mathrm{OPT}(F_1)$. Every occurrence count is multiplied by six. The fraction of satisfied equations and the fraction with each right-hand side are unchanged.*

Proof. Every assignment satisfies all six copies of an equation or none. Thus its number of satisfied equations is multiplied by six, as are the total equation count, every variable's occurrence count, and the number of equations with either right-hand side. Taking a maximum over the same assignments proves the optimum claim. □

Theorem 5.1. *For every fixed $0 < \epsilon < 1/4$, there is a constant $b_\epsilon \geq 6$ depending only on ϵ such that the following promise problem is NP-hard. The input is an unweighted system F of $m \geq 1$ equations $x + y + z = a \pmod 2$, where $a \in \{0, 1\}$ and x, y, z are distinct Boolean variables. Equations may be repeated. Let $\mathcal{X}(F)$ be its finite Boolean variable set. Every variable occurs at most b_ϵ times, with at most two possible occurrence counts. For $x \in \mathcal{X}(F)$, write d_x for the number of equations containing x. For $1 \leq j \leq m$, write a_j for the right-hand side of equation j, and let $\mathrm{OPT}(F)$ count the maximum number of satisfied equations, including repetitions. The instances satisfy*

$$d_x \in 6\mathbb{Z}_{>0}, \qquad \sum_{x \in \mathcal{X}(F)} d_x = 3m,$$

$$|\{j \in \{1, \ldots, m\} : a_j = 0\}| = |\{j \in \{1, \ldots, m\} : a_j = 1\}| = m/2.$$

YES: $\mathrm{OPT}(F) \geq (1 - \epsilon)m$.

NO: $\mathrm{OPT}(F) \leq (1/2 + \epsilon)m$.

Proof of the normalized form in Theorem 5.1. Hazan, Safra, and Schwartz state the required source gap as Theorem 10 of [HSS03], attributing it to Håstad. Apply their Max-3-LIN-q theorem with $q = 2$ and error $\epsilon/2$, and call its unweighted output F_{raw}. There is a fixed integer $B_\epsilon \geq 1$ bounding its occurrence counts; each variable's count is chosen from at most two values, and no variable repeats within an equation. Thus each equation has three distinct Boolean variables. The cited YES threshold $1 - \epsilon/2$ is at least $1 - \epsilon$, and its NO bound $< 1/2 + \epsilon/2$ implies the required bound $\leq 1/2 + \epsilon$. We retain this source promise and impose only the additional balance and divisibility conditions.

Delete unused variables from F_{raw}, calling the result F_0 and its equation count $m_0 \geq 1$. This changes neither the equations nor the optimum, and it removes only zero occurrence counts. Let $F_1 = F_0 \dot\cup F_0'$ be the union supplied by Lemma A.12, and let F be the sixfold repetition of F_1 from Lemma A.13. Record the variable and equation pairing between F_0 and its complemented copy, pairing corresponding repeated equations as well. This construction data remains available to the later reduction; it imposes no additional condition on the original source promise. The normalization is summarized by

$$\text{NP-hard gap} \xrightarrow{\text{[HSS03, Theorem 10], } q=2} F_{\mathrm{raw}} \xrightarrow{\text{delete unused variables}} F_0,$$

$$F_0 \xrightarrow{\text{Lemma A.12}} F_1 \xrightarrow{\text{Lemma A.13}} F.$$

The two lemmas give

$$m \overset{\text{Lems. A.12, A.13}}{=} 12m_0, \quad \mathrm{OPT}(F) \overset{\text{Lems. A.12, A.13}}{=} 12\,\mathrm{OPT}(F_0),$$
$$\frac{\mathrm{OPT}(F)}{m} = \frac{\mathrm{OPT}(F_0)}{m_0}.$$

Thus F retains the required YES and NO fractions.

Each remaining variable occurred at least once before repetition, and its final occurrence count is six times a positive integer. Set $b_\epsilon = 6B_\epsilon \geq 6$. Copying preserves occurrence counts, and repetition multiplies them by six, so there are still at most two possible positive counts and all are at most b_ϵ. Thus $d_x \in 6\mathbb{Z}_{>0}$. Counting the three variable occurrences in every equation gives $\sum_{x\in\mathcal{X}(F)} d_x = 3m$. The repetition creates $6m_0 = m/2$ equations with each right-hand side, so the required balance also holds and m is even. These operations increase the number of equations by the fixed factor twelve, preserve the distinct variables within each equation, and are deterministic and polynomial time. This proves all the conditions in (28). □

B Auxiliary proofs

B.1 The branching inequality

Lemma 3.3. *For a positive integer ν, positive reals $q_1, \ldots, q_\nu$, and nonnegative reals $n_1, \ldots, n_\nu$,*

$$3\sum_{j=1}^{\nu} q_j n_j - 2\max_{1\leq j\leq \nu}(q_j n_j) \leq \left(\sum_{j=1}^{\nu} q_j\right)\left(\sum_{j=1}^{\nu} n_j\right).$$

Proof of Lemma 3.3. Put $x_j = \sqrt{q_j n_j}$ for $1 \leq j \leq \nu$, and choose $k \in \{1, \ldots, \nu\}$ with $x_k = \max_{1\leq j\leq\nu} x_j$. Cauchy–Schwarz gives the first inequality below. All cross products are nonnegative, and $x_k x_j \geq x_j^2$ because $0 \leq x_j \leq x_k$. Therefore

$$\begin{aligned}
\left(\sum_{j=1}^{\nu} q_j\right)\left(\sum_{j=1}^{\nu} n_j\right) &\overset{\text{Cauchy–Schwarz}}{\geq} \left(\sum_{j=1}^{\nu} x_j\right)^2 \\
&= \sum_{j=1}^{\nu} x_j^2 + 2\sum_{1\leq i<j\leq\nu} x_i x_j \\
&\geq \sum_{j=1}^{\nu} x_j^2 + 2x_k \sum_{1\leq j\leq\nu,\, j\neq k} x_j \\
&\geq \sum_{j=1}^{\nu} x_j^2 + 2\sum_{1\leq j\leq\nu,\, j\neq k} x_j^2 \\
&= 3\sum_{j=1}^{\nu} x_j^2 - 2x_k^2.
\end{aligned}$$

Substituting $x_j^2 = q_j n_j$ proves the claim. □

B.2 The quadratic branching inequality

Lemma 3.12. *For a positive integer ν and nonnegative reals $W_1, \dots, W_\nu$,*

$$\sum_{j=1}^{\nu} W_j^2 + 14 \sum_{1 \le i < j \le \nu} W_i W_j \le \left(\sum_{j=1}^{\nu} \sqrt{W_j} \right)^4 .$$

Proof of Lemma 3.12. Put $x_j = \sqrt{W_j} \ge 0$ for $1 \le j \le \nu$. The fourth-power expansion has pure terms x_j^4 and, for each pair $1 \le i < j \le \nu$, the three mixed terms $4x_i^3 x_j + 6x_i^2 x_j^2 + 4x_i x_j^3$. All terms involving at least three distinct indices are nonnegative. For each pair,

$$\begin{aligned} 4x_i^3 x_j + 6x_i^2 x_j^2 + 4x_i x_j^3 &= 14 x_i^2 x_j^2 + 4x_i x_j (x_i - x_j)^2 \\ &\ge 14 x_i^2 x_j^2 . \end{aligned}$$

Sum these bounds, add the pure terms, and discard the terms with three or more indices. Substituting $x_j^2 = W_j$ proves the claim. □

B.3 Derivation of the anchor-length corollary

We prove Corollary 5.4 for the anchor graph $H_\ell(G)$, its cliques K_X, and vertices z_X from Section 5.1.4. Fix a tripartite graph G with nonempty specified independent parts A, B, C and an integer $\ell \ge 1$. Put

$$\mathrm{vc}_2(G) = \min_{X,Y \in \{A,B,C\},\ X \ne Y} \mathrm{vc}(G[X \cup Y]).$$

The following published identity supplies the case of long anchors.

Lemma B.1 (BNS anchor identity [BNS25, Lemma 6])**.** *Let G be a tripartite graph with specified nonempty independent parts A, B, C. For every positive integer $L \ge \mathrm{vc}(G)$, $\mathrm{td}(H_L(G)) = L + \mathrm{vc}(G) + 1$.*

For an elimination forest F of $H_\ell(G)$, write $h(F)$ for its height and define, for $X \in \{A, B, C\}$,

$$v_X = \arg\max_{v \in K_X} \mathrm{depth}_F(v).$$

This vertex is unique because the vertices of a clique form an ancestor chain.

Lemma B.2 (A path through an anchor [BNS25, Claim 7 in the proof of Lemma 6])**.** *Let $X \in \{A, B, C\}$. If every vertex of $R \subseteq V(H_\ell(G)) \setminus (K_X \cup \{z_X\})$ is ancestral to v_X, then*

$$h(F) \ge |R| + \ell + 1. \tag{175}$$

Proof. Since K_X is a clique, all its vertices are ancestral to its deepest vertex v_X. Thus $R \cup K_X$ lies on the ancestor path ending at v_X. The edge $z_X v_X$ makes z_X comparable with v_X, so $R \cup K_X \cup \{z_X\}$ lies on one root-to-leaf path. The sets are disjoint and contain $|R| + \ell + 1$ vertices in total, proving the bound. This argument applies to every elimination forest F and every $\ell \ge 1$. □

The next lemma adapts the argument of [BNS25, Claim 9 in the proof of Lemma 6]. Only the two specified anchors need to be incomparable; no condition on the third anchor is required.

Lemma B.3 (A cover from two incomparable anchors)**.** *Let $X, Y \in \{A, B, C\}$ be distinct and define*

$$S_{X,Y} = \{v \in X \cup Y : v \text{ is ancestral to both } v_X \text{ and } v_Y \text{ in } F\}.$$

If v_X, v_Y are incomparable, then $S_{X,Y}$ is a vertex cover of $G[X \cup Y]$.

Proof. For an edge $xy \in E(G[X \cup Y])$ with $x \in X$ and $y \in Y$, the four vertices v_X, x, y, v_Y induce a connected subgraph, since $v_X x$, xy, and $y v_Y$ are edges of $H_\ell(G)$. Lemma 2.2 gives one of these four vertices ancestral to all four. It cannot be v_X or v_Y, since those anchors are incomparable. Hence x or y is ancestral to both anchors. Thus $\{x, y\} \cap S_{X,Y} \neq \varnothing$ for every $xy \in E(G[X \cup Y])$, as required. □

Lemma B.4 (Extending incomparable anchors)**.** *If v_A, v_B, v_C are pairwise incomparable, then $h(F) \geq \ell + \mathrm{vc}(G) + 1$.*

Proof. Set $L = \max\{\ell, \mathrm{vc}(G)\}$ and $d = L - \ell$. In each K_X, insert a chain of d new vertices immediately above v_X. Together with K_X, these vertices form the size-L anchor clique in $H_L(G)$, which is joined completely to $X \cup \{z_X\}$. Every vertex of $X \cup \{z_X\}$ is comparable with v_X, since it is adjacent to v_X. Hence it is also comparable with the inserted chain. The resulting forest is therefore an elimination forest of $H_L(G)$. Since the three anchors are pairwise incomparable, each root-to-leaf path meets at most one inserted chain, so its height is at most $h(F) + d$. We now have $h(F) + d \overset{\text{Lem. B.1}}{\geq} L + \mathrm{vc}(G) + 1$, and thus $h(F) \geq L + \mathrm{vc}(G) + 1 - d = \ell + \mathrm{vc}(G) + 1$. □

We now distinguish the possible ancestor relations among the three anchors to obtain a bound for every anchor length.

Lemma B.5 (Bounds for arbitrary anchor length)**.** *For every tripartite graph G with nonempty specified independent parts A, B, C and every integer $\ell \geq 1$,*

$$1 + \min\{\ell + \mathrm{vc}(G),\ 2\ell + \mathrm{vc}_2(G),\ 3\ell\} \leq \mathrm{td}(H_\ell(G)) \leq \ell + \mathrm{vc}(G) + 1.$$

Proof. Upper bound. We follow the cover-first forest in the proof of [BNS25, Lemma 6]. Choose a vertex cover $C \subseteq V(G)$ with $|C| = \mathrm{vc}(G)$, place C on a top chain, then place each anchor clique on its own chain below it and attach the uncovered vertices of that part and its associated z_X as leaves below the anchor chain. If the cover is empty, use the three anchor chains as separate roots. Every source edge meets the top cover, and every anchor edge has comparable endpoints. A root-to-leaf path therefore uses at most $\mathrm{vc}(G) + \ell + 1$ vertices. The construction does not require $\ell \geq \mathrm{vc}(G)$, so it gives the upper bound for every $\ell \geq 1$.

Lower bound. Fix an elimination forest F and use the anchor notation above. There are three cases.

1. If v_A, v_B, v_C are pairwise incomparable, $h(F) \overset{\text{Lem. B.4}}{\geq} \ell + \mathrm{vc}(G) + 1$.
2. Suppose all three anchors are comparable, and let v_X be the deepest, with $\{Y, Z\} = \{A, B, C\} \setminus \{X\}$. Every vertex of the other two anchor cliques lies above v_X. Set $R = K_Y \cup K_Z$, so $|R| = 2\ell$. Then $h(F) \overset{(175)}{\geq} |R| + \ell + 1 = 3\ell + 1$.
3. Otherwise some pair is comparable, but the three anchors do not form a chain. Relabel so that v_A is an ancestor of v_B and v_B is incomparable with v_C. This includes the case where v_A is above both other anchors. Set $S = S_{B,C}$. The cover bound is

$$|S| \overset{\text{Lem. B.3}}{\geq} \mathrm{vc}(G[B \cup C]) \geq \mathrm{vc}_2(G). \tag{176}$$

Every vertex of K_A is an ancestor of v_A, and hence of v_B. Therefore $R = S \cup K_A$ is disjoint from $K_B \cup \{z_B\}$ and consists of ancestors of v_B. Since $S \subseteq V(G)$ and K_A is disjoint from $V(G)$, $|R| = |S| + |K_A| = |S| + \ell$. Applying the path bound with anchor B gives

$$h(F) \overset{(175)}{\geq} |R| + \ell + 1 = 2\ell + |S| + 1 \overset{(176)}{\geq} 2\ell + \mathrm{vc}_2(G) + 1.$$

Taking the weakest of these three bounds proves the lemma. □

Corollary 5.4. *If G is tripartite with specified nonempty independent parts A, B, C and $\ell \geq \max\{|A|, |B|, |C|\}$, then $\mathrm{td}(H_\ell(G)) = \ell + \mathrm{vc}(G) + 1$.*

Proof of Corollary 5.4. Since each part has size at most ℓ, any two parts together cover every edge and $\mathrm{vc}(G) \leq 2\ell$. Choose a pair X, Y attaining $\mathrm{vc}_2(G)$ and let Z be the omitted part. A minimum vertex cover of $G[X \cup Y]$, together with all of Z, covers G. Therefore $\mathrm{vc}(G) \leq \mathrm{vc}_2(G) + |Z| \leq \mathrm{vc}_2(G) + \ell$. The two cover bounds give $\ell + \mathrm{vc}(G) \leq 3\ell$ and $\ell + \mathrm{vc}(G) \leq 2\ell + \mathrm{vc}_2(G)$, respectively. Thus the minimum is $\ell + \mathrm{vc}(G)$, and

$$\mathrm{td}(H_\ell(G)) \overset{\text{Lem. B.5}}{=} \ell + \mathrm{vc}(G) + 1.$$

□

B.4 Total-depth normalization and the three paths

Let G be a simple tripartite graph with specified nonempty independent parts A, B, C, and put $N = |V(G)|$. Complete each part to a clique by setting

$$\widehat{G} = \left(V(G),\ E(G) \cup \bigcup_{X \in \{A,B,C\}} \binom{X}{2} \right).$$

For an integer $M \geq 1$, obtain $H_M^\Sigma(G)$ from $\widehat{G}$ by adding disjoint independent sets T_A, T_B, T_C, with

$$|T_X| = M, \qquad N_{H_M^\Sigma(G)}(t) = X \qquad (X \in \{A, B, C\},\ t \in T_X).$$

Depths in a rooted forest start at one, and ancestor includes the vertex itself. An elimination forest has one node per graph vertex and makes the endpoints of every edge comparable in the ancestor order. Write $\mathrm{totd}(F)$ for the sum of vertex depths in F, and $\mathrm{totd}(J)$ for its minimum over elimination forests of a graph J.

In any elimination forest F_G of $\widehat{G}$, the vertices of a part are pairwise comparable because they form a clique. They lie on one ancestor path and have a unique deepest vertex y_X. Define

$$W(F_G) = \sum_{X \in \{A,B,C\}} \big(\mathrm{depth}_{F_G}(y_X) + 1\big), \qquad W(G) = \min_{F_G} W(F_G).$$

For a choice of part X, let Y, Z be the other two parts. Write $\mathrm{vc}(J)$ for the minimum cardinality of a vertex cover of J, and define

$$\mathrm{cost}_X(G) = \min_{S_0} \left(2|S_0| + \mathrm{vc}\big(G[(Y \cup Z) \setminus S_0]\big)\right),$$

where $S_0 \subseteq V(G)$ covers every X–$(Y \cup Z)$ edge. All cover graphs below refer to G, before clique completion.

Lemma B.6 (Moving one added vertex)**.** *For any $X \in \{A, B, C\}$ and $u \in T_X$, every elimination forest of $H_M^\Sigma(G)$ can be changed so that u is a leaf child of the deepest vertex in X, without increasing total depth or changing the ancestor order among vertices of G.*

Proof. Let y_X be the deepest vertex of X. The edge uy_X makes u and y_X comparable. Delete u, attaching its children to its parent, or making them roots if u was a root. Reinsert u as a leaf child of y_X. Deletion preserves the ancestor order among the remaining vertices. Every neighbor of u belongs to X and is ancestral to y_X, so reinsertion preserves every edge as well.

Write $d = \mathrm{depth}_F(u)$ and $e = \mathrm{depth}_F(y_X)$, and let $D_F(u)$ be the set of proper descendants of u before the move. Every vertex of $D_F(u)$ moves up one level when u is deleted; reinserting u as a leaf does not change any other vertex's depth. If u was below y_X, its new depth is $e + 1 \le d$, so the total-depth change is

$$(e + 1 - d) - |D_F(u)| \le 0.$$

Otherwise u was a proper ancestor of y_X. Deletion moves y_X up one level, so the new depth of u is e. The old path strictly below u through y_X contains $e - d$ proper descendants of u. Consequently the total-depth change is

$$(e - d) - |D_F(u)| \le 0.$$

Thus total depth does not increase, and the ancestor order among vertices of G is unchanged. □

Lemma 5.6. *Every elimination forest of $H_M^\Sigma(G)$ can be transformed without increasing total depth so that, for every $X \in \{A, B, C\}$, all vertices of T_X are leaf children of the deepest vertex in X. Consequently,*

$$\mathrm{totd}(H_M^\Sigma(G)) = \min_{F_G} \big(\mathrm{totd}(F_G) + MW(F_G)\big),$$

and

$$MW(G) \le \mathrm{totd}(H_M^\Sigma(G)) \le MW(G) + \frac{N(N+1)}{2}.$$

Proof of Lemma 5.6. Apply Lemma B.6 to every added vertex. Previously moved vertices remain leaf children of their respective deepest vertices, so this gives the stated normalization.

Deleting the added leaves gives an elimination forest F_G of $\widehat{G}$, and their total contribution is $MW(F_G)$. Conversely, every F_G extends by these leaves to an elimination forest of $H_M^\Sigma(G)$. This proves the identity.

The lower bound follows from $\mathrm{totd}(F_G) \ge 0$ and $W(F_G) \ge W(G)$. For the upper bound, choose F_G minimizing W. Its total depth equals N plus the number of strict ancestor pairs, so $\mathrm{totd}(F_G) \le N + \binom{N}{2} = N(N+1)/2$. Substituting this forest in the minimization identity proves the upper bound. □

Fix an elimination forest F_G of $\widehat{G}$. For $v \in V(G)$, put

$$c(v) = |\{X \in \{A, B, C\} : v \text{ is ancestral to } y_X\}|,$$

and define

$$S_0 = \{v \in V(G) : c(v) = 3\}, \qquad S_1 = \{v \in V(G) : c(v) = 2\}.$$

These sets record the shared prefixes of the three clique paths.

Lemma B.7 (The same pair of descendants)**.** *If S_1 is nonempty, all its vertices are ancestral to the same two vertices among y_A, y_B, y_C.*

Proof. For two vertices of a rooted forest, their sets of descendants among y_A, y_B, y_C are disjoint or one contains the other. The sets for vertices in S_1 have size two, so any two intersect and must be equal. □

When S_1 is nonempty, write X for the part outside this common pair and Y, Z for the other parts. When S_1 is empty, choose any part as X and let Y, Z be the other two parts.

For a choice of X, Y, Z and sets S_0, S_1, write

$$\Gamma_X(S_0, S_1) = 2|S_0| + |S_1|.$$

This is the cost of a feasible pair of covers. Equivalently, $\mathrm{cost}_X(G)$ is the minimum of $\Gamma_X(S_0, S_1)$ over sets S_0 covering the X–$(Y \cup Z)$ edges and sets $S_1 \subseteq (Y \cup Z) \setminus S_0$ covering $G[(Y \cup Z) \setminus S_0]$.

Lemma B.8 (Covers from the three paths)**.** *The set S_0 covers the X–$(Y \cup Z)$ edges of G, and $S_1 \subseteq Y \cup Z$ covers $G[(Y \cup Z) \setminus S_0]$. Moreover,*

$$\begin{aligned} W(F_G) &= N + 3 + \Gamma_X(S_0, S_1) \\ &\geq N + 3 + \mathrm{cost}_X(G). \end{aligned}$$

Proof. Every vertex of a part is ancestral to its deepest vertex. A vertex of S_1 has precisely y_Y, y_Z as descendants among the three deepest vertices, so it cannot belong to X. A vertex of $X \setminus S_0$ is ancestral only to y_X, whereas a vertex of $(Y \cup Z) \setminus S_0$ has a nonempty descendant set contained in $\{y_Y, y_Z\}$. These sets are disjoint, so the vertices are incomparable and cannot be adjacent. Hence S_0 covers the first set of edges. Likewise, a YZ-edge outside S_0 with neither endpoint in S_1 would join vertices ancestral only to y_Y and only to y_Z, which are incomparable. This proves the second cover claim.

The sum of the three depths counts each vertex v exactly $c(v)$ times. Every vertex belongs to its own part's path, so $c(v) \geq 1$. The definitions of S_0, S_1 therefore give

$$\sum_{X \in \{A,B,C\}} \mathrm{depth}_{F_G}(y_X) = \sum_{v \in V(G)} c(v) = N + 2|S_0| + |S_1|.$$

Adding the three terminal offsets gives the identity; the cover claims give the inequality. □

Lemma 5.7. *For every tripartite graph G with nonempty specified parts,*

$$W(G) = N + 3 + \min\{\mathrm{cost}_A(G), \mathrm{cost}_B(G), \mathrm{cost}_C(G)\}.$$

Proof of Lemma 5.7. The lower bound is

$$W(G) = \min_{F_G} W(F_G) \overset{\text{Lem. B.8}}{\geq} N + 3 + \min_{X \in \{A,B,C\}} \mathrm{cost}_X(G).$$

For the converse, choose a part X attaining the minimum, and let Y, Z be the other two parts. Choose a cover S_0 attaining $\mathrm{cost}_X(G)$ and a minimum-cardinality cover S_1 of $G[(Y \cup Z) \setminus S_0]$. Put S_0 on a common chain. Below it, put the remaining X vertices on one chain. On the other branch put S_1 on a chain, followed by separate chains for the remaining Y and Z vertices. Empty chains are omitted. Each part lies on one ancestor path. Every cross-part edge either meets S_0 or is a remaining YZ-edge meeting S_1, so this is an elimination forest of $\widehat{G}$.

The path through S_0 to the X-chain and the two paths through S_0, S_1 to the Y- and Z-chains contain every vertex of their respective parts. If a final chain is empty, end its path at the preceding shared chain. The depth of each part's deepest vertex is at most its path length. Counting these three path lengths counts every vertex once, every vertex of S_0 twice more, and every vertex of S_1 once more. Hence

$$W(F_G) \le N + 3 + \Gamma_X(S_0, S_1).$$

The last term equals $\mathrm{cost}_X(G)$, proving the upper bound. □

Lemma B.9 (Normalizing successive covers)**.** *Let X, Y, Z be the specified parts of G. Suppose that S_0 covers the X–$(Y \cup Z)$ edges and that $S_1 \subseteq (Y \cup Z) \setminus S_0$ covers $G[(Y \cup Z) \setminus S_0]$. An independent block of vertices in one part with identical neighborhoods can be made wholly contained in S_0, wholly contained in S_1, or disjoint from both, without increasing $\Gamma_X(S_0, S_1)$. The S_1 state is unavailable for a block in X.*

Moreover, if $v \in S_0$ and every neighbor of v across the X–$(Y \cup Z)$ cut already belongs to S_0, then v can be removed from S_0 without increasing the cost. If $v \in Y \cup Z$, it suffices to add v to S_1, decreasing the cost by one. If $v \in X$, deleting it decreases the cost by two.

Proof. Give each vertex state 0, 1, or 2 according as it belongs to S_0, to S_1, or to neither; these states cost 2, 1, and 0. For a nonempty block D, choose a vertex of minimum state cost in D, and give every vertex of D its state. This cannot increase Γ_X. It also preserves both covers: every edge from D to a vertex outside D now has the same endpoint states as the existing edge from the chosen vertex to that neighbor. The first cover requires a state-0 endpoint on each first-cut edge; the second requires a state-0 or state-1 endpoint on each YZ-edge. These conditions held on the representative edge, and D has no internal edges. A block in X cannot acquire state 1, because none of its vertices had that state. An empty block needs no change.

For the last assertion, deleting v preserves the first cover by the stated condition. If $v \in X$, the residual graph is unchanged and the cost decreases by two. Otherwise, every new residual edge is incident with v, so moving v to S_1 covers all such edges and replaces cost two by cost one. □

B.5 Sparse replacements on prescribed vertex sets

We use the Margulis–Gabber–Galil family from Theorem 6.15 and the expander mixing lemma (Lemma 6.14).

Fix $\zeta > 0$ and $\kappa \ge 1$ as in Lemma 6.16. We map the vertices of a powered expander onto the prescribed sets and verify the degree, no-hole, and connectedness guarantees.

Use $\lambda_0 = 5\sqrt{2}/8$ from Theorem 6.15. Fix an integer $h \ge 1$ with $\lambda_0^h < \zeta/4$, and set

$$d_{\zeta,\kappa} = \lceil 4\kappa \rceil 8^h.$$

Use this same walk length for both constructions. For each square order $M = \nu^2$, take $\mathcal{E} = \mathcal{G}_\nu^{[h]}$ from that theorem. It has degree $d = 8^h$ and normalized nonconstant norm at most λ_0^h, so its adjacency eigenvalue bound is $d\lambda_0^h$. Lemma 6.14 gives

$$e_{\mathcal{E}}(S,T) \overset{\text{Lem. 6.14}}{\ge} d\sqrt{|S||T|}\left(\frac{\sqrt{|S||T|}}{M} - \lambda_0^h\right). \tag{177}$$

For an integer $t \ge 1$, if $|S|, |T| \ge t > \lambda_0^h M$, then $\sqrt{|S||T|}/M \ge t/M > \lambda_0^h$, so the right-hand side is positive. Hence a step joins S to T.

We accommodate arbitrary sizes by grouping auxiliary vertices and assigning each group to a prescribed vertex. Auxiliary steps give edges between distinct images; group sizes bound the output degrees.

For a map $\pi : V(\mathcal{E}) \to U$ and a subset $X \subseteq U$, the *preimage* is $\pi^{-1}(X) = \{x \in V(\mathcal{E}) : \pi(x) \in X\}$. The map is *onto* if every vertex of U has at least one preimage. When $|U| = n \le M$, partition the M auxiliary vertices into n groups. Write $M = n\lfloor M/n \rfloor + r$, where r is an integer with $0 \le r < n$. Give r groups size $\lfloor M/n \rfloor + 1$ and the other groups size $\lfloor M/n \rfloor$. Since $M \ge n$, every group is nonempty. Assign one group to each vertex of U. This gives

$$\lfloor M/n \rfloor \le |\pi^{-1}(\{u\})| \le \lceil M/n \rceil \qquad (u \in U). \tag{178}$$

Use the following two constructions.

1. *One prescribed set.* Let U have size $n \ge 1$, put $M = \lceil \sqrt{n} \rceil^2$, and take the M-vertex graph $\mathcal{E}$ above. Choose an onto map $\pi : V(\mathcal{E}) \to U$ with the group sizes just specified. Define the replacement R_U by

$$\begin{aligned} V(R_U) &= U, \\ E(R_U) &= \{uv \in \tbinom{U}{2} : e_{\mathcal{E}}(\pi^{-1}(\{u\}), \pi^{-1}(\{v\})) > 0\}. \end{aligned}$$

   Thus equal images give no edge, and repeated pairs give a single edge.

2. *Two prescribed sides.* Let A, B be disjoint sets of sizes $n_A, n_B \ge 1$, put $M = \lceil \sqrt{\max\{n_A, n_B\}} \rceil^2$, and take the M-vertex graph $\mathcal{E}$ above. Choose onto maps $\pi_A : V(\mathcal{E}) \to A$ and $\pi_B : V(\mathcal{E}) \to B$, with preimage sizes $\lfloor M/n_A \rfloor$ or $\lceil M/n_A \rceil$ for the first map, and $\lfloor M/n_B \rfloor$ or $\lceil M/n_B \rceil$ for the second. Define the replacement $R_{A,B}$ by

$$\begin{aligned} V(R_{A,B}) &= A \dot\cup B, \\ E(R_{A,B}) &= \{uv : u \in A,\ v \in B, e_{\mathcal{E}}(\pi_A^{-1}(\{u\}), \pi_B^{-1}(\{v\})) > 0\}. \end{aligned}$$

   This gives one edge for each distinct pair and no edge within either side. The two maps have the same domain, and their preimage groups may differ.

A walk is *closed* if its first and last vertices coincide.

Lemma 6.16. *For every fixed $\zeta > 0$ and $\kappa \ge 1$, there is a constant $d_{\zeta,\kappa}$ such that the following connected simple graphs, each of maximum degree at most $d_{\zeta,\kappa}$, can be constructed deterministically in polynomial time.*

1. *For any specified nonempty finite set U, there is a graph on U in which every two disjoint sets $X, Y \subseteq U$ with $|X|, |Y| \ge \lceil \zeta |U| \rceil$ have an edge between them.*

2. *For any specified disjoint nonempty finite sets A, B with $\max\{|A|, |B|\} / \min\{|A|, |B|\} \le \kappa$, there is a bipartite graph on $A \dot\cup B$ with parts A, B in which every $X \subseteq A$, $Y \subseteq B$ with $|X|, |Y| \ge \lceil \zeta(|A| + |B|) \rceil$ have an edge between them.*

Proof of Lemma 6.16. Each auxiliary vertex has $d = 8^h$ outgoing and incoming steps, because its adjacency matrix is symmetric with row sum d. For any output vertex v, let ρ be its map (π, π_A, or π_B), and let n_ρ be the size of that map's prescribed set. Count outgoing steps from its preimage,

except on side B, where we count incoming steps. Each output neighbor uses at least one such step, so the same bound covers all three cases:

$$\deg(v) \le 8^h |\rho^{-1}(\{v\})| \overset{(178)}{\le} 8^h \lceil M/n_\rho \rceil.$$

For every integer $n \ge 1$,

$$M = \lceil \sqrt{n} \rceil^2 \le (\sqrt{n}+1)^2 \le 4n. \tag{179}$$

Hence $\lceil M/n \rceil \le 4$. For the bipartite construction, the same estimate gives $M \overset{(179)}{\le} 4\max\{n_A, n_B\}$. The side-size ratio is at most κ, so $M/n_A, M/n_B \le 4\kappa$. Both constructions therefore have degree at most $\lceil 4\kappa \rceil 8^h = d_{\zeta,\kappa}$.

We next verify the no-hole conditions at the thresholds in the lemma. In the one-set case, put $t = \lceil \zeta n \rceil$. Then $\lambda_0^h M < (\zeta/4)M \overset{(179)}{\le} \zeta n \le t$. In the bipartite case, put $t = \lceil \zeta(n_A + n_B) \rceil$. Then $\lambda_0^h M < (\zeta/4)M \overset{(179)}{\le} \zeta \max\{n_A, n_B\} \le t$. We apply the mixing bound to preimages of t-sets. If no such admissible sets exist, the corresponding no-hole condition is vacuous. Otherwise, any larger admissible sets contain admissible t-element subsets. For the one-set construction, take disjoint t-sets $X, Y \subseteq U$ and put $S = \pi^{-1}(X)$, $T = \pi^{-1}(Y)$. For the bipartite construction, take t-sets $X \subseteq A$, $Y \subseteq B$ and put $S = \pi_A^{-1}(X)$, $T = \pi_B^{-1}(Y)$. In both cases surjectivity gives $|S|, |T| \ge t > \lambda_0^h M$, so (177) supplies a step from S to T. Its endpoint images give an edge between X and Y. In the one-set case the preimages are disjoint and their images are distinct. In the bipartite case the preimages may overlap, and even a loop step gives a valid edge because its images lie in the disjoint sets A, B.

It remains to prove connectedness. If $\mathcal{E}$ had a nonempty proper component C, then $x = \mathbf{1}_C - (|C|/M)\mathbf{1}$ would be a nonzero zero-sum vector with $A_{\mathcal{E}} x = dx$, contradicting its nonconstant eigenvalue bound $d\lambda_0^h < d$. Thus $\mathcal{E}$ is connected. It also has a loop at $(0,0)$: repeat h times a loop of the Margulis–Gabber–Galil graph at that vertex. This explicitly supplies the odd closed walk needed below.

In the one-set construction, applying π to an auxiliary walk gives a sequence of equal or adjacent output vertices. Deleting consecutive repetitions gives an output walk. Connectedness of $\mathcal{E}$ and surjectivity of π thus prove connectedness of the output.

For the bipartite construction we need auxiliary walks of both parities. From any x, walk to $(0,0)$, traverse its loop, and return along the reversed walk. Prepending this odd closed walk to a walk from x to y changes its parity. Map a walk $x = x_0, x_1, \dots, x_\ell = y$ to the alternating sequence $\pi_A(x_0), \pi_B(x_1), \pi_A(x_2), \dots$ in the output. Each consecutive pair is an edge by construction, using the reverse auxiliary step when the sequence goes from B to A. An even-length walk reaches $\pi_A(y)$, and an odd-length walk reaches $\pi_B(y)$. Thus $\pi_A(x)$ is connected to both images of every y, and surjectivity gives connectedness.

Both outputs are simple by their edge definitions. The auxiliary order M is at most four times the largest prescribed set size. Theorem 6.15 constructs its walk-power graph in polynomial time for fixed h. Map the endpoints as specified above and sort the resulting pairs to merge repeated edges, discarding equal images in the one-set construction. Since h depends only on ζ, these operations take polynomial time for the fixed parameters. □

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The author directed repeated investigations using up to ten parallel AI agents. Agents explored stronger bounds, compared candidate constructions, checked algebra and case analyses, and independently reviewed overlapping parts of the arguments. They also searched for missing hypotheses, undefined notation, unsupported uses of source results, and opportunities to simplify proofs. AI assistance included locating and inspecting source papers, comparing theorem statements and versions, checking citations, and spelling out the changes to constructions inherited from the literature.

Codex generated and executed scripts for finite gadget checks, algebraic and parameter calculations, and manuscript consistency checks. It also edited the LaTeX sources, reorganized proofs and appendices, standardized notation, prepared reduction diagrams, compiled the manuscript, and inspected rendered pages. The mathematical checks consisted of model reasoning, source inspection, and finite computational checks; they did not constitute formal verification of the proofs.

The author supplied the research direction, selected the questions and target bounds, reviewed and questioned arguments during development, and decided which results and presentation changes to retain. Responsibility for the manuscript, its mathematical claims, and its references rests with the author.